

STATISTICS PAPER II · 2027 FORECAST

Statistical Inference — 2027 Forecast

405 forecast questions across Estimation and Hypothesis Testing, with exam shortcut and answer key

405

Forecast Questions

17

Topics Covered

FORECAST · AI-GENERATED

No problem is copied from a past paper. Every explanatory note is an AI-derived shortcut, independently generated and verified by the builder of this platform, not an official answer key.

Contents

Point Estimation	4 questions
Methods of Estimation	5 questions
Sufficiency and Completeness	5 questions
Optimal Estimation	6 questions
Interval Estimation	5 questions
Unbiasedness, Consistency, Efficiency, Sufficiency & Completeness — Foundations	40 questions
Cramér–Rao Lower Bound & Fisher Information	40 questions
Factorization Theorem, Sufficiency & Completeness — Worked Families	30 questions
Maximum Likelihood Estimation	50 questions
Rao–Blackwell, UMVUE & Method of Moments	40 questions
Confidence Intervals	50 questions
Hypothesis Testing	15 questions
Simple and Composite Hypotheses, Two Kinds of Error, Size and Power	1 questions
UMP, MLR, Unbiased and Similar Tests	1 questions
Sequential and Decision Theory	8 questions
Neyman–Pearson Lemma, UMP Tests & Power Functions	50 questions
Likelihood Ratio Tests, SPRT & Decision Theory	55 questions

Point Estimation

4 forecast questions in this topic

Estimation

Characteristics of a Good Estimator (Unbiasedness, Consistency, Efficiency, MSE)

Easy

Q1.

An estimator T_n of θ is said to be (weakly) consistent if

A. $\mathcal{P}(|T_n - \theta| > \epsilon) \rightarrow 0$ as $n \rightarrow \infty$, for every $\epsilon > 0$

B. $\text{Var}(T_n) = 0$ for every n

C. $E(T_n) = \theta$ for every n

D. $T_n = \theta$ exactly for all sufficiently large n

Correct answer: A

EXAM SHORTCUT

Consistency is convergence in probability of T_n to θ - a large-sample property, distinct from (and not implied by) unbiasedness.

Estimation

Characteristics of a Good Estimator (Unbiasedness, Consistency, Efficiency, MSE)

Easy

Q2.

For an estimator T of θ , the mean squared error $\text{MSE}(T) = E[(T - \theta)^2]$ decomposes as

A. $\text{Var}(T) - [\text{Bias}(T)]^2$

B. $\text{Var}(T) / \text{Bias}(T)$

C. $[\text{Var}(T)]^2 + \text{Bias}(T)$

D. $\text{Var}(T) + [\text{Bias}(T)]^2$

Correct answer: D

EXAM SHORTCUT

The bias-variance decomposition: $\text{MSE} = \text{Var}(T) + [\text{Bias}(T)]^2$.

Q3.

Regarding minimization of MSE over ALL estimators (not just unbiased ones), which statement is correct?

- A. The sample mean always has the smallest MSE among all estimators of the population mean
- B. Minimizing MSE and requiring unbiasedness are equivalent conditions
- C. The MLE always has the smallest MSE for every θ
- D. **No single estimator achieves uniformly minimum MSE over all θ among all estimators - restricting to unbiased estimators is necessary for a meaningful 'best estimator' theory**

Correct answer: D

EXAM SHORTCUT

The trivial constant estimator $T \equiv \theta_0$ has $\text{MSE} = (\theta_0 - \theta)^2 = 0$ exactly at $\theta = \theta_0$, so no estimator can beat it everywhere - this is why UMVUE theory restricts attention to unbiased estimators.

Q4.

For iid observations from a standard Cauchy distribution, the sample mean $\bar{X}$ as an estimator of the location parameter

- A. **is inconsistent - $\bar{X}$ has exactly the same Cauchy distribution regardless of n**
- B. equals the maximum likelihood estimator
- C. is unbiased with variance decreasing as $1/n$
- D. is consistent and efficient

Correct answer: A

EXAM SHORTCUT

The Cauchy distribution has no finite mean or variance; remarkably, $\bar{X}$ for n iid standard Cauchy variables has EXACTLY the same standard Cauchy distribution as a single observation - it never concentrates.

Methods of Estimation

5 forecast questions in this topic

Estimation

Maximum Likelihood Estimation

Medium

Q1.

For $X_1, \dots, X_n$ iid $U(0, \theta)$, the maximum likelihood estimator of θ is

A. $X_{(1)}$, the smallest order statistic

B. ~~$X_{(n)}$~~ , the largest order statistic

C. $2\bar{X}$

D. $\bar{X}$

Correct answer: B

EXAM SHORTCUT

Likelihood $= \theta^{-n}$ for $\theta \geq X_{(n)}$, strictly decreasing in θ - so the MLE is the smallest admissible θ , namely $X_{(n)}$.

Estimation

Maximum Likelihood Estimation

Hard

Q2.

For $X_1, \dots, X_n$ iid $U(\theta - \frac{1}{2}, \theta + \frac{1}{2})$, the maximum likelihood estimator of θ

A. is uniquely $\bar{X}$

B. does not exist

C. ✓ is not unique - any value in $[X_{(n)} - \frac{1}{2}, X_{(1)} + \frac{1}{2}]$ maximizes the likelihood equally

D. is uniquely the midrange $(X_{(1)} + X_{(n)})/2$

Correct answer: C

EXAM SHORTCUT

The likelihood is constant ($=1$) for every θ satisfying $X_{(n)} - \frac{1}{2} \leq \theta \leq X_{(1)} + \frac{1}{2}$, so the MLE is non-unique - the midrange is merely one conventional choice among infinitely many.

Estimation

Method of Moments

Medium

Q3.

An iid sample from $U(a,b)$ has $\bar{X}=10$ and $\frac{1}{n} \sum (X_i - \bar{X})^2 = 12$. The method-of-moments estimates of a and b are

- A. $a=0, b=20$
- B. ~~$a=4, b=16$~~
- C. $a=8, b=12$
- D. $a=-2, b=22$

Correct answer: B

EXAM SHORTCUT

$a+b=2\bar{X}=20$; $b-a=2\sqrt{3 \times 12} = 2\sqrt{36} = 12$; solving gives $a=4, b=16$.

Estimation

Method of Moments

Easy

Q4.

For $X_1, \dots, X_n$ iid Exponential with rate θ (mean $1/\theta$), do the method-of-moments and maximum-likelihood estimators of θ coincide?

- A. No, they differ by a factor of $n/(n-1)$
- B. Only asymptotically, not for finite n
- C. No: MLE $= 1/\bar{X}$ but MoM $= \bar{X}$
- D. ~~Yes~~, both equal $1/\bar{X}$

Correct answer: D

EXAM SHORTCUT

Both MLE and MoM for the exponential rate reduce to $\hat{\theta} = 1/\bar{X}$ - a rare case of exact coincidence for finite n .

Q5.

If $\hat{\theta}$ is the maximum likelihood estimator of θ , and g is any function, the MLE of $g(\theta)$ is

- A. $g^{-1}(\hat{\theta})$
- B. equal to $\hat{\theta}$ only if g is the identity function
- C. undefined unless g is linear
- D. $g(\hat{\theta})$ - the invariance property of MLEs

Correct answer: D

EXAM SHORTCUT

The MLE invariance property: for ANY function g (not just one-to-one ones, via the induced-likelihood extension), the MLE of $g(\theta)$ is simply $g(\hat{\theta})$.

Sufficiency and Completeness

5 forecast questions in this topic

Estimation

Sufficient Estimator and Factorization Theorem

Medium

Q1.

For $X_1, \dots, X_n$ iid with pdf $f(x, \theta) = e^{-(x-\theta)}$ for $x > \theta$ (shifted exponential), a sufficient statistic for θ is

- A. $\sum X_i^2$
- B. $X_{(1)}$, the minimum order statistic
- C. $X_{(n)}$, the maximum order statistic
- D. $\bar{X}$

Correct answer: B

EXAM SHORTCUT

Since the support $x > \theta$ depends on θ through the SMALLEST value needed, the binding constraint - and hence the sufficient statistic - is $X_{(1)}$.

Estimation

Sufficient Estimator and Factorization Theorem

Hard

Q2.

For X_1, X_2 iid Poisson(θ), why does $T = X_1 + 2X_2$ fail to be sufficient for θ , while $T' = X_1 + X_2$ succeeds?

- A. ✓ The conditional distribution of (X_1, X_2) given $X_1 + X_2 = t$ is Binomial($t, \frac{1}{2}$), free of θ ; the conditional distribution given $X_1 + 2X_2$ still involves θ
- B. $X_1 + 2X_2$ need not be an integer
- C. $X_1 + 2X_2$ has larger variance than $X_1 + X_2$
- D. Sufficiency requires equal coefficients only for continuous distributions, not discrete ones

Correct answer: A

EXAM SHORTCUT

The joint pmf depends on the data only through $x_1 + x_2$ (the exponent of θ); a differently-weighted combination like $x_1 + 2x_2$ does not capture this and leaves θ in the conditional distribution.

Q3.

For $X \sim N(0, \theta)$ (mean known = 0, θ the unknown variance), which statement is correct regarding completeness?

- A. X is complete for θ , but X^2 is not
- B. X alone is not complete for θ ; but X^2 is complete for θ**
- C. Neither X nor X^2 is complete for θ
- D. Both X and X^2 are complete for θ

Correct answer: B

EXAM SHORTCUT

$g(X)=X$ has $E_\theta[X]=0$ for every θ without $X \equiv 0$ a.s., so X is not complete; but $X^2/\theta \sim \chi^2_1$ scaling makes X^2 complete.

Q4.

For iid observations from $U(\theta - \frac{1}{2}, \theta + \frac{1}{2})$, the range $R = X_{(n)} - X_{(1)}$ is

- A. Ancillary - its distribution does not depend on θ at all**
- B. distributed as $\text{Poisson}(\theta)$
- C. a complete sufficient statistic for θ
- D. an unbiased estimator of θ

Correct answer: A

EXAM SHORTCUT

Shifting θ shifts the whole sample by a constant, leaving the range unchanged in distribution - the textbook example of an ancillary statistic.

Q5.

For $X_1, \dots, X_n$ iid $N(\mu, \sigma^2)$ with both parameters unknown, a complete sufficient statistic for (μ, σ^2) is

- A. $\bar{X}$ alone
- B. (sample median, sample range) jointly
- C. $(\bar{X}, s^2)$ jointly
- D. s^2 alone

Correct answer: C**EXAM SHORTCUT**

The two-parameter normal family has $(\bar{X}, s^2)$ (equivalently $(\sum X_i, \sum X_i^2)$) as its complete sufficient statistic - neither component alone suffices for both parameters.

Optimal Estimation

6 forecast questions in this topic

Estimation

Rao-Blackwell Theorem and UMVUE Construction

Medium

Q1.

If T is sufficient for θ and U is any unbiased estimator of θ , then $\phi(T) = E[U | T]$ satisfies

- A. $\phi(T)$ always equals U
- B. $\phi(T)$ is free of θ , unbiased for θ , with $\text{Var}(\phi(T)) \leq \text{Var}(U)$ for every θ
- C. $\phi(T)$ may still depend on θ
- D. $\phi(T)$ is unbiased but always has strictly larger variance than U

Correct answer: B

EXAM SHORTCUT

This is exactly the Rao-Blackwell theorem: conditioning on a sufficient statistic never increases variance, and sufficiency guarantees the conditional expectation is a genuine (data-only) statistic.

Estimation

Rao-Blackwell Theorem and UMVUE Construction

Medium

Q2.

Rao-Blackwellizing an unbiased estimator using a statistic T produces the UMVUE if and only if T is

- A. complete only
- B. ancillary
- C. sufficient only
- D. **complete and sufficient**

Correct answer: D

EXAM SHORTCUT

Lehmann-Scheffe: completeness (in addition to sufficiency) is exactly what makes the Rao-Blackwellized estimator unique, and hence the UMVUE.

Estimation

Cramer-Rao Inequality, MVB and Fisher Information

Medium

Q3.

For $X \sim \text{Poisson}(\theta)$, based on a sample of size n , the Cramer-Rao lower bound for the variance of any unbiased estimator of θ is

- A. θ^2/n
- B. $1/(n\theta^2)$
- C. n/θ
- D. θ/n

Correct answer: D

EXAM SHORTCUT

Single-observation Fisher information for Poisson is $I(\theta)=1/\theta$; for n iid observations, $I_n(\theta)=n/\theta$, so CRLB $=\theta/n$.

Estimation

Cramer-Rao Inequality, MVB and Fisher Information

Hard

Q4.

For a single observation from $N(\mu, \theta)$ with μ known and $\theta=\sigma^2$, the Fisher information $I(\theta)$ equals

- A. $1/\theta^2$
- B. $1/(2\theta^2)$
- C. $n/(2\theta^2)$
- D. $2\theta^2$

Correct answer: B

EXAM SHORTCUT

$\log f = -\frac{1}{2} \log(2\pi\theta) - (x-\mu)^2/(2\theta)$; differentiating twice and taking $E[\cdot]$ gives $I(\theta)=1/(2\theta^2)$ for one observation.

Estimation

Bhattacharya Bounds

Medium

Q5.

The Bhattacharya bound generalizes the Cramer-Rao bound by

- A. applying only to biased estimators
- B. ignoring the likelihood function entirely
- C. **Incorporating higher-order derivatives of the log-likelihood to give a bound at least as sharp (tight) as the CRLB**
- D. requiring the estimator to be sufficient

Correct answer: C

EXAM SHORTCUT

Bhattacharya bounds use second- and higher-order score derivatives, giving a sequence of bounds that are never looser than, and often strictly sharper than, the first-order Cramer-Rao bound.

Estimation

Rao-Blackwell Theorem and UMVUE Construction

Hard

Q6.

For $X_1, \dots, X_n$ iid $\text{Poisson}(\theta)$, with $T = \sum X_i$ complete sufficient, the UMVUE of θ^2 is

- A. **$T(T-1)/n^2$**
- B. T^2/n^2
- C. $T(T+1)/n^2$
- D. $T(T-1)/n$

Correct answer: A

EXAM SHORTCUT

Using the Poisson factorial-moment identity $E[T(T-1)] = (n\theta)^2$, dividing by n^2 gives an exactly unbiased estimator of θ^2 built from the complete sufficient T .

Interval Estimation

5 forecast questions in this topic

Estimation

Confidence Intervals, Pivots and Optimum Bounds

Medium

Q1.

For $X_1, \dots, X_n$ iid Exponential with mean θ , a pivotal quantity for constructing a confidence interval for θ is

A. $\bar{X} - \theta \sim \chi^2_1$

B. $n\bar{X}^2/\theta \sim F$

C. $2n\bar{X}/\theta \sim \chi^2_{2n}$

D. $\bar{X}/\theta \sim N(0,1)$

Correct answer: C

EXAM SHORTCUT

Sums of exponential variables relate to the gamma/chi-square family: $2n\bar{X}/\theta \sim \chi^2_{2n}$ exactly, for every n .

Estimation

Confidence Intervals, Pivots and Optimum Bounds

Medium

Q2.

A 95% confidence interval for μ (with $\sigma=4$ known), based on a sample of size n , is reported as (18.04, 21.96).

The sample mean and sample size are

A. $\bar{X}=20, n=16$

B. $\bar{X}=20, n=64$

C. $\bar{X}=20, n=4$

D. $\bar{X}=19, n=16$

Correct answer: A

EXAM SHORTCUT

Midpoint $= 20 = \bar{X}$; half-width $= 1.96 = 1.96(4)/\sqrt{n} \Rightarrow \sqrt{n} = 4 \Rightarrow n = 16$.

Estimation

Resampling: Bootstrap and Jackknife

Easy

Q3.

The bootstrap method estimates the sampling distribution of a statistic by

- A. assuming normality and computing exact moments
- B. requiring a known parametric family for the population
- C. **resampling with replacement from the observed data**
- D. resampling without replacement, taking exactly half the data each time

Correct answer: C

EXAM SHORTCUT

Bootstrap = resample WITH replacement from the empirical distribution (the observed sample) to approximate the true sampling distribution non-parametrically.

Estimation

Resampling: Bootstrap and Jackknife

Medium

Q4.

The jackknife estimator of the bias of a statistic T_n is based on

- A. bootstrap resampling with replacement
- B. **leave-one-out recomputation of the statistic**
- C. a single random split of the data into two halves
- D. repeating the full-sample estimate n times

Correct answer: B

EXAM SHORTCUT

Jackknife systematically deletes one observation at a time, recomputes $T_{(i)}$ each time, and uses the spread of these leave-one-out values to estimate bias (and variance).

Estimation

Confidence Intervals, Pivots and Optimum Bounds

Easy

Q5.

To halve the width of a 95% confidence interval for μ (with σ known), the sample size must be

- A. doubled
- B. increased by 50%
- C. **quadrupled**
- D. halved

Correct answer: C

EXAM SHORTCUT

CI width $\propto 1/\sqrt{n}$; halving the width requires $\sqrt{n}$ to double, i.e. n to quadruple.

Unbiasedness, Consistency, Efficiency, Sufficiency & Completeness — Foundations

40 forecast questions in this topic

Estimation

Unbiasedness

Medium

Q1.

An estimator $\hat{\theta}$ is unbiased if:

- A. $MSE(\hat{\theta}) = 0$
- B. $E[\hat{\theta}] = \theta$ for all θ
- C. $Var(\hat{\theta}) = 0$
- D. $\hat{\theta} \rightarrow \theta$ in probability

Correct answer: B

EXAM SHORTCUT

D is consistency, not unbiasedness; A implies both bias=0 AND var=0 simultaneously

Estimation

MSE Decomposition

Medium

Q2.

$MSE(\hat{\theta})$ equals:

- A. $Var(\hat{\theta})$ only
- B. $[Bias(\hat{\theta})]^2$ only
- C. $Var(\hat{\theta}) + [Bias(\hat{\theta})]^2$
- D. $Var(\hat{\theta}) - [Bias(\hat{\theta})]^2$

Correct answer: C

EXAM SHORTCUT

A holds only when estimator is unbiased; never subtract bias²

Estimation

Consistency

Medium

Q3.

$\hat{\theta}_n$ is consistent for θ if:

- A. $\hat{\theta}_n \rightarrow \theta$ almost surely
- B. $E[\hat{\theta}_n] \rightarrow \theta$ and $\text{Var}(\hat{\theta}_n) \rightarrow 0$
- C. Both A and B
- D. $P(|\hat{\theta}_n - \theta| > \epsilon) \rightarrow 0$ for every $\epsilon > 0$

Correct answer: D

EXAM SHORTCUT

B is sufficient (by Chebyshev) but not necessary; A is strong consistency (stricter)

Estimation

Efficiency

Medium

Q4.

The relative efficiency of $\hat{\theta}_1$ w.r.t. $\hat{\theta}_2$ is:

- A. $\text{Bias}(\hat{\theta}_1)/\text{Bias}(\hat{\theta}_2)$
- B. $\text{MSE}(\hat{\theta}_1)/\text{MSE}(\hat{\theta}_2)$
- C. $\text{Var}(\hat{\theta}_1)/\text{Var}(\hat{\theta}_2)$
- D. $\text{Var}(\hat{\theta}_2)/\text{Var}(\hat{\theta}_1)$

Correct answer: D

EXAM SHORTCUT

Efficiency of $\hat{\theta}_1 = \text{CRLB}/\text{Var}(\hat{\theta}_1)$; relative efficiency = Var of denominator over numerator

Estimation

Sufficiency Definition

Medium

Q5.

A statistic $T(X)$ is sufficient for θ if:

- A. The conditional distribution of X given T does not depend on θ
- B. T has minimum variance among all unbiased estimators
- C. T attains the Cramér-Rao lower bound
- D. T is a complete statistic

Correct answer: A

EXAM SHORTCUT

Sufficiency is about conditional distribution, not variance; completeness is a separate concept

Estimation

Admissibility

Hard

Q6.

An estimator $\hat{\theta}$ is inadmissible if:

A. ~~There exists another estimator with smaller MSE for ALL θ~~

B. It is not a function of sufficient statistic

C. It is biased

D. It does not attain the CRLB

Correct answer: A

EXAM SHORTCUT

Biased estimators can be admissible (e.g., James-Stein); CRLB non-attainment $\neq$ inadmissibility

Estimation

Ancillary Stat

Hard

Q7.

A statistic $A(X)$ is ancillary if:

A. $E[A(X)] = 0$ for all θ

B. It is independent of the sufficient statistic

C. ~~Its distribution does not depend on θ~~

D. It has minimum variance

Correct answer: C

EXAM SHORTCUT

B is Basu's theorem consequence (ancillary $\perp$ complete sufficient), not the definition

Estimation

Completeness Def

Hard

Q8.

A statistic T is complete if:

A. ~~$E_{\theta}[g(T)] = 0$ for all θ implies $g(T) = 0$ a.s.~~

B. T achieves minimum variance

C. The family of distributions of T is full-rank exponential

D. T is sufficient and unbiased

Correct answer: A

EXAM SHORTCUT

C is sufficient for completeness (exponential family with full rank) but not the definition itself

Estimation

Consistent Unbiased

Medium

Q9.

Sample mean $\bar{X}$ from $N(\mu, \sigma^2)$ is:

- A. **Unbiased and consistent for μ**
- B. Unbiased but inconsistent for μ
- C. Consistent but biased for μ
- D. Neither unbiased nor consistent

Correct answer: A

EXAM SHORTCUT

$E[\bar{X}] = \mu$ (unbiased); $\text{Var}(\bar{X}) = \sigma^2/n \rightarrow 0$ (consistent)

Estimation

Best Linear Unbiased

Medium

Q10.

By Gauss-Markov theorem, OLS estimator is BLUE meaning:

- A. Best (minimum variance) among all unbiased estimators
- B. Best among all estimators including biased ones
- C. Minimum MSE among linear estimators
- D. **Best (minimum variance) among all LINEAR unbiased estimators**

Correct answer: D

EXAM SHORTCUT

“Linear” is crucial; MLE may be better overall; Gauss-Markov does NOT require normality

Estimation

Asymptotic Efficiency

Hard

Q11.

An estimator is asymptotically efficient if:

- A. $\text{Var}(\hat{\theta}) = 1/(nI(\theta))$ exactly
- B. $\hat{\theta}$ is consistent
- C. $\hat{\theta}$ is unbiased for all n
- D. **$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, 1/I(\theta))$**

Correct answer: D

EXAM SHORTCUT

A is exact CRLB attainment (finite n); D is the asymptotic version; B alone is insufficient

Estimation

Pitman Closeness

Hard

Q12.

Pitman closeness criterion prefers $\hat{\theta}_1$ over $\hat{\theta}_2$ if:

- A. $\text{Bias}(\hat{\theta}_1) < \text{Bias}(\hat{\theta}_2)$
- B. $\text{MSE}(\hat{\theta}_1) < \text{MSE}(\hat{\theta}_2)$
- C. $\text{Var}(\hat{\theta}_1) < \text{Var}(\hat{\theta}_2)$
- D. $P(|\hat{\theta}_1 - \theta| < |\hat{\theta}_2 - \theta|) > 1/2$

Correct answer: D

EXAM SHORTCUT

Pitman closeness can contradict MSE ordering — classic ISS trap

Estimation

Invariance MLE

Medium

Q13.

If $\hat{\theta}$ is MLE of θ , then MLE of $g(\theta)$ is:

- A. $g(\hat{\theta})$ only if g is monotone
- B. Obtained by separate maximization
- C. $g(\hat{\theta})$ always
- D. $g(\hat{\theta})$ only if g is differentiable

Correct answer: C

EXAM SHORTCUT

Invariance of MLE holds for ANY function g , not just monotone or differentiable ones

Estimation

Unbiased Estimator Unique

Hard

Q14.

If a complete sufficient statistic exists, then:

- A. All unbiased estimators are functions of it
- B. There is at most one unbiased estimator based on it
- C. There is exactly one UMVUE
- D. The MLE is always unbiased

Correct answer: C

EXAM SHORTCUT

B is wrong (there can be many unbiased estimators); C follows from Lehmann- Scheffé theorem

Estimation

Fisher Information Additive

Medium

Q15.

For i.i.d. sample of size n , $ln(\theta)$ equals:

- A. $l_1(\theta)/n$
- B. $\sqrt{n} \cdot l_1(\theta)$
- C. $\sqrt{n} \cdot l_1(\theta)$
- D. $l_1(\theta)$

Correct answer: C

EXAM SHORTCUT

Fisher information is ADDITIVE for independent observations; $CRLB = 1/(n l_1(\theta))$

Estimation

Median Unbiased

Hard

Q16.

An estimator $\hat{\theta}$ is median-unbiased if:

- A. $P(\hat{\theta} \leq \theta) = P(\hat{\theta} \geq \theta) = 1/2$
- B. $\hat{\theta}$ has symmetric distribution
- C. $E[\hat{\theta}] = \theta$
- D. Median of $\hat{\theta}$ equals $E[\hat{\theta}]$

Correct answer: A

EXAM SHORTCUT

Mean-unbiasedness $\neq$ median-unbiasedness; for skewed distributions they differ

Estimation

Cramer Rao Regularity

Hard

Q17.

CRLB requires regularity conditions including:

- A. The distribution must be normal
- B. The sample size must be large
- C. θ must be a location parameter
- D. Support of $f(x; \theta)$ must NOT depend on θ

Correct answer: D

EXAM SHORTCUT

For $U(0, \theta)$, $\text{Exp}(\theta)$ shifted — support depends on θ , so CRLB does NOT apply

Estimation

Sufficiency Minimal

Hard

Q18.

A minimal sufficient statistic is:

A. ~~Both A and B~~

B. A sufficient statistic of smallest possible dimension

C. A function of every other sufficient statistic

D. The complete sufficient statistic

Correct answer: A

EXAM SHORTCUT

Minimal sufficient = coarsest sufficient partition; complete sufficient $\subseteq$ minimal sufficient (when it exists)

Estimation

Equivariant Estimator

Hard

Q19.

A location equivariant estimator $\hat{\theta}$ satisfies:

A. $E[\hat{\theta}] = \theta$

B. $\hat{\theta}(cX) = \hat{\theta}(X)$ for all $c > 0$

C. ~~$\hat{\theta}(X+c) = \hat{\theta}(X) + c$~~ for all c

D. $\hat{\theta}$ is a function of order statistics

Correct answer: C

EXAM SHORTCUT

B is scale equivariance; both together give affine equivariance

Estimation

Consistent Estimator Function

Medium

Q20.

If $\hat{\theta}_n \xrightarrow{p} \theta$ and g is continuous, then:

A. $E[g(\hat{\theta}_n)] \rightarrow g(\theta)$

B. $g(\hat{\theta}_n)$ may not be consistent

C. $g(\hat{\theta}_n) \xrightarrow{p} g(\theta)$ only if g is monotone

D. ~~$g(\hat{\theta}_n) \xrightarrow{p} g(\theta)$~~

Correct answer: D

EXAM SHORTCUT

Continuous mapping theorem guarantees D; A requires uniform integrability (stronger condition)

Estimation

Bias Variance Tradeoff

Medium

Q21.

Shrinkage estimators like James-Stein are preferred because:

- A. They satisfy CRLB
- B. They are unbiased with smaller variance
- C. **They introduce bias but reduce variance enough to lower total MSE**
- D. They are consistent while OLS is not

Correct answer: C

EXAM SHORTCUT

James-Stein is BIASED; its advantage is total MSE reduction in $p \geq 3$ dimensions

Estimation

Order Stat Distribution

Medium

Q22.

PDF of k -th order statistic $X(k)$ from $f(x)$ with CDF $F(x)$ is:

- A. $n! k!(n-k)! [F(x)]^k [1-F(x)]^{n-k} f(x)$
- B. $n [1-F(x)]^{n-1} f(x)$
- C. **$n! (k-1)!(n-k)! [F(x)]^{k-1} [1-F(x)]^{n-k} f(x)$**
- D. $n [F(x)]^{n-1} f(x)$

Correct answer: C

EXAM SHORTCUT

D is PDF of maximum ($k = n$); B is PDF of minimum ($k = 1$); check factorial carefully

Estimation

Exponential Family Complete

Hard

Q23.

For a full-rank exponential family $f(x; \theta) = h(x) \exp[\eta(\theta)T(x) - B(\theta)]$:

- A. $T(x)$ is sufficient but not necessarily complete
- B. Completeness depends on sample size
- C. **$T(x)$ is complete sufficient**
- D. $T(x)$ is complete but not sufficient

Correct answer: C

EXAM SHORTCUT

Full-rank (open set in natural parameter space) is the key condition; curved exponential families may lack completeness

Estimation

Ancillary Basu

Hard

Q24.

By Basu's theorem, if T is complete sufficient and A is ancillary, then:

- A. T and A are independent
- B. A is a function of T
- C. T and A have the same distribution
- D. A is also sufficient

Correct answer: A

EXAM SHORTCUT

This is Basu's theorem — independence, not functional relationship; requires completeness

Estimation

Unbiased Estimator Existence

Hard

Q25.

For $U(0, \theta)$, an unbiased estimator of θ^{-1} based on $X(n)$:

- A. $(n-1)/(n \cdot X_{(n)})$
- B. $1/X_{(n)}$
- C. $1/((n+1)/n \cdot X_{(n)})$
- D. Does not exist

Correct answer: A

EXAM SHORTCUT

$E[X_{(n)}] = n\theta/(n+1)$; by direct integration $E[1/X_{(n)}] = n/((n-1)\theta)$ for $n > 1$, so $(n-1)/(n \cdot X_{(n)})$ is unbiased for $1/\theta$.

Estimation

Lehmann Scheffe

Hard

Q26.

Lehmann-Scheffé theorem states:

- A. Complete sufficient statistic is always MLE
- B. Rao-Blackwell improvement always yields UMVUE
- C. **Any unbiased estimator based on complete sufficient statistic is UMVUE**
- D. UMVUE always attains CRLB

Correct answer: C

EXAM SHORTCUT

D is false (UMVUE may not attain CRLB); B requires completeness for uniqueness

Estimation

Fisher Consistency

Hard

Q27.

An estimator T_n is Fisher consistent if:

- A. $T_n \xrightarrow{p} \theta$ as $n \rightarrow \infty$
- B. **The functional $T(F\theta) = \theta$ where $F\theta$ is the true CDF**
- C. $E[T_n] = \theta$ for all n
- D. T_n is a function of sufficient statistic

Correct answer: B

EXAM SHORTCUT

Fisher consistency is a property of the functional form, not a sequence property; different from ordinary consistency

Estimation

Superefficiency

Hard

Q28.

Hodges' superefficient estimator demonstrates:

- A. **An estimator can have asymptotic variance below CRLB at isolated points**
- B. CRLB can be beaten uniformly
- C. Superefficiency implies admissibility
- D. MLE is never asymptotically efficient

Correct answer: A

EXAM SHORTCUT

Superefficiency occurs only on a set of measure zero; Le Cam showed it cannot hold on a dense set

Estimation

Stein Paradox

Hard

Q29.

James-Stein estimator dominates sample mean in terms of MSE when:

- A. $p \geq 1$ (any dimension)
- B. **$p \geq 3$**
- C. $p \geq 2$
- D. $p \geq 4$

Correct answer: B

EXAM SHORTCUT

For $p = 1, 2$ sample mean is admissible; Stein paradox kicks in at $p \geq 3$

Estimation

Moment Estimator Consistency

Medium

Q30.

Method of moments estimators are generally:

- A. Consistent and efficient
- B. Neither consistent nor efficient
- C. Efficient but not consistent
- D. **Consistent but not necessarily efficient**

Correct answer: D

EXAM SHORTCUT

MOM estimators are consistent (by LLN) but typically less efficient than MLE

Estimation

CRLB Multiparameter

Hard

Q31.

For multiparameter case, CRLB for $g(\theta)$ involves:

- A. **Fisher information matrix $I(\theta)$ and gradient ∇g**
- B. Scalar Fisher information
- C. Only diagonal elements of $I(\theta)$
- D. Determinant of $I(\theta)$

Correct answer: A

EXAM SHORTCUT

$\text{Var}(g) \geq [\nabla g]^T I(\theta)^{-1} [\nabla g]$; off-diagonal elements matter

Estimation

Truncated Distribution MLE

Hard

Q32.

For truncated normal $N(\mu, 1)$ truncated to $(0, \infty)$, MLE of μ :

- A. Is $\bar{X}$ (same as untruncated)
- B. Is the sample median
- C. **Requires solving a nonlinear equation**
- D. Does not exist

Correct answer: C

EXAM SHORTCUT

Truncation changes the likelihood; MLE satisfies $X = \mu + \phi(\mu)/\Phi(\mu)$ (Mills ratio equation)

Estimation

Robust Estimator

Medium

Q33.

The sample median is preferred over sample mean when:

- A. Distribution is symmetric
- B. **Distribution has heavy tails or outliers**
- C. Sample size is large
- D. Distribution is normal

Correct answer: B

EXAM SHORTCUT

For normal distribution, mean is more efficient; median's advantage is robustness to outliers

Estimation

Unbiased Variance Estimator

Medium

Q34.

$S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$ is unbiased for σ^2 because:

- A. $\bar{X}$ is independent of S^2 under normality
- B. **$E[\sum (X_i - \bar{X})^2] = (n-1)\sigma^2$**
- C. $E[\sum (X_i - \bar{X})^2] = n\sigma^2$
- D. Both A and C

Correct answer: B

EXAM SHORTCUT

The $(n-1)$ denominator corrects for one degree of freedom lost by estimating μ with $\bar{X}$

Estimation

Jackknife Bias

Medium

Q35.

The jackknife bias estimate for $\hat{\theta}$ is:

- A. $n(\hat{\theta} - \hat{\theta}(\cdot))$
- B. $\hat{\theta} - \hat{\theta}(\cdot)$
- C. **$(n-1)(\hat{\theta}(\cdot) - \hat{\theta})$**
- D. $(n-1)(\hat{\theta} - \hat{\theta}(\cdot))$

Correct answer: C

EXAM SHORTCUT

$\hat{\theta}(\cdot) = (1/(n-1)) \sum \hat{\theta}_i$ is mean of leave-one-out estimates; bias = $(n-1)(\hat{\theta}(\cdot) - \hat{\theta})$

Estimation

Bootstrap Principle

Medium

Q36.

Bootstrap estimates sampling distribution of $\hat{\theta}$ by:

- A. Generating data from fitted parametric model only
- B. ☒ Resampling WITH replacement from observed data
- C. Analytical approximation using CLT
- D. Resampling WITHOUT replacement from observed data

Correct answer: B

EXAM SHORTCUT

With replacement is key; without replacement is subsampling (different method); parametric bootstrap uses fitted model

Estimation

Delta Method

Medium

Q37.

By delta method, asymptotic variance of $g(\bar{X})$ is:

- A. ☒ $[g'(\mu)]^2 \cdot \sigma^2/n$
- B. $[g''(\mu)]^2 \cdot \sigma^2/n$
- C. $[g(\mu)]^2 \cdot \sigma^2/n$
- D. $g'(\mu) \cdot \sigma^2/n$

Correct answer: A

EXAM SHORTCUT

First derivative squared, not second; applies when $g'(\mu) \neq 0$

Estimation

Variance Stabilizing

Hard

Q38.

Variance stabilizing transformation for $\text{Poisson}(\lambda)$ is:

- A. $\log X$
- B. ☒ $\sqrt{X}$
- C. X^2
- D. $1/X$

Correct answer: B

EXAM SHORTCUT

For Poisson, $\text{Var}(X) = \lambda = E[X]$; VST satisfies $g'(\mu) \propto 1/\sqrt{\mu}$, giving $g(\mu) = 2\sqrt{\mu}$; $\sqrt{X}$ is the answer

Estimation

Ancillary Example

Hard

Q39.

For location family $f(x-\theta)$, which is ancillary?

- A. Sample maximum $X(n)$
- B. Sample variance S^2 (when θ is location only)
- C. Sample range $X(n) - X(1)$
- D. Sample mean $\bar{X}$

Correct answer: C

EXAM SHORTCUT

Range and all spacings are ancillary for location families; S^2 is ancillary only if scale is known

Estimation

Completeness Bernoulli

Medium

Q40.

For $X_1, \dots, X_n \sim \text{Bernoulli}(p)$, $T = \sum X_i$ is complete because:

- A. $E_p[g(T)] = 0$ for all $p \in (0, 1)$ implies $g = 0$ (polynomial argument)
- B. It is sufficient
- C. It has binomial distribution
- D. It is the MLE

Correct answer: A

EXAM SHORTCUT

Completeness proof: $E[g(T)] = \sum_{t=0}^n g(t) \binom{n}{t} p^t (1-p)^{n-t} = 0$ for all p implies all $g(t) = 0$.

Cramér–Rao Lower Bound & Fisher Information

40 forecast questions in this topic

Estimation

CRLB Formula

Medium

Q1.

For unbiased estimator $\hat{\theta}$, CRLB states:

A. $\text{Var}(\hat{\theta}) \geq n I(\theta)$

B. $\text{Var}(\hat{\theta}) \geq n I(\theta)$

C. $\text{Var}(\hat{\theta}) \geq I(\theta)$

D. $\text{Var}(\hat{\theta}) \geq I(\theta)$

Correct answer: A

EXAM SHORTCUT

For i.i.d. sample of size n : $\text{CRLB} = 1/(n I(\theta))$ where I is single-observation Fisher info

Estimation

Fisher Info Normal

Medium

Q2.

For $X \sim N(\mu, \sigma^2)$ with σ^2 known, $I(\mu)$ equals:

A. $2/\sigma^4$

B. $1/(2\sigma^2)$

C. $1/\sigma^2$

D. σ^2

Correct answer: C

EXAM SHORTCUT

$\log f = -((x-\mu)^2)/(2\sigma^2) + C$; $\partial^2 \log f / \partial \mu^2 = -1/\sigma^2$; $I(\mu) = 1/\sigma^2$

Estimation

Fisher Info Bernoulli

Medium

Q3.

For $X \sim \text{Bernoulli}(p)$, $I(p)$ equals:

A. $1/p + 1/(1-p)$

B. $1/[p(1-p)]$

C. $p(1-p)$

D. $p/(1-p)$

Correct answer: B

EXAM SHORTCUT

$I(p) = E[(\partial \log f / \partial p)^2] = 1/[p(1-p)]$; CRLB for $\bar{X}$ is $p(1-p)/n$ — verify: $\text{Var}(\bar{X}) = p(1-p)/n$ attains CRLB

Estimation

Fisher Info Poisson

Medium

Q4.

For $X \sim \text{Poisson}(\lambda)$, $I(\lambda)$ equals:

A. $1/\lambda^2$ B. $1/\lambda$ C. λ^2 D. λ

Correct answer: B

EXAM SHORTCUT

$\log f = x \log \lambda - \log(x!); \partial^2 \log f / \partial \lambda^2 = -x/\lambda^2; I(\lambda) = E[X]/\lambda^2 = 1/\lambda$

Estimation

CRLB Function Parameter

Medium

Q5.

CRLB for estimating $g(\theta)$ (unbiased) is:

A. $[g'(\theta)]^2 I(\theta)$ B. $(g'(\theta))/(nI(\theta))$ C. $[g(\theta)]^2 nI(\theta)$ D. $[g'(\theta)]^2 nI(\theta)$

Correct answer: D

EXAM SHORTCUT

Chain rule applies: $[g'(\theta)]^2$ in numerator; A forgets the n factor

Estimation

CRLB Attainment Condition

Hard

Q6.

CRLB is attained if and only if:

A. $\partial \log f / \partial \theta = A(\theta)[T(x) - g(\theta)]$ for some functions A, g

B. The sample size is large

C. The estimator is MLE

D. The distribution is normal

Correct answer: A

EXAM SHORTCUT

This is the exact attainment condition; it implies $T(x)$ is UMVUE for $g(\theta)$; only exponential family members typically satisfy this

Estimation

CRLB Exponential

Medium

Q7.

For $X \sim \text{Exp}(\theta)$ (rate θ), CRLB for $\hat{\theta}$ (unbiased) is:

- A. n/θ^2
- B. θ/n
- C. $1/(n\theta^2)$
- D. θ^2/n

Correct answer: D

EXAM SHORTCUT

$l(\theta) = 1/\theta^2$; CRLB = θ^2/n ; MLE $1/\hat{X}$ attains this bound

Estimation

CRLB Uniform DNE

Hard

Q8.

For $X \sim U(0, \theta)$, CRLB for θ :

- A. Equals $4\theta^2/n$
- B. Equals $\theta^2/(3n)$
- C. Does not exist
- D. Equals θ^2/n

Correct answer: C

EXAM SHORTCUT

Support $(0, \theta)$ depends on θ ; regularity condition violated; CRLB framework inapplicable; UMVUE variance is $\theta^2/[n(n+2)]$

Estimation

Information Inequality Biased

Hard

Q9.

For biased estimator with bias $b(\theta)$, the generalized CRLB is:

- A. $\text{Var}(\hat{\theta}) \geq ([b'(\theta)]^2)/(nI(\theta))$
- B. $\text{MSE}(\hat{\theta}) \geq nI(\theta)$
- C. $\text{Var}(\hat{\theta}) \geq ([1 + b'(\theta)]^2)/(nI(\theta))$
- D. $\text{Var}(\hat{\theta}) \geq nI(\theta)$

Correct answer: C

EXAM SHORTCUT

$[1 + b'(\theta)]^2$ not $[b'(\theta)]^2$; when $b(\theta) = 0$ (unbiased), reduces to standard CRLB

Estimation

Bhattacharya Bound

Hard

Q10.

Bhattacharya bounds improve upon CRLB by:

- A. Using larger sample sizes
- B. Restricting to exponential families
- C. Using sufficient statistics
- D. Using higher-order score functions (derivatives of log-likelihood)**

Correct answer: D

EXAM SHORTCUT

Bhattacharya uses $\partial^k \log f / \partial \theta^k$ for $k = 1, 2, \dots$; first-order Bhattacharya = CRLB; higher orders give tighter bounds

Estimation

Bhattacharya Normal Theta Sq

Hard

Q11.

For $X \sim N(\theta, 1)$, estimating θ^2 , Bhattacharya bound (2nd order) is:

- A. $2/n$
- B. $4\theta^2/n + 2/n^2$**
- C. $4\theta^2/n - 2/n^2$
- D. $4\theta^2/n$

Correct answer: B

EXAM SHORTCUT

CRLB gives $4\theta^2/n$; 2nd-order Bhattacharya adds $2/n^2$ term; UMVUE $X^2 - 1/n$ has variance $4\theta^2/n + 2/n^2$ — attains 2nd-order bound

Estimation

Fisher Info Gamma

Hard

Q12.

For $X \sim \Gamma(\alpha, \beta)$ with α known, $I(\beta)$ equals:

- A. β^2/α
- B. α/β^2**
- C. α^2/β^2
- D. $1/(\alpha\beta^2)$

Correct answer: B

EXAM SHORTCUT

$E[X] = \alpha\beta$, $\text{Var}(X) = \alpha\beta^2$; $I(\beta) = \alpha/\beta^2$; CRLB for $\hat{\beta} = X/\alpha$ is $\beta^2/(n\alpha)$ ✓

Estimation

CRLB Location Family

Hard

Q13.

For location family $f(x-\theta)$, Fisher information is:

- A. $I(\theta) = \int [(f'(x))/(f(x))]^2 f(x) dx$ (independent of θ)
- B. $I(\theta)$ depends on θ
- C. $I(\theta) = \theta^2 \int [(f'(x))/(f(x))]^2 f(x) dx$
- D. $I(\theta) = 1/\sigma^2$ always

Correct answer: A

EXAM SHORTCUT

For location families, Fisher information is constant (does not depend on θ); this is a key property

Estimation

CRLB Scale Family

Hard

Q14.

For scale family $(1/(\theta f))(x/\theta)$, $I(\theta)$ equals:

- A. θ^2 always
- B. $1/\theta^2$ times location Fisher information
- C. $\theta^2 \int [(f'(x))/(f(x))]^2 f(x) dx$
- D. $\theta^2 \int [1 + (xf'(x))/(f(x))]^2 f(x) dx$

Correct answer: D

EXAM SHORTCUT

Score for scale family: $\partial \log f / \partial \theta = -1/\theta - (x/\theta^2) f'(x/\theta)/f(x/\theta)$; careful substitution needed

Estimation

Efficient Estimator Definition

Medium

Q15.

An unbiased estimator is efficient if:

- A. Its variance equals CRLB
- B. **Both A and B (they are equivalent when CRLB is attained)**
- C. Its variance is less than all other unbiased estimators
- D. It is consistent

Correct answer: B

EXAM SHORTCUT

When CRLB is attained, the estimator is UMVUE; A and C are equivalent in this context

Estimation

CRLB Cauchy

Hard

Q16.

For $X \sim \text{Cauchy}(\theta, 1)$, $I(\theta)$ equals:

- A. Does not exist
- B. 2
- C. $1/2$
- D. 1

Correct answer: C

EXAM SHORTCUT

$f(x; \theta) = \pi[1+(x-\theta)^2]^{-1}$; $I(\theta) = \int \frac{1}{4(x-\theta)^2} \pi^2[1+(x-\theta)^2]^{-4} \cdot \pi[1+(x-\theta)^2] dx = 1/2$; sample mean is inconsistent for Cauchy

Estimation

Information Matrix Bivariate

Hard

Q17.

For $N(\mu, \sigma^2)$ with both parameters unknown, Fisher information matrix is:

- A. $\text{diag}(1/\sigma^2, 2\sigma^4)$
- B. $\text{diag}(1/\sigma^2, 1/(2\sigma^4))$
- C. $\text{diag}(n/\sigma^2, n/(2\sigma^4))$
- D. Non-diagonal matrix

Correct answer: B

EXAM SHORTCUT

Off-diagonal is zero (score functions are uncorrelated); single-observation matrix; multiply by n for sample

Estimation

CRLB Beta Distribution

Hard

Q18.

For $X \sim \text{Beta}(\theta, 1)$, $f(x; \theta) = \theta x^{\theta-1}$, $I(\theta)$ equals:

- A. θ
- B. θ^2
- C. $1/\theta$
- D. $1/\theta^2$

Correct answer: D

EXAM SHORTCUT

$\log f = \log \theta + (\theta-1) \log x$; $\partial \log f / \partial \theta = 1/\theta + \log x$; $I(\theta) = \text{Var}(1/\theta + \log X) = \text{Var}(\log X) = 1/\theta^2$ (since $-\log X \sim \text{Exp}(\theta)$)

Estimation

CRLB Truncated

Hard

Q19.

For $X \sim \text{Exp}(\theta)$ truncated to $(0, c)$, CRLB:

- A. Is larger than θ^2/n (less information)
- B. Does not exist
- C. **Is smaller than θ^2/n (more information)**
- D. Is same as untruncated: θ^2/n

Correct answer: C

EXAM SHORTCUT

Truncation adds information (we know $X < c$); Fisher information increases; CRLB decreases

Estimation

Bhattacharya First Order

Medium

Q20.

The first-order Bhattacharya bound coincides with:

- A. **CRLB**
- B. Rao-Blackwell bound
- C. Chapman-Robbins bound
- D. Hammersley-Chapman-Robbins bound

Correct answer: A

EXAM SHORTCUT

Bhattacharya hierarchy: 1st order = CRLB; higher orders give tighter bounds; Chapman-Robbins is a different (non-differentiability-requiring) bound

Estimation

Fisher Info Geometric

Medium

Q21.

For $X \sim \text{Geometric}(p)$, $P(X = k) = (1-p)^{k-1}p$, $I(p)$ equals:

- A. **$p^2(1-p)$**
- B. $p(1-p)$
- C. p^2
- D. $(1-p)^2$

Correct answer: A

EXAM SHORTCUT

$E[X] = 1/p$; $\text{Var}(X) = (1-p)/p^2$; $I(p) = 1/[p^2(1-p)]$; CRLB for $\hat{p} = 1/X$ is $p^2(1-p)/n$

Estimation

CRLB Negative Binomial

Hard

Q22.

For $X \sim \text{NegBin}(r, p)$ with r known, $I(p)$ equals:

- A. $p^2(1-p)$
- B. $r^2 p^2(1-p)$
- C. $r p(1-p)$
- D. $\checkmark p^2(1-p)$

Correct answer: D

EXAM SHORTCUT

NegBin is exponential family; $I(p) = r/[p^2(1-p)]$; for $r=1$ reduces to geometric case

Estimation

Attainment Exponential Family

Hard

Q23.

In exponential family $f(x; \theta) = h(x) \exp[\eta(\theta)T(x) - B(\theta)]$, CRLB for $E\theta[T(X)]$ is attained by:

- A. $-T=(1)/(n)\sum T(X_i)$
- B. $T(X)$ itself
- C. $\checkmark$ All of the above
- D. MLE of $E\theta[T(X)]$

Correct answer: C

EXAM SHORTCUT

In exponential family, T is UMVUE for $E[T]$, equals MLE (by invariance), and attains CRLB — all three coincide

Estimation

CRLB Double Exponential

Hard

Q24.

For $X \sim \text{Laplace}(\theta, 1)$, $f(x; \theta) = (1)/(2e^{-|x-\theta|})$, $I(\theta)$ equals:

- A. 2
- B. Does not exist
- C. 1/2
- D. $\checkmark$ 1

Correct answer: D

EXAM SHORTCUT

$\partial \log f / \partial \theta = \text{sgn}(x-\theta)$; $I(\theta) = E[\text{sgn}(X-\theta)^2] = 1$; sample median (not mean) is MLE and UMVUE

Estimation

CRLB Weibull

Hard

Q25.

For Weibull with shape k known and scale λ : $f(x; \lambda) = (k)/(\lambda)(x/\lambda)^{k-1}e^{-(x/\lambda)^k}$, $I(\lambda)$ equals:

- A. k^2/λ
- B. k^2/λ^2
- C. k/λ^2
- D. $1/\lambda^2$

Correct answer: B

EXAM SHORTCUT

Score: $\partial \log f / \partial \lambda = -k/\lambda + k(x/\lambda)^k/\lambda$; $I(\lambda) = k^2/\lambda^2$; CRLB = $\lambda^2/(nk^2)$

Estimation

Information Reparametrization

Hard

Q26.

If $\phi = g(\theta)$ and g is differentiable, then $I(\phi)$ equals:

- A. $I(\theta) \cdot [g'(\theta)]^2$
- B. $I(\theta)/[g'(\theta)]^2$
- C. $I(\theta) \cdot g'(\theta)$
- D. $I(\theta)/g'(\theta)$

Correct answer: B

EXAM SHORTCUT

Fisher information transforms inversely to variance; CRLB for $g(\theta)$ is $[g'(\theta)]^2/(nI(\theta))$ which equals $1/(nI(\phi))$ ✓

Estimation

CRLB Mixture

Hard

Q27.

For mixture $f(x; \theta) = \theta f_1(x) + (1 - \theta)f_2(x)$, $I(\theta)$ equals:

- A. $I_1(\theta) + I_2(\theta)$
- B. $\theta I_1(\theta) + (1 - \theta)I_2(\theta)$
- C. $\theta^2 I_1(\theta) + (1 - \theta)^2 I_2(\theta)$
- D. $\int [f_1(x) - f_2(x)]^2 \theta f_1(x) + (1 - \theta)f_2(x) dx$

Correct answer: D

EXAM SHORTCUT

Score = $[f_1(x) - f_2(x)]/f(x; \theta)$; $I(\theta) = E[\text{score}^2]$ gives D; B is wrong — mixture Fisher info is NOT weighted average

Estimation

CRLB Censored

Hard

Q28.

For Type-I right-censored exponential data, Fisher information compared to complete data:

A. Is smaller (censoring loses information)

B. Is larger (censoring adds information about censoring mechanism)

C. Is the same

D. Depends on censoring time

Correct answer: A

EXAM SHORTCUT

Censoring always loses information; $I_{\text{censored}} < I_{\text{complete}}$; CRLB is larger for censored data

Estimation

Bhattacharya Poisson

Hard

Q29.

For $X \sim \text{Poisson}(\lambda)$, estimating $e^{-2\lambda}$, CRLB equals:

A. $4\lambda e^{-4\lambda}/n$

B. $4\lambda^2 e^{-4\lambda}/n$

C. $e^{-4\lambda}/n$

D. $4e^{-4\lambda}/n$

Correct answer: A

EXAM SHORTCUT

$g(\lambda) = e^{-2\lambda}$; $g'(\lambda) = -2e^{-2\lambda}$; $I(\lambda) = 1/\lambda$; $\text{CRLB} = [g'(\lambda)]^2 / (nI(\lambda)) = 4e^{-4\lambda} \cdot \lambda / n$

Estimation

CRLB Pareto

Hard

Q30.

For Pareto $f(x; \theta) = \theta/x^{\theta+1}$, $x > 1$, $I(\theta)$ equals:

A. $1/\theta^2$

B. θ^2

C. θ

D. $1/\theta$

Correct answer: A

EXAM SHORTCUT

$\log f = \log \theta - (\theta + 1) \log x$; $\partial \log f / \partial \theta = 1/\theta - \log x$; $I(\theta) = \text{Var}(\log X) = 1/\theta^2$ (since $\log X \sim \text{Exp}(\theta)$)

Estimation

CRLB Inverse Gaussian

Hard

Q31.

For Inverse Gaussian $IG(\mu, \lambda)$ with λ known, $I(\mu)$ equals:

- A. $1/\mu^2$
- B. μ^3/λ
- C. λ/μ^3
- D. λ^2/μ^3

Correct answer: C

EXAM SHORTCUT

$I(\mu) = \lambda/\mu^3$; CRLB for $\hat{\mu}$ is $\mu^3/(n\lambda)$; verify: $\text{Var}(X) = \mu^3/(n\lambda)$ ✓ attains CRLB

Estimation

CRLB Logistic

Hard

Q32.

For $X \sim \text{Logistic}(\theta, 1)$, $I(\theta)$ equals:

- A. 3
- B. $3/\pi^2$
- C. $\pi/3$
- D. $\pi^2/3$

Correct answer: C

EXAM SHORTCUT

$f(x; \theta) = e^{-(x-\theta)/[1+e^{-(x-\theta)}]^2}$; $I(\theta) = 1/3$; CRLB = $3/n$; sample mean has variance $\pi^2/(3n) > 3/n$ — not efficient

Estimation

CRLB Binomial P

Medium

Q33.

For $X \sim \text{Bin}(n, p)$ (single observation), $I(p)$ equals:

- A. $np(1-p)$
- B. $1/[p(1-p)]$
- C. $p(1-p)/n$
- D. $n/[p(1-p)]$

Correct answer: D

EXAM SHORTCUT

$\text{Binomial}(n, p) = \text{sum of } n \text{ Bernoulli}(p)$; $I_{\text{Bin}}(n, p) = n \cdot I_{\text{Bern}}(p) = n/[p(1-p)]$

Estimation

CRLB Multivariate Normal

Hard

Q34.

For $X \sim N(\mu, \Sigma)$ with Σ known, $I(\mu)$ (as matrix) equals:

- A. Σ
- B. Σ^{-1}
- C. $|\Sigma|^{-1/p}$
- D. $\text{tr}(\Sigma^{-1})/p$

Correct answer: B

EXAM SHORTCUT

Fisher information matrix for location in multivariate normal is Σ^{-1} ; CRLB matrix is Σ/n ; $\bar{X}$ attains it

Estimation

CRLB Curved Exponential

Hard

Q35.

For curved exponential family (e.g., $N(\theta, \theta^2)$), CRLB:

- A. Is attained by MLE asymptotically
- B. **Is never attained exactly for finite n**
- C. Is attained by method of moments
- D. Does not exist

Correct answer: B

EXAM SHORTCUT

Curved exponential families generally cannot attain CRLB for finite n ; MLE is asymptotically efficient

Estimation

Information Processing Inequality

Hard

Q36.

Data processing inequality for Fisher information states:

- A. $I(T(\theta)) \leq I(X(\theta))$ for any statistic $T = T(X)$
- B. $I(T(\theta)) \geq I(X(\theta))$ for sufficient T
- C. **$I(T(\theta)) = I(X(\theta))$ if and only if T is sufficient**
- D. $I(T(\theta)) = I(X(\theta))$ always

Correct answer: C

EXAM SHORTCUT

A is true in general; C is the equality condition (sufficiency); B is wrong — sufficient statistic preserves, not increases, information

Estimation

CRLB Sequential

Hard

Q37.

In sequential sampling with random sample size N , CRLB:

- A. Does not apply
- B. Is $1/(N \cdot I(\theta))$ with N fixed
- C. ~~Is~~ $1/(E[N] \cdot I(\theta))$
- D. Requires Wald's identity

Correct answer: C

EXAM SHORTCUT

For sequential procedures, expected Fisher information is $E[N] \cdot I(\theta)$; CRLB uses expected sample size

Estimation

Bhattacharya Normal Variance

Hard

Q38.

For $X \sim N(0, \sigma^2)$, estimating σ^2 , CRLB equals:

- A. σ^4/n
- B. σ^2/n
- C. $4\sigma^4/n$
- D. ~~$2\sigma^4/n$~~

Correct answer: D

EXAM SHORTCUT

$I(\sigma^2) = 1/(2\sigma^4)$; CRLB = $2\sigma^4/n$; $n\sum X_i^2$ is UMVUE and attains CRLB (since $\mu = 0$ known)

Estimation

CRLB Attained Examples

Hard

Q39.

Which distribution's MLE does NOT attain CRLB for finite n ?

- A. Bernoulli(p), estimating p
- B. ~~$N(\mu, \sigma^2)$~~ with both unknown, estimating σ^2
- C. $N(\mu, \sigma^2)$ with σ^2 known, estimating μ
- D. Poisson(λ), estimating λ

Correct answer: B

EXAM SHORTCUT

S^2 (with $n-1$ denominator) is UMVUE but does NOT attain CRLB; $\hat{\sigma}^2$ (with n denominator) is biased; neither attains CRLB

Q40.

Which condition is NOT required for CRLB?

- A. Support of $f(x; \theta)$ independent of θ
- B. $f(x; \theta)$ twice differentiable in θ
- C. Interchange of differentiation and integration
- D. Distribution must be from exponential family**

Correct answer: D

EXAM SHORTCUT

CRLB applies to any regular family; exponential family is sufficient for attainment but NOT required for the bound itself

Factorization Theorem, Sufficiency & Completeness — Worked Families

30 forecast questions in this topic

Estimation

Factorization Theorem

Medium

Q1.

By Fisher-Neyman factorization theorem, $T(X)$ is sufficient for θ iff:

A. $f(x; \theta) = g(T(x); \theta) \cdot h(x)$

B. $f(x; \theta) = g(T(x); \theta) + h(x)$

C. $f(x; \theta) = g(x) \cdot h(T(x); \theta)$

D. $f(x; \theta) = g(T(x)) \cdot h(x; \theta)$

Correct answer: A

EXAM SHORTCUT

g depends on x ONLY through $T(x)$ and on θ ; h depends on x but NOT on θ ; product not sum

Estimation

Sufficient Stat Normal

Medium

Q2.

For $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ with both unknown, minimal sufficient statistic is:

A. **All of the above are equivalent**

B. $(X, \sum X_i^2)$

C. (X, S^2)

D. $(\sum X_i, \sum X_i^2)$

Correct answer: A

EXAM SHORTCUT

All three are equivalent (one-to-one functions of each other); minimal sufficient statistic is not unique in representation

Estimation

Sufficient Stat Uniform

Medium

Q3.

For $X_1, \dots, X_n \sim U(0, \theta)$, sufficient statistic is:

- A. $\bar{X}$
- B. $X_{(1)}$ (minimum)
- C. $\sum X_i$
- D. $X_{(n)}$ (maximum)

Correct answer: D

EXAM SHORTCUT

$f(x; \theta) = \theta^{-n} I(X_{(n)} \leq \theta)$; factorization gives $T = X_{(n)}$; $\bar{X}$ is NOT sufficient here

Estimation

Sufficient Stat Exponential

Medium

Q4.

For $X_1, \dots, X_n \sim \text{Exp}(\theta)$, sufficient statistic is:

- A. $\sum X_i$
- B. $\bar{X}$
- C. Both B and C
- D. $X_{(1)}$

Correct answer: C

EXAM SHORTCUT

$\bar{X}$ and $\sum X_i$ are one-to-one functions of each other; both are sufficient; $X_{(1)}$ is NOT sufficient for rate parameter

Estimation

Sufficient Stat Cauchy

Hard

Q5.

For $X_1, \dots, X_n \sim \text{Cauchy}(\theta, 1)$, sufficient statistic is:

- A. $X_{(1)}$ and $X_{(n)}$
- B. $(X_{(1)}, X_{(2)}, \dots, X_{(n)})$ — all order statistics
- C. $\bar{X}$
- D. Sample median

Correct answer: B

EXAM SHORTCUT

Cauchy is NOT exponential family; likelihood does not factor to reduce dimension; full order statistics are minimal sufficient

Estimation

Complete Sufficient Uniform

Hard

Q6.

For $U(0, \theta)$, $X(n)$ is complete sufficient. UMVUE of θ is:

- A. $n+1X(n)$
- B. $(n+1)/(nX(n))$
- C. $X(n)$
- D. $2X$

Correct answer: B

EXAM SHORTCUT

$E[X(n)] = n\theta/(n+1)$ (unbiased correction: $(n+1)/n X(n)$); $2X$ is unbiased but NOT UMVUE (not function of complete sufficient stat)

Estimation

Factorization Beta

Medium

Q7.

For $X_1, \dots, X_n \sim \text{Beta}(\theta, 1)$, $f(x; \theta) = \theta x^{\theta-1}$, sufficient statistic is:

- A. $\sum X_i$
- B. $\sum \log X_i$
- C. $\prod X_i$
- D. **Both B and C**

Correct answer: D

EXAM SHORTCUT

$\log f = n \log \theta + (\theta-1) \sum \log x_i$; $T = \sum \log X_i$ is sufficient; $\prod X_i = \exp(\sum \log X_i)$ is equivalent

Estimation

Complete Stat Uniform Both

Hard

Q8.

For $X_1, \dots, X_n \sim U(\alpha, \beta)$ with both unknown, complete sufficient statistic is:

- A. (X, S^2)
- B. $X(n)$ alone
- C. $X(1)$ alone
- D. **$(X(1), X(n))$**

Correct answer: D

EXAM SHORTCUT

Both endpoints needed; $(X(1), X(n))$ is complete sufficient for (α, β) ; neither alone is sufficient

Estimation

Factorization Shifted Exp

Hard

Q9.

For $X_1, \dots, X_n \sim f(x; \theta) = e^{-(x-\theta)}$, $x > \theta$, sufficient statistic is:

- A. $\sum X$
- B. $\sum X_i$
- C. $X(1)$ (minimum)
- D. $X(n)$ (maximum)

Correct answer: C

EXAM SHORTCUT

$f(x; \theta) = e^{-\sum (x_i - \theta)} I(X(1) > \theta) = e^{-n\theta} e^{-\sum x_i} I(X(1) > \theta)$; $T = X(1)$ is sufficient

Estimation

Completeness Uniform Both

Hard

Q10.

For $U(\theta, \theta + 1)$, complete sufficient statistic is:

- A. No complete sufficient statistic exists
- B. $\sum X$
- C. $X(n) - X(1)$ (range)
- D. $(X(1), X(n))$

Correct answer: D

EXAM SHORTCUT

Range $X(n) - X(1)$ is ancillary (not sufficient); $(X(1), X(n))$ is minimal sufficient and complete

Estimation

Factorization Twoparam

Hard

Q11.

For $N(\mu, \sigma^2)$ with both unknown, joint likelihood factors through:

- A. $(\sum X_i, \sum X_i^2)$
- B. $(\bar{X}, \sum (X_i - \bar{X})^2)$
- C. $(\bar{X}, S^2)$ where S^2 uses $n-1$
- D. All of the above

Correct answer: D

EXAM SHORTCUT

All are equivalent representations of the same sufficient statistic; factorization theorem doesn't specify unique form

Estimation

Sufficient Not Complete

Hard

Q12.

An example of sufficient but NOT complete statistic:

- A. $X(n)$ for $U(0, \theta)$
- B. $(X(1), X(n))$ for $N(\mu, 1)$
- C. $\bar{X}$ for $N(\mu, 1)$
- D. $\sum X_i$ for $\text{Poisson}(\lambda)$

Correct answer: B

EXAM SHORTCUT

$(X(1), X(n))$ is sufficient for μ in $N(\mu, 1)$ but NOT complete; $E[X(n) + X(1) - 2\mu] = 0$ but $X(n) + X(1) - 2\mu \neq 0$ a.s.

Estimation

Basu Theorem Application

Hard

Q13.

For $N(\mu, \sigma^2)$ with σ^2 known, by Basu's theorem:

- A. $\bar{X}$ and S^2 are independent
- B. $\bar{X}$ and sample range are independent
- C. All order statistics are independent of $\bar{X}$
- D. S^2 and $\bar{X}$ are independent (since $\bar{X}$ is complete sufficient and S^2 is ancillary)

Correct answer: B

EXAM SHORTCUT

$\bar{X}$ is complete sufficient for μ (with σ^2 known); range is ancillary for μ ; by Basu, they're independent

Estimation

Factorization Pareto

Medium

Q14.

For $X_1, \dots, X_n \sim \text{Pareto}(\theta)$, $f(x; \theta) = \theta/x^{\theta+1}$, $x > 1$, sufficient statistic is:

- A. $\prod X_i$
- B. $\sum X_i$
- C. $\sum \log X_i$
- D. $X(1)$

Correct answer: C

EXAM SHORTCUT

$\log f = n \log \theta - (\theta + 1) \sum \log x_i$; $T = \sum \log X_i$ is sufficient; $\prod X_i = \exp(\sum \log X_i)$ equivalent

Estimation

Complete Stat Exponential Family

Hard

Q15.

For k -parameter exponential family with natural parameters $\eta_1, \dots, \eta_k$ varying freely, the natural sufficient statistic $(T_1, \dots, T_k)$ is:

- A. Complete but not sufficient
- B. Minimal sufficient only
- C. **Complete sufficient**
- D. Sufficient only

Correct answer: C

EXAM SHORTCUT

Full-rank exponential family: natural sufficient statistic is BOTH complete AND minimal sufficient; “full-rank” = natural parameter space has non-empty interior

Estimation

Factorization Geometric

Medium

Q16.

For $X_1, \dots, X_n \sim \text{Geometric}(p)$, sufficient statistic is:

- A. **Both B and C**
- B. $\sum X_i$
- C. $\sum X_i$
- D. $X_{(1)}$

Correct answer: A

EXAM SHORTCUT

$\log f = \sum x_i \log(1-p) + n \log p$; $T = \sum X_i$ is sufficient; X is equivalent

Estimation

Sufficiency Independence

Hard

Q17.

If T is sufficient for θ and U is independent of T , then:

- A. U is ancillary
- B. U contains no information about θ
- C. (T, U) is sufficient for θ
- D. **All of the above**

Correct answer: D

EXAM SHORTCUT

Independence from sufficient statistic implies ancillarity; (T, U) is trivially sufficient; ancillary statistics carry no θ -information

Estimation

Minimal Sufficient Ratio

Hard

Q18.

Lehmann-Scheffé method finds minimal sufficient statistic via:

- A. Rao-Blackwell theorem
- B. Maximizing likelihood over θ
- C. Finding complete statistic
- D. Ratio $f(x; \theta)/f(y; \theta)$ constant in θ iff $T(x) = T(y)$

Correct answer: D

EXAM SHORTCUT

The ratio method: T is minimal sufficient iff the ratio of likelihoods is free of θ exactly when $T(x) = T(y)$

Estimation

Complete Stat Discrete

Hard

Q19.

For $X \sim \text{Bin}(n, p)$ with n known, X itself is complete sufficient because:

- A. $E[g(X)] = \sum_{x=0}^n g(x) \binom{n}{x} p^x (1-p)^{n-x} = 0$ for all p implies $g=0$
- B. It is the MLE
- C. It is a function of order statistics
- D. It attains CRLB

Correct answer: A

EXAM SHORTCUT

Completeness proof uses the fact that a polynomial in $p/(1-p)$ that is identically zero must have all coefficients zero

Estimation

Factorization Hypergeometric

Hard

Q20.

For Hypergeometric(N, M, n) with M unknown, sufficient statistic is:

- A. $\sum X$
- B. No reduction possible
- C. $\sum X_i$ (number of successes in sample)
- D. $X_{(1)}$

Correct answer: C

EXAM SHORTCUT

Hypergeometric is NOT exponential family but $\sum X_i$ is still sufficient; factorization applies directly

Estimation

Sufficiency Transformation

Medium

Q21.

If T is sufficient for θ and $\phi = g(T)$ is one-to-one, then:

- A. ϕ is also sufficient for θ
- B. ϕ is sufficient only if g is linear
- C. ϕ is sufficient only if g is monotone
- D. ϕ is not necessarily sufficient

Correct answer: A

EXAM SHORTCUT

One-to-one functions of sufficient statistics are sufficient; the factorization still holds through the bijection

Estimation

Complete Stat Location

Hard

Q22.

For location family $f(x-\theta)$ with $\theta \in \mathbb{R}$, the order statistics $(X(1), \dots, X(n))$ are:

- A. Minimal sufficient but NOT complete
- B. Complete sufficient
- C. Sufficient but not minimal
- D. Neither sufficient nor complete

Correct answer: A

EXAM SHORTCUT

For location families, order statistics are minimal sufficient but NOT complete; spacings $X(i+1) - X(i)$ are ancillary, providing non-trivial functions with zero expectation

Estimation

Factorization Multinomial

Hard

Q23.

For Multinomial($n; p_1, \dots, p_k$) with $\sum p_i = 1$, sufficient statistic is:

- A. $(n_1, \dots, n_{k-1})$ (cell counts, last determined by constraint)
- B. $n_1/n, \dots, n_{k-1}/n$ (proportions)
- C. All of the above
- D. $(n_1, \dots, n_k)$

Correct answer: C

EXAM SHORTCUT

All are equivalent representations; $(n_1, \dots, n_{k-1})$ determines $n_k = n - \sum_{i=1}^{k-1} n_i$

Estimation

Sufficiency Nuisance

Hard

Q24.

When $\theta = (\psi, \lambda)$ with λ nuisance, T is sufficient for ψ in the presence of λ if:

- A. T is ancillary for λ
- B. T is sufficient for $\theta = (\psi, \lambda)$ jointly
- C. T is complete for ψ
- D. Conditional distribution of X given T is free of ψ (not necessarily λ)

Correct answer: B

EXAM SHORTCUT

Joint sufficiency implies marginal sufficiency; conditional sufficiency (given λ) is a weaker concept

Estimation

Complete Stat Two sample

Hard

Q25.

For two independent samples $X_i \sim N(\mu_1, \sigma^2)$ and $Y_j \sim N(\mu_2, \sigma^2)$ with common σ^2 , complete sufficient statistic is:

- A. $(\bar{X}, \bar{Y}, S_p^2)$ where S_p^2 is pooled variance
- B. $(\bar{X}, \bar{Y}, \sum (X_i - \bar{X})^2 + \sum (Y_j - \bar{Y})^2)$
- C. $(\sum X_i, \sum Y_j, \sum X_i^2 + \sum Y_j^2)$
- D. All of the above

Correct answer: D

EXAM SHORTCUT

All are equivalent; three-dimensional sufficient statistic for three parameters (μ_1, μ_2, σ^2)

Estimation

Factorization Inverse Gaussian

Hard

Q26.

For $X \sim IG(\mu, \lambda)$ with μ known, sufficient statistic for λ is:

- A. $\sum (1/X_i - 1/\mu)$
- B. $\sum X_i$
- C. $\sum (X_i - \mu)^2 / X_i$
- D. $\sum X_i$

Correct answer: C

EXAM SHORTCUT

$\log f \propto \lambda \sum [(x_i - \mu)^2 / x_i]$; $T = \sum (X_i - \mu)^2 / X_i$ is sufficient for λ

Estimation

Completeness Truncated Family

Hard

Q27.

For Poisson(λ) truncated to $\{1, 2, 3, \dots\}$ (zero-truncated), $\sum X_i$ is:

- A. Complete but not sufficient
- B. **✓Complete sufficient**
- C. Sufficient but not complete
- D. Neither

Correct answer: B

EXAM SHORTCUT

Zero-truncated Poisson is still exponential family (with modified normalizing constant); $\sum X_i$ remains complete sufficient

Estimation

Sufficient Stat Regression

Hard

Q28.

In simple linear regression $Y_i = \alpha + \beta x_i + \epsilon_i$, $\epsilon_i \sim N(0, \sigma^2)$, sufficient statistic for $(\alpha, \beta, \sigma^2)$ is:

- A. $(\hat{\alpha}, \hat{\beta}, \hat{\sigma}^2)$ — OLS estimates
- B. $(Y, \sum (x_i - \bar{x})Y_i, \text{RSS})$
- C. $(\sum Y_i, \sum x_i Y_i, \sum Y_i^2)$
- D. **✓All of the above**

Correct answer: D

EXAM SHORTCUT

All are equivalent three-dimensional sufficient statistics; OLS estimates are one-to-one functions of the natural sufficient statistics

Estimation

Basu Theorem Proof Idea

Hard

Q29.

Basu's theorem proof relies on:

- A. **✓Completeness of T and ancillarity of A to show $P(A \in B | T = t) = P(A \in B)$**
- B. Factorization theorem
- C. Independence of order statistics
- D. Rao-Blackwell theorem

Correct answer: A

EXAM SHORTCUT

Key step: $E_\theta[P(A \in B | T) - P_\theta(A \in B)] = 0$ for all θ ; by completeness, $P(A \in B | T) = P_\theta(A \in B)$ a.s., giving independence

Estimation

Factorization Logistic Regression

Hard

Q30.

In logistic regression with $P(Y_i = 1) = \frac{e^{\beta x_i}}{1 + e^{\beta x_i}}$, sufficient statistic for β is:

A. $\sum x_i Y_i$ **B.** $-Y$ **C.** $(\sum Y_i, \sum x_i Y_i)$ **D.** $\sum Y_i$ **Correct answer: A****EXAM SHORTCUT**

$\log L = \beta \sum x_i y_i - \sum \log(1 + e^{\beta x_i})$; $T = \sum x_i Y_i$ is sufficient for β (single parameter)

Maximum Likelihood Estimation

50 forecast questions in this topic

Estimation

MLE Normal Both

Medium

Q1.

For $X_1, \dots, X_n \sim N(\mu, \sigma^2)$, MLE of σ^2 is:

A. $\frac{1}{n+1} \sum (X_i - \bar{X})^2$

B. $\frac{1}{n} \sum (X_i - \mu)^2$

C. $S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$

D. $\hat{\sigma}^2 = \frac{1}{n} \sum (X_i - \bar{X})^2$

Correct answer: D

EXAM SHORTCUT

MLE uses n denominator (biased); unbiased estimator uses $n-1$; B is MLE only when μ is known

Estimation

MLE Uniform

Medium

Q2.

For $X_1, \dots, X_n \sim U(0, \theta)$, MLE of θ is:

A. $\bar{X}$

B. $X(n)$

C. $2\bar{X}$

D. $(n+1)/(n\bar{X})$

Correct answer: B

EXAM SHORTCUT

Likelihood $L(\theta) = \theta^{-n} I(\theta \geq X(n))$ is decreasing in θ ; minimized at $\theta = X(n)$; D is UMVUE not MLE

Estimation

MLE Shifted Exponential

Hard

Q3.

For $X_1, \dots, X_n \sim e^{-(x-\theta)}, x > \theta$, MLE of θ is:

A. $X(1)$

B. $\bar{X} - 1$

C. $X(1) - 1/n$

D. $\bar{X}$

Correct answer: A

EXAM SHORTCUT

Likelihood $L(\theta) = e^{-\sum (x_i - \theta)} I(\theta \leq X(1)) = e^{n\theta} e^{-\sum x_i} I(\theta \leq X(1))$; increasing in θ ; maximized at $\theta = X(1)$

Estimation

MLE Cauchy

Hard

Q4.

For $X_1, \dots, X_n \sim \text{Cauchy}(\theta, 1)$, MLE of θ :

- A. Is $X_{(n/2)}$ (middle order statistic)
- B. Is $\bar{X}$
- C. Is sample median
- D. **Has no closed form; requires numerical optimization**

Correct answer: D

EXAM SHORTCUT

Score equation $\sum 2(x_i - \theta) = 0$ has no closed form; sample median is a good starting point but MLE $\neq$ median exactly

Estimation

MLE Invariance Example

Medium

Q5.

If $\hat{\lambda} = \bar{X}$ is MLE of λ for Poisson, MLE of $P(X=0) = e^{-\lambda}$ is:

- A. $1 - \bar{X}$
- B. **$e^{-\bar{X}}$**
- C. $(1/n) \sum I(X_i = 0)$
- D. $e^{-\lambda}$ evaluated at UMVUE

Correct answer: B

EXAM SHORTCUT

By invariance of MLE: $\hat{e^{-\lambda}} = e^{-\hat{\lambda}} = e^{-\bar{X}}$; C is unbiased but not MLE; note $e^{-\bar{X}} \neq$ UMVUE

Estimation

MLE Consistency

Medium

Q6.

Under regularity conditions, $\hat{\theta}_n$ satisfies:

- A. Both A and B
- B. **$\sqrt{n}(\hat{\theta}_n - \theta_0) \rightarrow \text{true value}$**
- C. $\hat{\theta}_n$ is unbiased for all n
- D. $\hat{\theta}_n$ a.s. $\rightarrow \theta_0$

Correct answer: B

EXAM SHORTCUT

MLE is consistent (in probability) under regularity; strong consistency (a.s.) requires additional conditions; C is generally false

Estimation

MLE Asymptotic Normality

Medium

Q7.

Under regularity conditions, $\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} :$

A. $N(0, 1/I(\theta))$ B. ~~A~~Both B and C (same thing for scalar θ)C. $N(0, I(\theta)^{-1})$ D. $N(0, I(\theta))$

Correct answer: B

EXAM SHORTCUT

For scalar θ : $1/I(\theta)$ and $I(\theta)^{-1}$ are identical; for vector θ : $I(\theta)^{-1}$ is the matrix inverse

Estimation

MLE Multimodal

Hard

Q8.

When log-likelihood has multiple local maxima, MLE is:

A. The average of all local maxima

B. The local maximum closest to MOM estimate

C. Undefined

D. ~~A~~The global maximum

Correct answer: D

EXAM SHORTCUT

MLE is defined as the GLOBAL maximum; in practice, numerical methods may find local maxima — this is a computational challenge, not a definitional one

Estimation

MLE Boundary

Hard

Q9.

For $X_1, \dots, X_n \sim N(\mu, 1)$ with constraint $\mu \geq 0$, MLE of μ is:

A. $\bar{X}$ alwaysB. $\bar{X}$ if $\bar{X} > 0$, undefined otherwiseC. $|\bar{X}|$ D. ~~A~~ $\max(\bar{X}, 0)$

Correct answer: D

EXAM SHORTCUT

Constrained MLE: if unconstrained MLE $\bar{X} \geq 0$, use it; if $\bar{X} < 0$, boundary value $\mu = 0$ maximizes constrained likelihood

Estimation

MLE Discrete Uniform

Hard

Q10.

For $X_1, \dots, X_n \sim \text{DiscreteUniform}\{1, 2, \dots, N\}$ with N unknown, MLE of N is:

- A. $X(n)$
- B. $\lceil 2X \rceil$
- C. $X(n) + 1$
- D. $2X - 1$

Correct answer: A

EXAM SHORTCUT

$L(N) = N - nI(N \geq X(n))$; decreasing in N ; MLE = $X(n)$ (smallest valid N); note MLE is biased

Estimation

MLE Mixture EM

Hard

Q11.

EM algorithm for MLE in mixture models:

- A. **E-step: compute expected complete-data log-likelihood; M-step: maximize it**
- B. E-step: maximize; M-step: compute expectation
- C. Directly maximizes observed-data likelihood
- D. Requires complete data

Correct answer: A

EXAM SHORTCUT

EM: E-step computes $Q(\theta|\theta(t)) = E[\log L_C | \text{observed}, \theta(t)]$; M-step maximizes Q ; guaranteed to increase observed likelihood

Estimation

MLE Exponential Family

Medium

Q12.

In exponential family, MLE satisfies:

- A. $E\theta[T(X)] = T$ (observed sufficient statistic equals expected)
- B. Score equation: $\sum \partial \log f / \partial \theta = 0$
- C. Neither; MLE requires numerical methods
- D. **Both A and B**

Correct answer: D

EXAM SHORTCUT

In exponential family, score equation reduces to moment equation $E\theta[T] = T$; both characterizations are equivalent

Estimation

MLE Regularity Failure

Hard

Q13.

MLE may fail to be consistent when:

- A. Distribution is normal
- B. Likelihood is unimodal
- C. **Number of parameters grows with sample size (Neyman-Scott problem)**
- D. Sample size is large

Correct answer: C

EXAM SHORTCUT

Neyman-Scott: incidental parameters problem; MLE of variance in panel data with individual fixed effects is inconsistent

Estimation

MLE Poisson Truncated

Hard

Q14.

For zero-truncated Poisson, MLE $\hat{\lambda}$ satisfies:

- A. $\hat{\lambda} = \bar{X}$
- B. **$\bar{X} = \lambda / (1 - e^{-\lambda})$**
- C. $\hat{\lambda} = \bar{X} / (1 - e^{-\bar{X}})$
- D. $\hat{\lambda} = \bar{X} - 1$

Correct answer: B

EXAM SHORTCUT

Truncated mean $E[X|X > 0] = \lambda / (1 - e^{-\lambda})$; setting equal to $\bar{X}$ gives implicit equation; no closed form

Estimation

MLE Gamma Shape

Hard

Q15.

For $\Gamma(\alpha, \beta)$ with both unknown, MLE of α satisfies:

- A. **$\log \hat{\alpha} - \psi(\hat{\alpha}) = \log \bar{X} - \log \bar{X}$**
- B. $\hat{\alpha} = n / \sum \log X_i$
- C. $\hat{\alpha} = \bar{X} / \beta$
- D. No closed form exists; use Newton-Raphson

Correct answer: A

EXAM SHORTCUT

ψ is digamma function; A is the score equation for α ; D is correct in spirit but A specifies the equation; C gives $\hat{\alpha}$ in terms of $\hat{\beta}$

Estimation

MLE Logistic

Hard

Q16.

For logistic regression, MLE:

A. Both B and C

B. Is always unique

C. Requires iterative methods (IRLS or Newton-Raphson)

D. Has closed form via OLS

Correct answer: A

EXAM SHORTCUT

Logistic MLE: no closed form; IRLS converges to unique global maximum (loglikelihood is concave); uniqueness requires full rank design matrix

Estimation

MLE Censored Exponential

Hard

Q17.

For Type-II censored sample (smallest r out of n observed) from $\text{Exp}(\theta)$, MLE of θ is:

A. $r/\sum_{i=1}^r X_{(i)}$

B. $1/X_{(r)}$

C. $r/[\sum_{i=1}^r X_{(i)} + (n-r)X_{(r)}]$

D. $1/\bar{X}_r$ where $\bar{X}_r$ is mean of r observed

Correct answer: C

EXAM SHORTCUT

Total time on test = $\sum_{i=1}^r X_{(i)} + (n-r)X_{(r)}$; MLE = $r/(\text{total time on test})$; A ignores the censored contribution.

Estimation

MLE Weibull Shape

Hard

Q18.

For Weibull(k, λ) with λ known, MLE of k satisfies:

A. $\hat{k} = 1/\log \bar{X}$

B. $\hat{k} = n/\sum \log X_i$

C. $\hat{k} = \bar{X}/\lambda$

D. $1/\hat{k} + (1/n)\sum \log X_i - (\sum X_i^{\hat{k}} \log X_i)/(\sum X_i^{\hat{k}}) = 0$

Correct answer: D

EXAM SHORTCUT

Score equation for Weibull shape is transcendental; B is MLE for Beta($\theta, 1$) not Weibull; requires numerical solution

Estimation

MLE Dirichlet

Hard

Q19.

For Dirichlet($a_1, \dots, a_k$), MLE of a_j satisfies:

A. $\psi(a_j) - \psi(a_0) = \log X_j$ where $a_0 = \sum a_j$

B. $\hat{a}_j = X_j$

C. No MLE exists

D. $\hat{a}_j = \bar{n} X_j$

Correct answer: A

EXAM SHORTCUT

ψ is digamma; system of equations must be solved simultaneously; B/D are MOM not MLE

Estimation

MLE Nonregular Rate

Hard

Q20.

For $X_1, \dots, X_n \sim U(\theta - 1/2, \theta + 1/2)$, MLE of θ :

A. Is $X(1) + X(n)$

B. Is X

C. Is any value in $[X(n) - 1/2, X(1) + 1/2]$

D. Does not exist

Correct answer: C

EXAM SHORTCUT

Likelihood is constant for $\theta \in [X(n) - 1/2, X(1) + 1/2]$; MLE is NOT unique; $X(1) + X(n)$ is a natural choice but not the unique MLE

Estimation

MLE Convergence Rate

Hard

Q21.

Under regularity, MLE converges at rate:

A. n^{-1}

B. $n^{-1/3}$

C. $n^{-1/2}$ (parametric rate)

D. $(\log n)^{-1}$

Correct answer: C

EXAM SHORTCUT

Parametric MLE: $\hat{\theta} - \theta = O_p(n^{-1/2})$; nonparametric rates are slower; $n^{-1/2}$ is the hallmark of parametric efficiency

Estimation

MLE Profile Likelihood

Hard

Q22.

Profile likelihood for ψ (with nuisance λ) is:

- A. $L_p(\psi) = \max_{\lambda} L(\psi, \lambda)$
- B. $L_p(\psi) = L(\psi, \hat{\lambda})$ where $\hat{\lambda}$ is marginal MLE
- C. $L_p(\psi) = L(\psi, \hat{\lambda}(\psi))$ where $\hat{\lambda}(\psi) = \arg \max_{\lambda} L(\psi, \lambda)$
- D. $L_p(\psi) = \int L(\psi, \lambda) d\lambda$

Correct answer: C

EXAM SHORTCUT

Profile likelihood: maximize over nuisance at each fixed ψ ; A and C are equivalent; D is marginal likelihood (Bayesian); B uses wrong $\hat{\lambda}$

Estimation

MLE Second Order

Hard

Q23.

Bartlett correction improves likelihood ratio test by:

- A. Using F -distribution instead of χ^2
- B. Scaling $-2 \log \Lambda$ by factor $1 + b/n$ to achieve χ^2 distribution to order $O(n^{-2})$
- C. Adjusting degrees of freedom
- D. Adding b/n to the test statistic

Correct answer: B

EXAM SHORTCUT

Bartlett correction: divide $-2 \log \Lambda$ by $(1 + b/n)$; improves χ^2 approximation from $O(n^{-1})$ to $O(n^{-2})$ error

Estimation

MLE Penalized

Hard

Q24.

Penalized MLE (e.g., LASSO) maximizes:

- A. $\log L(\theta) + \lambda \|\theta\|_1$
- B. $\log L(\theta) - \lambda \|\theta\|_2^2$
- C. A only (LASSO uses L_1 penalty)
- D. $\log L(\theta) - \lambda \|\theta\|_1$

Correct answer: D

EXAM SHORTCUT

LASSO: L_1 penalty subtracted; Ridge: L_2^2 penalty subtracted; both are D-type (subtract penalty); C is correct but incomplete

Estimation

MLE Robust

Hard

Q25.

M-estimators generalize MLE by solving:

A. Both A and C (equivalent formulations)

B. $\min \sum \rho(X_i; \theta)$ for loss ρ

C. $\sum \partial \log f / \partial \theta = 0$ only

D. $\sum \psi(X_i; \theta) = 0$ for some function ψ

Correct answer: A

EXAM SHORTCUT

M-estimators: $\psi = \partial \rho / \partial \theta$; MLE is special case with $\rho = -\log f$; Huber Mestimator uses bounded ψ

Estimation

MLE Exponential Rate

Medium

Q26.

For $X_1, \dots, X_n \sim \text{Exp}(\lambda)$ (rate parametrization), MLE of λ is:

A. $1/\bar{X}$

B. Both B and C

C. $n / \sum X_i$

D. $\bar{X}$

Correct answer: B

EXAM SHORTCUT

$1/\bar{X} = n / \sum X_i$ — identical; both are correct expressions for MLE of rate λ

Estimation

MLE Beta Both

Hard

Q27.

For $X \sim \text{Beta}(\alpha, \beta)$ with both unknown, MLE satisfies:

A. $\psi(\alpha) - \psi(\alpha + \hat{\beta}) = \log \bar{X}$ and $\psi(\beta) - \psi(\alpha + \hat{\beta}) = \log(1 - \bar{X})$

B. No MLE exists for Beta with both parameters unknown

C. $\hat{\alpha} / (\hat{\alpha} + \hat{\beta}) = \bar{X}$

D. $\hat{\alpha} = \bar{X}, \hat{\beta} = 1 - \bar{X}$

Correct answer: A

EXAM SHORTCUT

System of two digamma equations; C gives one equation (moment condition) but not MLE; requires numerical solution

Estimation

MLE Nonidentifiable

Hard

Q28.

MLE fails when model is non-identifiable because:

- A. Score equation has no solution
- B. Fisher information is zero
- C. Likelihood is unbounded
- D. Multiple parameter values give same likelihood; MLE is not unique**

Correct answer: D

EXAM SHORTCUT

Non-identifiability: $L(\theta_1) = L(\theta_2)$ for $\theta_1 \neq \theta_2$; MLE set is not a singleton; Fisher information matrix is singular

Estimation

MLE Separated Data

Hard

Q29.

In logistic regression, MLE does not exist when:

- A. Response is binary
- B. Predictors are correlated
- C. Sample size is small
- D. Data is perfectly separated (one predictor perfectly predicts outcome)**

Correct answer: D

EXAM SHORTCUT

Perfect separation: likelihood is monotone increasing; $\text{MLE} \rightarrow \pm\infty$; Firth's penalized likelihood is a remedy

Estimation

MLE Consistency Proof

Hard

Q30.

Wald's proof of MLE consistency uses:

- A. Rao-Blackwell theorem
- B. Law of large numbers applied to log-likelihood ratio**
- C. Factorization theorem
- D. Central limit theorem

Correct answer: B

EXAM SHORTCUT

Key: $n(\log L(\theta)/(L)(\theta_0) - \log L(\theta)/(L)(\theta_0)) \xrightarrow{p} E_{\theta_0}[\log(f(X;\theta)/(f)(X;\theta_0))] \leq 0$ by Jensen (Kullback-Leibler divergence ≥ 0)

Estimation

MLE Poisson Regression

Medium

Q31.

In Poisson regression with $\log \mu_i = x_i(T)/(i\beta)$, score equation for β is:

- A. $\sum x_i Y_i = \sum x_i \mu_i$
- B. **All of the above**
- C. $X^T(Y - \mu) = 0$
- D. $\sum x_i (Y_i - \mu_i) = 0$

Correct answer: B

EXAM SHORTCUT

All three are equivalent forms of the Poisson GLM score equation; IRLS solves this iteratively

Estimation

MLE Negative Binomial

Hard

Q32.

For $\text{NegBin}(r, p)$ with r known, MLE of p is:

- A. **$r/(r + X)$**
- B. r/X
- C. $X/(r + X)$
- D. $1/X$

Correct answer: A

EXAM SHORTCUT

$E[X] = r(1 - p)/p$; solving $X = r(1 - \hat{p})/\hat{p}$ gives $\hat{p} = r/(r + X)$; C is MLE of $1 - p$

Estimation

MLE Zero Inflated

Hard

Q33.

Zero-inflated Poisson (ZIP) has $P(X = 0) = \pi + (1 - \pi)e^{-\lambda}$. MLE requires:

- A. Bayesian methods
- B. **EM algorithm (latent class indicator)**
- C. Method of moments only
- D. Direct score equations with closed form

Correct answer: B

EXAM SHORTCUT

ZIP is a mixture model; EM treats class membership as latent; E-step computes posterior probability of structural zero; M-step updates π and λ

Estimation

MLE Survival Kaplan Meier

Hard

Q34.

Kaplan-Meier estimator is:

- A. Bayesian estimator
- B. **MLE of survival function under right censoring (nonparametric)**
- C. Method of moments estimator
- D. MLE of exponential survival function

Correct answer: B

EXAM SHORTCUT

KM is the nonparametric MLE of $S(t)$; it maximizes the nonparametric likelihood over all step functions; not restricted to parametric family

Estimation

MLE Information Matrix Equality

Hard

Q35.

Fisher information matrix equality states:

- A. $-E[\partial^2 \log L / \partial \theta \partial \theta^T] = E[(\partial \log L / \partial \theta)(\partial \log L / \partial \theta)^T]$
- B. $I(\theta) = \text{Var}(\partial \log L / \partial \theta)$
- C. $E[\partial \log L / \partial \theta] = 0$
- D. **Both A and B**

Correct answer: D

EXAM SHORTCUT

C is the score equation property; A is the information matrix equality; B follows from C (since $E[\text{score}] = 0$, variance = second moment); all three are related

Estimation

MLE Restricted Parameter

Hard

Q36.

For $X \sim N(\theta, 1)$ with $\theta \in \{0, 1\}$ (discrete parameter space), MLE is:

- A. 0 always (conservative)
- B. **0 if $X < 1/2$, 1 if $X \geq 1/2$**
- C. $\bar{X}$ always
- D. $\bar{X}$ rounded to nearest integer

Correct answer: B

EXAM SHORTCUT

Compare $L(0) = e^{-n\bar{X}^2/2}$ vs $L(1) = e^{-n(\bar{X}-1)^2/2}$; $L(1) > L(0)$ iff $\bar{X} > 1/2$

Estimation

MLE Curved Exponential Normal

Hard

Q37.

For $X_1, \dots, X_n \sim N(\theta, \sigma^2)$ (curved exponential family), MLE of θ is:

A. $\bar{X}$ B. $\hat{\theta} = \bar{X} - (\sigma^2 / S^2)$ approximatelyC. $\sqrt{X^2 + S^2}$ D. S (sample SD)**Correct answer: B****EXAM SHORTCUT**

Score equations for $\mu = \theta$ and $\sigma^2 = \theta^2$ must be combined; leads to quadratic in θ ; no simple closed form

Estimation

MLE Poisson UMVUE Comparison

Hard

Q38.

For $\text{Poisson}(\lambda)$, MLE of $e^{-\lambda}$ is $e^{-\bar{X}}$. UMVUE is:

A. $e^{-\bar{X}}$ B. $(1/n) \sum I(X_i = 0)$ C. $e^{-\bar{X}}(1 + 1/(2n))$ (bias-corrected)D. $(1 - 1/n) \sum X_i$ **Correct answer: D****EXAM SHORTCUT**

UMVUE of $e^{-\lambda} = P(X=0)$: use Rao-Blackwell with $T = \sum X_i$; $E[I(X_1=0)|T=t] = (1-1/n)t$; so UMVUE = $(1-1/n)T$

Estimation

MLE Asymptotic Variance

Medium

Q39.

Asymptotic variance of $\hat{\theta}$ is:

A. $nI(\theta)$ B. $I(\theta)/n$ C. $1/(nI(\theta))$ D. $1/I(\theta)$ **Correct answer: C****EXAM SHORTCUT**

$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, 1/I(\theta))$; so $\text{AsyVar}(\hat{\theta}) = 1/(nI(\theta))$; D is for single observation

Estimation

MLE Score Test

Hard

Q40.

Rao's score test statistic is:

A. **All equivalent asymptotically**B. $(\theta - \theta_0)^2 / (nI(\theta)) \xrightarrow{d} \chi^2_1$ C. $[\partial \log L / \partial \theta | \theta = \theta_0]^2 / nI(\theta_0) \xrightarrow{d} \chi^2_1$ under H_0 D. $-2 \log \Lambda \xrightarrow{d} \chi^2_1$

Correct answer: A

EXAM SHORTCUT

Wald (C), Score/Rao (A), and LRT (B) are asymptotically equivalent under H_0 ; they differ in finite samples and under alternatives

Estimation

MLE Exponential Mean

Medium

Q41.

For $X_1, \dots, X_n \sim \text{Exp}(\mu)$ (mean parametrization), MLE of μ is:A. **$\bar{X}$** B. $\bar{X}$ (same as B)C. $n / \sum X_i$ D. $1/\bar{X}$

Correct answer: A

EXAM SHORTCUT

Mean parametrization: $f(x; \mu) = (1/\mu)e^{-x/\mu}$; MLE = $\bar{X}$; rate parametrization: MLE = $1/\bar{X}$; know which parametrization is used

Estimation

MLE Multivariate Normal

Hard

Q42.

For $X_1, \dots, X_n \sim N_p(\mu, \Sigma)$ with both unknown, MLE of Σ is:A. **$(1/n) \sum (X_i - \bar{X})(X_i - \bar{X})^T$** B. $(n-1) \sum (X_i - \bar{X})(X_i - \bar{X})^T$ C. $(n-p) \sum (X_i - \bar{X})(X_i - \bar{X})^T$

D. Sample covariance matrix

Correct answer: A

EXAM SHORTCUT

MLE uses n denominator (biased); unbiased estimator uses $n-1$; $n-p$ appears in regression residual variance

Estimation

MLE Dirichlet Multinomial

Hard

Q43.

For Dirichlet-Multinomial model, MLE of Dirichlet parameters:

A. ~~Requires solving system of digamma equations iteratively~~

B. Does not exist

C. Has closed form via method of moments

D. Is simply cell proportions

Correct answer: A

EXAM SHORTCUT

Same structure as Dirichlet MLE; digamma equations; Minka's fixed-point iteration is commonly used

Estimation

MLE Nonparametric

Hard

Q44.

Nonparametric MLE of CDF F based on i.i.d. sample is:

A. Kernel density estimate

B. Smoothed empirical CDF

C. Normal CDF fitted to data

D. ~~Empirical CDF~~ $\hat{F}_n(x) = (1/n) \sum I(X_i \leq x)$

Correct answer: D

EXAM SHORTCUT

NPMLE of CDF is the empirical CDF (Glivenko-Cantelli); it maximizes the nonparametric likelihood $\prod f(X_i)$ over all distributions

Estimation

MLE Exponential Family Canonical

Hard

Q45.

In canonical exponential family, MLE $\hat{\eta}$ of natural parameter satisfies:

A. $\hat{\eta} = T$

B. $\hat{\eta}$ minimizes KL divergence from empirical distribution

C. $\nabla A(\eta) = T$ where A is log-partition function

D. ~~Both A and C~~

Correct answer: D

EXAM SHORTCUT

$\nabla A(\eta) = E[T(X)]$; setting equal to T gives MLE; this is equivalent to minimizing KL divergence (maximum entropy principle)

Estimation

MLE Robust Breakdown

Hard

Q46.

Breakdown point of MLE for normal location parameter is:

- A. $1/4$
- B. $1/n$
- C. **0 (single outlier can make MLE arbitrarily large)**
- D. $1/2$ (50% contamination)

Correct answer: C

EXAM SHORTCUT

Sample mean (MLE for normal location) has breakdown point 0; one extreme outlier destroys it; median has breakdown point $1/2$

Estimation

MLE Conditional

Hard

Q47.

Conditional MLE (conditioning on sufficient statistic for nuisance) is used to:

- A. Handle missing data
- B. Reduce computation
- C. Increase efficiency
- D. **Eliminate nuisance parameters from likelihood**

Correct answer: D

EXAM SHORTCUT

Conditional likelihood eliminates nuisance parameters; used in matched casecontrol studies, logistic regression with fixed effects

Estimation

MLE Marginal

Hard

Q48.

Marginal MLE (integrating out random effects) is used in:

- A. Fixed effects models
- B. **Mixed effects models (REML is a variant)**
- C. Saturated models
- D. Nonparametric models

Correct answer: B

EXAM SHORTCUT

REML (Restricted MLE) maximizes marginal likelihood after integrating out fixed effects; used for variance component estimation in mixed models

Estimation

MLE Semiparametric

Hard

Q49.

In Cox proportional hazards model, partial likelihood MLE:

- A. Estimates baseline hazard and regression coefficients jointly
- B. Requires parametric baseline hazard
- C. **Estimates regression coefficients without specifying baseline hazard**
- D. Is equivalent to full MLE

Correct answer: C

EXAM SHORTCUT

Cox partial likelihood: baseline hazard cancels; only regression coefficients estimated; semiparametric efficiency theory applies

Estimation

MLE Information Observed

Hard

Q50.

Observed Fisher information $I(\theta) = -\partial^2 \log L / \partial \theta^2$ vs expected $I(\theta)$:

- A. Expected is always larger
- B. They are always equal
- C. Observed is always larger
- D. **They are asymptotically equivalent; observed often preferred in practice**

Correct answer: D

EXAM SHORTCUT

Both are consistent for $I(\theta_0)$; Efron-Hinkley showed observed information is preferred for conditional inference; neither dominates in finite samples

Rao–Blackwell, UMVUE & Method of Moments

40 forecast questions in this topic

Estimation

Rao Blackwell Statement

Medium

Q1.

Rao-Blackwell theorem states: if T_0 is unbiased for $g(\theta)$ and S is sufficient, then $T^* = E[T_0|S]$:

A. **All of the above**

B. Is unbiased for $g(\theta)$

C. Has smaller or equal variance than T_0

D. Is a function of S

Correct answer: A

EXAM SHORTCUT

All three properties hold: unbiasedness (tower property), variance reduction (conditional variance formula), and T^* is a function of S by definition

Estimation

Rao Blackwell Example Bernoulli

Medium

Q2.

For $X_1, \dots, X_n \sim \text{Bernoulli}(p)$, improve $T_0 = X_1$ (unbiased for p) using $S = \sum X_i$:

A. $E[X_1 | \sum X_i = t] = t(t-1)/[n(n-1)]$

B. $E[X_1 | \sum X_i = t] = 1/n$

C. **$E[X_1 | \sum X_i = t] = t/n$**

D. $E[X_1 | \sum X_i = t] = t^2/n^2$

Correct answer: C

EXAM SHORTCUT

$P(X_1 = 1 | \sum X_i = t) = ((n-1)/(t-1))/(n-1) = t/n$; result is X — the UMVUE

Estimation

UMVUE Poisson Zero

Hard

Q3.

For $X_1, \dots, X_n \sim \text{Poisson}(\lambda)$, UMVUE of $P(X=0) = e^{-\lambda}$ is:

- A. $(1 - 1/n)^{\sum X_i}$
- B. $(1/n)^{\sum I(X_i=0)}$
- C. $e^{-\bar{X}}$
- D. $e^{-\sum X_i/n}$

Correct answer: A

EXAM SHORTCUT

$E[I(X_1=0)|T=t]$ where $T = \sum X_i \sim \text{Poisson}(n\lambda)$: $P(X_1=0|T=t) = (1 - 1/n)t$; so UMVUE = $(1 - 1/n)T$

Estimation

UMVUE Normal Sigma

Hard

Q4.

For $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ with μ known, UMVUE of σ^2 is:

- A. $S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$
- B. $(1/n) \sum (X_i - \mu)^2$
- C. $(1/n) \sum (X_i - \bar{X})^2$
- D. $\frac{1}{n-1} \sum (X_i - \mu)^2$

Correct answer: B

EXAM SHORTCUT

When μ is KNOWN, $\sum (X_i - \mu)^2/n$ is UMVUE (attains CRLB); when μ unknown, UMVUE is S^2 with $n-1$ denominator

Estimation

UMVUE Uniform Theta Sq

Hard

Q5.

For $X_1, \dots, X_n \sim U(0, \theta)$, UMVUE of θ^2 is:

- A. $4X^2/3$
- B. $(n+1)(n+2)/n^2 \sum X_i^2$
- C. $(n+2)/(n \sum X_i^2)$
- D. $\sum X_i^2/n$

Correct answer: B

EXAM SHORTCUT

$E[X(k)/(n)] = n\theta^k/n+k$; for $k=2$: $E[\sum X_i^2/n] = n\theta^2/n+2$; UMVUE = $(n+1)(n+2)/(n^2) \sum X_i^2$

Estimation

UMVUE Exponential Function

Hard

Q6.

For $X_1, \dots, X_n \sim \text{Exp}(\lambda)$, UMVUE of λ^k is:

- A. $(1/\bar{X})^k$
- B. $n^{-k} \bar{X}^{-k}$
- C. $\frac{\Gamma(n)}{\Gamma(n-k)} \cdot (\sum X_i)^k$
- D. $n! / (n-k)! n^k \cdot \bar{X}^k$ (for $k < n$)

Correct answer: C

EXAM SHORTCUT

$T = \sum X_i \sim \Gamma(n, 1/\lambda)$; $E[T-k] = (\lambda^k \Gamma(n-k)) / (\Gamma(n))$; UMVUE of $\lambda^k = \frac{\Gamma(n)}{\Gamma(n-k)} T^{-k}$

Estimation

UMVUE Bernoulli P Sq

Hard

Q7.

For $X_1, \dots, X_n \sim \text{Bernoulli}(p)$, UMVUE of p^2 is:

- A. T^2/n^2
- B. $\bar{X}^2$
- C. $\frac{T(T-1)}{n(n-1)}$ where $T = \sum X_i$
- D. $\bar{X}(X-1/n)$

Correct answer: C

EXAM SHORTCUT

$E[X_1 X_2 | T = t] = \frac{t(t-1)}{n(n-1)}$ ((hypergeometric argument); UMVUE = $\frac{T(T-1)}{n(n-1)}$); note $\bar{X}^2$ is biased for p^2

Estimation

UMVUE Poisson Lambda Sq

Hard

Q8.

For $X_1, \dots, X_n \sim \text{Poisson}(\lambda)$, UMVUE of λ^2 is:

- A. $\bar{X}^2$
- B. $\frac{T(T-1)}{n^2}$ where $T = \sum X_i$
- C. T^2/n^2
- D. $\bar{X}^2 - \bar{X}/n$

Correct answer: B

EXAM SHORTCUT

$E[T(T-1)] = E[T^2] - E[T] = n^2 \lambda^2 + n \lambda - n \lambda = n^2 \lambda^2$; so UMVUE = $\frac{T(T-1)}{n^2} = \bar{X}^2 - \bar{X}/n$; D and B are equivalent

Estimation

Method Of Moments Gamma

Medium

Q9.

For $X \sim \Gamma(\alpha, \beta)$, MOM estimators using first two moments:

A. $\hat{\alpha} = S^2/X^2, \hat{\beta} = X/S^2$

B. $\hat{\alpha} = X/S^2, \hat{\beta} = S^2$

C. $\hat{\alpha} = X^2/S^2, \hat{\beta} = S^2/X$

D. $\hat{\alpha} = X, \hat{\beta} = S^2/X$

Correct answer: C

EXAM SHORTCUT

$$E[X] = \alpha\beta, \text{Var}(X) = \alpha\beta^2; \beta = \text{Var}/E = S^2/X; \alpha = E/\beta = X^2/S^2$$

Estimation

Method Of Moments Beta

Medium

Q10.

For $X \sim \text{Beta}(\alpha, \beta)$, MOM estimators:

A. $\hat{\alpha} = X[X(1-X)/S^2 - 1], \hat{\beta} = (1-X)[X(1-X)/S^2 - 1]$

B. $\hat{\alpha} = X, \hat{\beta} = 1 - X$

C. $\hat{\alpha} = X^2/S^2, \hat{\beta} = (1-X)^2/S^2$

D. $\hat{\alpha} = X/S^2, \hat{\beta} = (1-X)/S^2$

Correct answer: A

EXAM SHORTCUT

$$E[X] = \alpha/(\alpha + \beta), \text{Var}(X) = \alpha\beta/[(\alpha + \beta)^2(\alpha + \beta + 1)]; \text{ let } m = X, v = S^2; \alpha = m[m(1-m)/v - 1]$$

Estimation

Rao Blackwell Improvement

Hard

Q11.

Rao-Blackwell improvement is strict (strict inequality in variance) unless:

A. T_0 is the UMVUE

B. **Both A and C (equivalent)**

C. T_0 is the MLE

D. T_0 is already a function of S

Correct answer: B

EXAM SHORTCUT

Variance reduction is strict unless T_0 is already a function of S ; if $T_0 = h(S)$, then $E[T_0|S] = T_0$ and no improvement occurs

Estimation

UMVUE Existence

Hard

Q12.

UMVUE exists if and only if:

- A. The distribution is from exponential family
- B. A complete sufficient statistic exists
- C. An unbiased estimator exists
- D. Both A and B**

Correct answer: D

EXAM SHORTCUT

Need both: (1) an unbiased estimator must exist, and (2) a complete sufficient statistic must exist; then Lehmann-Scheffé gives unique UMVUE

Estimation

MOM Consistency

Medium

Q13.

Method of moments estimators are consistent because:

- A. They are functions of sufficient statistics
- B. Sample moments converge to population moments by LLN**
- C. They attain CRLB
- D. They are always unbiased

Correct answer: B

EXAM SHORTCUT

LLN: $\bar{X}(k)/n \rightarrow E[Xk]$; continuous mapping theorem: MOM estimators (continuous functions of sample moments) are consistent

Estimation

UMVUE Normal Absolute Deviation

Hard

Q14.

For $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ with σ^2 known, UMVUE of $|\mu|$:

- A. Does not exist (no unbiased estimator of $|\mu|$)
- B. $|X|$
- C. $\sqrt{n} \Phi(|X|/\sqrt{n}/\sigma) - \sigma \phi(|X|/\sqrt{n}/\sigma) / \sqrt{n}$ (complex expression)**
- D. $|X|$ corrected for bias

Correct answer: C

EXAM SHORTCUT

UMVUE of $|\mu|$ exists but has complex form involving normal CDF and PDF; $|X|$ is biased for $|\mu|$; this is a classic hard ISS question

Estimation

UMVUE Shifted Exponential

Hard

Q15.

For $X_1, \dots, X_n \sim e^{-(x-\theta)}, x > \theta$, UMVUE of θ is:

- A. $\bar{X} - 1$
- B. $X_{(1)} + 1/n$
- C. $X_{(1)} - 1/n$
- D. $X_{(1)}$

Correct answer: C

EXAM SHORTCUT

$E[X_{(1)}] = \theta + 1/n$; unbiased: $X_{(1)} - 1/n$; $X_{(1)}$ is complete sufficient; so $X_{(1)} - 1/n$ is UMVUE

Estimation

MOM Negative Binomial

Hard

Q16.

For $X \sim \text{NegBin}(r, p)$ with r known, MOM estimator of p :

- A. r/X
- B. $r/(r+X)$
- C. $X/(r+X)$
- D. $1/(1+X/r)$

Correct answer: B

EXAM SHORTCUT

$E[X] = r(1-p)/p$; solving: $\hat{p} = r/(r+X)$; same as MLE in this case; D is equivalent to B

Estimation

UMVUE Cauchy

Hard

Q17.

For $X_1, \dots, X_n \sim \text{Cauchy}(\theta, 1)$, UMVUE of θ :

- A. Is the MLE
- B. Is the sample median
- C. Is $\bar{X}$
- D. Does not exist (no complete sufficient statistic)

Correct answer: D

EXAM SHORTCUT

Cauchy is not exponential family; minimal sufficient statistic is all order statistics; no complete sufficient statistic exists; hence no UMVUE

Estimation

Rao Blackwell Poisson Indicator

Hard

Q18.

For $X_1, \dots, X_n \sim \text{Poisson}(\lambda)$, improve $T_0 = I(X_1 = 1)$ using $T = \sum X_i$:

A. $E[I(X_1=1)|T=t] = e^{-1/n} (1/n)^t / t!$ (Poisson conditional)

B. $E[I(X_1=1)|T=t] = C(t,1)(1/n)(1-1/n)^{t-1}$

C. $E[I(X_1=1)|T=t] = t/n$

D. $E[I(X_1=1)|T=t] = t(1-1/n)^{t-1}/n$

Correct answer: B

EXAM SHORTCUT

Given $T=t$, $X_1 \sim \text{Bin}(t, 1/n)$; $P(X_1=1|T=t) = C(t,1)(1/n)(1-1/n)^{t-1} = t(1-1/n)^{t-1}/n$; D and B are equivalent.

Estimation

UMVUE Uniform Probability

Hard

Q19.

For $X_1, \dots, X_n \sim U(0, \theta)$, UMVUE of $P(X \leq c) = c/\theta$ (for fixed $c < \theta$) is:

A. $\min(c/X(n), 1)$

B. $cn/[(n+1)X(n)]$

C. $c/X(n)$

D. $\frac{n+1}{n} \cdot \frac{c}{X(n)}$ if $X(n) > c$, else 1

Correct answer: D

EXAM SHORTCUT

Need to handle case $X(n) \leq c$ (then $P(X \leq c) = 1$ certainly); UMVUE must be bounded in $[0, 1]$; careful case analysis required

Estimation

MOM Lognormal

Medium

Q20.

For $X \sim \text{LogNormal}(\mu, \sigma^2)$, MOM estimators using $E[X] = e^{\mu + \sigma^2/2}$ and $\text{Var}(X) = e^{2\mu + \sigma^2}(e^{\sigma^2} - 1)$:

A. $\hat{\sigma}^2 = \log(1 + S^2/\bar{X}^2)$, $\hat{\mu} = \log \bar{X} - \hat{\sigma}^2/2$

B. Both A and B (A is MOM, B is MLE)

C. $\hat{\mu} = \log \bar{X}$, $\hat{\sigma}^2 = S^2_{\log}$

D. $\hat{\mu} = \log \bar{X}$, $\hat{\sigma}^2 = S^2/\bar{X}^2$

Correct answer: B

EXAM SHORTCUT

MOM uses raw moments (A); MLE uses log-transformed data (B); they differ; C is also the UMVUE approach

Estimation

UMVUE Bernoulli Variance

Hard

Q21.

For $X_1, \dots, X_n \sim \text{Bernoulli}(p)$, UMVUE of $p(1-p) = \text{Var}(X)$ is:

A. ~~Both B and C (equivalent)~~

B. $T(n-T)/(n(n-1))$ where $T = \sum X_i$

C. $S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$

D. $\bar{X}(1 - \bar{X})$

Correct answer: A

EXAM SHORTCUT

$S^2 = (T(n-T))/(n(n-1))$ for Bernoulli data; both B and C are the same expression; $\bar{X}(1 - \bar{X})$ uses n denominator (biased)

Estimation

UMVUE Exponential Survival

Hard

Q22.

For $X_1, \dots, X_n \sim \text{Exp}(\lambda)$, UMVUE of $S(t) = P(X > t) = e^{-\lambda t}$ is:

A. $(1)/(n) \sum I(X_i > t)$

B. $(1 - t/T)^{n-1} I(T > t)$ where $T = \sum X_i$

C. $e^{-t\lambda}$ where $\hat{\lambda} = 1/\bar{X}$

D. $e^{-t/\bar{X}}$

Correct answer: B

EXAM SHORTCUT

$E[I(X_1 > t) | T = \sum X_i]$: given T , $X_1/T \sim \text{Beta}(1, n-1)$; $P(X_1 > t | T) = (1 - t/T)^{n-1} I(T > t)$

Estimation

MOM Weibull

Hard

Q23.

For Weibull(k, λ) with both unknown, MOM requires:

A. $\hat{k} = 1/\log \bar{X}$, $\hat{\lambda} = \bar{X}$

B. ~~Solving two equations involving gamma functions: $E[X] = \lambda \Gamma(1 + 1/k)$, $E[X^2] = \lambda^2 \Gamma(1 + 2/k)$~~

C. $\hat{k} = \bar{X}/S$, $\hat{\lambda} = \bar{X}$

D. Closed form exists: $\hat{k} = \sqrt{6S/(\bar{r}\bar{X})}$

Correct answer: B

EXAM SHORTCUT

Weibull MOM: two equations with gamma functions; ratio $E[X^2]/(E[X])^2 = \Gamma(1 + 2/k)/[\Gamma(1 + 1/k)]^2$ depends only on k ; solve numerically

Estimation

UMVUE Normal Sigma Unknown Mu

Hard

Q24.

For $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ with both unknown, UMVUE of σ is:

A. ~~Both B and C (equivalent)~~

B. S/c_4 where $c_4 = \sqrt{2/(n-1)} \cdot \Gamma(n/2)/\Gamma((n-1)/2)$

C. $S = \sqrt{S^2}$

D. $S \cdot \sqrt{(n-1)/2} \cdot \Gamma((n-1)/2)/\Gamma(n/2)$

Correct answer: A

EXAM SHORTCUT

$E[S] = \sigma c_4$; UMVUE of σ is S/c_4 ; $c_4 < 1$ so UMVUE $> S$; for large n , $c_4 \approx 1$

Estimation

Rao Blackwell Geometric

Hard

Q25.

For $X_1, \dots, X_n \sim \text{Geometric}(p)$, improve $T_0 = I(X_1 = 1)$ using $S = \sum X_i$:

A. ~~$E[I(X_1 = 1)|S = s] = n/s$~~

B. $E[I(X_1 = 1)|S = s] = ((s-1)/(n-1))/(s/n)$ (negative hypergeometric)

C. $E[I(X_1 = 1)|S = s] = (n-1)/(s-1)$

D. $E[I(X_1 = 1)|S = s] = p$

Correct answer: A

EXAM SHORTCUT

Given $\sum X_i = s$, by symmetry $P(X_1 = 1|S = s) = P(X_j = 1|S = s)$ for all j ; sums to $n \cdot P(X_1 = 1|S = s)$; careful computation gives n/s

Estimation

UMVUE Pareto

Hard

Q26.

For $X_1, \dots, X_n \sim \text{Pareto}(\theta)$, $f(x; \theta) = \theta/x^{\theta+1}$, $x > 1$, UMVUE of θ is:

A. $-n / \sum \log X_i$

B. ~~$-(n-1) / \sum \log X_i$~~

C. $n / \sum \log X_i$

D. $(n+1) / \sum \log X_i$

Correct answer: B

EXAM SHORTCUT

$T = \sum \log X_i$; $-\log X_i \sim \text{Exp}(\theta)$; $T \sim \Gamma(n, 1/\theta)$; $E[1/T] = \theta/(n-1)$; UMVUE of θ : $E[-(n-1)/T] = \theta$ ✓

Estimation

MOM Inverse Gaussian

Hard

Q27.

For $X \sim IG(\mu, \lambda)$, MOM estimators:

- A. $\hat{\mu} = 1/\bar{X}, \hat{\lambda} = \bar{X}^2/S^2$
- B. $\hat{\mu} = \bar{X}, \hat{\lambda} = S^2/\bar{X}$
- C. $\hat{\mu} = \bar{X}, \hat{\lambda} = \bar{X}^2/S^2$
- D. $\hat{\mu} = \bar{X}, \hat{\lambda} = \bar{X}^3/S^2$

Correct answer: D

EXAM SHORTCUT

$$E[X] = \mu, \text{Var}(X) = \mu^3/\lambda; \hat{\mu} = \bar{X}; \hat{\lambda} = \bar{X}^3/S^2 = \bar{X}^3/S^2$$

Estimation

UMVUE Hypergeometric

Hard

Q28.

For Hypergeometric(N, M, n) with N known, UMVUE of M/N (population proportion) is:

- A. $(X/n) \cdot N/(N-1)$
- B. X/n is already UMVUE
- C. X/N
- D. X/n (sample proportion)

Correct answer: B

EXAM SHORTCUT

X/n is unbiased for M/N and is a function of the complete sufficient statistic X ; hence it is UMVUE

Estimation

MOM Truncated Normal

Hard

Q29.

For $N(\mu, 1)$ truncated to $(0, \infty)$, MOM estimator of μ satisfies:

- A. $\hat{\mu} = X \cdot \Phi(X)$
- B. $\hat{\mu} = X - 1$
- C. $X = \mu + \phi(\mu)/\Phi(\mu)$ (Mills ratio equation)
- D. $\hat{\mu} = X$

Correct answer: C

EXAM SHORTCUT

$E[X|X > 0] = \mu + \phi(\mu)/\Phi(\mu)$; setting equal to $\bar{X}$ gives implicit equation; same as MLE equation for truncated normal

Estimation

UMVUE Double Exponential

Hard

Q30.

For $X_1, \dots, X_n \sim \text{Laplace}(\theta, 1)$, UMVUE of θ :

A. Does not exist (no complete sufficient statistic)

B. Is $\bar{X}$

C. Is a weighted average of order statistics

D. Is sample median

Correct answer: A

EXAM SHORTCUT

Laplace is a location family; order statistics are minimal sufficient but NOT complete; no UMVUE exists; sample median is best equivariant

Estimation

Rao Blackwell Normal

Hard

Q31.

For $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ with σ^2 known, improve $T_0 = \bar{X}^2 - \sigma^2$ (unbiased for μ^2) using $S = \bar{X}$:

A. $\bar{X}^2 - \sigma^2$

B. $\bar{X}^2$

C. $\bar{X}^2 + \sigma^2/n$

D. $\bar{X}^2 - \sigma^2/n$

Correct answer: D

EXAM SHORTCUT

$E[\bar{X}^2 - \sigma^2 | \bar{X}] = E[\bar{X}^2 | \bar{X}] - \sigma^2$; $E[\bar{X}^2 | \bar{X}] = \bar{X}^2 + \sigma^2(n-1)/n$; so result = $\bar{X}^2 + \sigma^2(n-1)/n - \sigma^2 = \bar{X}^2 - \sigma^2/n$ ✓

Estimation

UMVUE Binomial Probability

Hard

Q32.

For $X_1, \dots, X_n \sim \text{Bin}(m, p)$ with m known, UMVUE of $P(X = k) = \binom{m}{k} p^k (1-p)^{m-k}$ is:

A. $\binom{m}{k} p^k (1-p)^{m-k}$ where $\hat{p} = \bar{X}/m$

B. $\binom{nm-T}{m-k} \binom{T-k}{k} / \binom{nm}{m}$ where $T = \sum X_i$

C. $\binom{m}{k} \binom{nm-m-T-k}{nm-T} / \binom{nm}{m}$ where $T = \sum X_i$

D. Does not exist in closed form

Correct answer: C

EXAM SHORTCUT

Hypergeometric-type conditional probability; $E[(X_1 = k) | T = \sum X_i]$ computed via hypergeometric distribution; C is the correct form

Estimation

MOM Generalized Pareto

Hard

Q33.

For Generalized Pareto $GP(\xi, \sigma)$, MOM estimators:

- A. $\hat{\xi} = 1/2 - X^2/S^2, \hat{\sigma} = X(1 - \hat{\xi})$
- B. $\hat{\xi} = 1 - X^2/(2S^2), \hat{\sigma} = X(1 - \hat{\xi})$ (Hosking-Wallis)
- C. $\hat{\xi} = X/S, \hat{\sigma} = S$
- D. $\hat{\xi} = S/X - 1, \hat{\sigma} = X$

Correct answer: B

EXAM SHORTCUT

$E[X] = \sigma/(1 - \xi)$, $\text{Var}(X) = \sigma^2/[(1 - \xi)^2(1 - 2\xi)]$; solving gives Hosking-Wallis MOM; valid only for $\xi < 1/2$

Estimation

UMVUE Geometric Probability

Hard

Q34.

For $X_1, \dots, X_n \sim \text{Geometric}(p)$, UMVUE of p is:

- A. $n / \sum X_i$
- B. $1/(X-1)$
- C. $1/X$
- D. $(n-1)/(\sum X_i - 1)$

Correct answer: D

EXAM SHORTCUT

$T = \sum X_i \sim \text{NegBin}(n, p)$; $E[T] = n/p$; unbiased estimator: n/T is NOT exactly unbiased; correct UMVUE: $(n-1)/(T-1)$ since $E[(n-1)/(T-1)] = p$ for NegBin

Estimation

Rao Blackwell Uniform

Hard

Q35.

For $X_1, \dots, X_n \sim U(0, \theta)$, improve $T_0 = 2X_1$ (unbiased for θ) using $S = X(n)$:

- A. $E[2X_1 | X(n) = t] = (n+1)/(nt)$
- B. $E[2X_1 | X(n) = t] = 2t/n+1$
- C. $E[2X_1 | X(n) = t] = t$
- D. $E[2X_1 | X(n) = t] = ((n+1)t)/(n)$

Correct answer: A

EXAM SHORTCUT

Given $X(n) = t$: $E[X_1 | X(n) = t] = (1)/(nt) + n-1 \cdot n \cdot (t)/(2) = t(n+1)/2n$; so $E[2X_1 | X(n) = t] = ((n+1)t)/(n)$ ✓

Estimation

MOM Compound Poisson

Hard

Q36.

For compound Poisson where $N \sim \text{Poisson}(\lambda)$ and $Y = \sum_{i=1}^N X_i$ with $X_i \sim \text{Exp}(\mu)$, MOM for (λ, μ) :

- A. $\hat{\mu} = 2Y/S^2, \hat{\lambda} = Y \mu$
- B. $\hat{\mu} = \sqrt{2Y/S^2}, \hat{\lambda} = Y \mu$
- C. $\hat{\lambda} = Y \mu, \hat{\mu} = Y/S^2$ solve simultaneously
- D. $E[Y] = \lambda/\mu, \text{Var}(Y) = 2\lambda/\mu^2$ (use $E[Y] = \lambda E[X], \text{Var}(Y) = \lambda E[X^2]$)

Correct answer: A**EXAM SHORTCUT**

$E[Y] = \lambda/\mu, \text{Var}(Y) = \lambda E[X^2] = \lambda \cdot 2/\mu^2$; ratio: $\text{Var}/E = 2/\mu$; $\hat{\mu} = 2Y/S^2$; then $\hat{\lambda} = Y \mu$

Estimation

UMVUE Normal Quantile

Hard

Q37.

For $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ with σ^2 known, UMVUE of $\xi_p = \mu + z_p \sigma$ (the p -th quantile) is:

- A. $\bar{X} + z_p \sigma$
- B. $\bar{X} + z_p S$
- C. $\bar{X} + z_p \sigma \sqrt{1 + 1/n}$
- D. $\bar{X} + z_p \sigma / \sqrt{n}$

Correct answer: A**EXAM SHORTCUT**

$\bar{X}$ is UMVUE of μ ; $z_p \sigma$ is a constant; UMVUE of $\mu + z_p \sigma$ is $\bar{X} + z_p \sigma$; note σ is KNOWN here

Estimation

MOM Mixture Normal

Hard

Q38.

For mixture $\pi N(\mu_1, \sigma^2) + (1 - \pi)N(\mu_2, \sigma^2)$ with σ^2 known, MOM requires:

- A. 2 moment equations
- B. Only first moment equation
- C. EM algorithm (MOM not applicable)
- D. **At least 3 moment equations for 3 unknowns (π, μ_1, μ_2)**

Correct answer: D**EXAM SHORTCUT**

3 unknown parameters need 3 equations; use $E[X], E[X^2], E[X^3]$; mixture MOM can have multiple solutions — EM preferred

Estimation

UMVUE Negative Binomial

Hard

Q39.

For $X_1, \dots, X_n \sim \text{NegBin}(r, p)$ with r known, UMVUE of p is:

- A. $nr / \sum X_i$
- B. $r / (X + r)$
- C. $(nr - 1) / (\sum X_i + nr - 1)$
- D. $nr / (\sum X_i + nr)$

Correct answer: C

EXAM SHORTCUT

$T = \sum X_i \sim \text{NegBin}(nr, p)$; UMVUE of p : $E[(nr - 1) / (T + nr - 1)] = p$ (analogous to geometric case); careful with exact formula

Estimation

MOM Generalized Extreme Value

Hard

Q40.

For $\text{GEV}(\mu, \sigma, \xi)$, MOM estimators:

- A. Require $\xi < 1/2$ (finite variance)
- B. Are always less efficient than MLE
- C. **Exist only for $\xi < 1/3$ (finite third moment required)**
- D. Exist in closed form for all ξ

Correct answer: C

EXAM SHORTCUT

GEV moments: $E[X]$ exists for $\xi < 1$; $\text{Var}(X)$ for $\xi < 1/2$; third moment for $\xi < 1/3$; MOM using first 3 moments requires $\xi < 1/3$

Confidence Intervals

50 forecast questions in this topic

Estimation

CI Normal Mean

Medium

Q1.

100(1 - α)% CI for μ when σ^2 known is:

A. $\bar{X} \pm z_{\alpha/2} S / \sqrt{n}$

B. $\bar{X} \pm t_{\alpha/2, n-1} S / \sqrt{n}$

C. $\bar{X} \pm z_{\alpha/2} \sigma$

D. $\bar{X} \pm z_{\alpha/2} \sigma / \sqrt{n}$

Correct answer: D

EXAM SHORTCUT

Known σ : use $z_{\alpha/2}$; unknown σ : use t_{n-1} ; never mix z with S or t with σ

Estimation

CI Normal Variance

Medium

Q2.

100(1 - α)% CI for σ^2 is:

A. $[(n-1)S^2 / (\chi^2_{\alpha/2, n-1}), (n-1)S^2 / \chi^2_{1-\alpha/2, n-1}]$

B. $[S^2 \pm z_{\alpha/2} \sqrt{2\sigma^4/n}]$

C. $[(n-1)S^2 / (\chi^2_{1-\alpha/2, n-1}), (n-1)S^2 / (\chi^2_{\alpha/2, n-1})]$

D. $[(nS^2) / (\chi^2_{\alpha/2, n}), (nS^2) / \chi^2_{1-\alpha/2, n}]$

Correct answer: C

EXAM SHORTCUT

χ^2 CI: smaller χ^2 value in denominator gives LARGER bound; lower bound uses LARGER χ^2 quantile; $(n-1)$ df not n

Estimation

CI Proportion

Medium

Q3.

Wald CI for proportion p is:

- A. $\hat{p} \pm z_{\alpha/2} \sqrt{\hat{p}(1-\hat{p})/n}$
- B. $\hat{p} \pm z_{\alpha/2} / \sqrt{n}$
- C. $\hat{p} \pm t_{\alpha/2, n-1} \sqrt{\hat{p}(1-\hat{p})/n}$
- D. $\hat{p} \pm z_{\alpha/2} \sqrt{p(1-p)/n}$

Correct answer: A

EXAM SHORTCUT

Wald uses estimated $\hat{p}$ in SE; Wilson CI is preferred for small n ; D uses true p (unknown); never use t for proportions

Estimation

CI Shortest

Hard

Q4.

For symmetric unimodal distributions, shortest CI is:

- A. Highest density region (HDR)
- B. One-sided CI
- C. Depends on sample size
- D. Equal-tails CI

Correct answer: A

EXAM SHORTCUT

For symmetric distributions, equal-tails = HDR; for skewed (e.g., χ^2), HDR is shorter than equal-tails; equal-tails is NOT always shortest

Estimation

CI Pivotal Quantity

Medium

Q5.

A pivotal quantity $Q(X, \theta)$ has distribution:

- A. That depends on θ
- B. That is always χ^2
- C. That is always normal
- D. That does not depend on θ

Correct answer: D

EXAM SHORTCUT

Pivot: distribution free of θ ; examples: $(\bar{X} - \mu)/(\sigma/\sqrt{n}) \sim N(0, 1)$; $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$

Estimation

CI Exponential

Medium

Q6.

For $X_1, \dots, X_n \sim \text{Exp}(\lambda)$, exact CI for λ uses pivot:

A. $\sqrt{n}(X - 1/\lambda) \sim N(0, 1/\lambda^2)$

B. $2\lambda \sum X_i \sim \chi^2_{2n}$

C. $n\lambda X \sim \chi^2_n$

D. $\lambda \sum X_i \sim \chi^2_n$

Correct answer: B

EXAM SHORTCUT

$2\lambda \sum X_i \sim \chi^2_{2n}$; sum of n independent χ^2_2 is χ^2_{2n} ; CI: $[\chi^2_{1-\alpha/2, 2n}/(2 \sum X_i), \chi^2_{\alpha/2, 2n}/(2 \sum X_i)]$

Estimation

CI Uniform

Hard

Q7.

For $X_1, \dots, X_n \sim U(0, \theta)$, exact one-sided CI for θ based on $X(n)$ is:

A. $[X(n)/\alpha^{1/n}, \infty)$

B. $[X(n), X(n) \cdot \alpha^{-1/n}]$

C. $[X(n) \cdot \alpha^{1/n}, \infty)$

D. $[X(n), X(n) \cdot \alpha^{1/n}]$

Correct answer: B

EXAM SHORTCUT

$U = X_{(n)}/\theta \sim \text{Beta}(n, 1)$; $P(U \geq \alpha^{1/n}) = 1 - \alpha$; so $P(\theta \leq X_{(n)}/\alpha^{1/n}) = 1 - \alpha$; one-sided CI: $[X_{(n)}, X_{(n)} \cdot \alpha^{-1/n}]$.

Estimation

CI Fiducial

Hard

Q8.

Fiducial interval (Fisher) differs from confidence interval in that:

A. It treats θ as random given observed data

B. It is always shorter

C. It requires prior distribution

D. It has different coverage probability

Correct answer: A

EXAM SHORTCUT

Fiducial: treats parameter as having distribution induced by data (without prior); numerically equals CI for location families; philosophically different from frequentist CI

Estimation

CI Bayesian Credible

Hard

Q9.

A 95% Bayesian credible interval differs from 95% frequentist CI in that:

- A. $P(\theta \in \text{CI} | X) = 0.95$ (posterior probability)
- B. It is always wider
- C. Coverage is 95% in repeated sampling
- D. $P(\theta \in \text{CI}) = 0.95$ (prior probability)

Correct answer: A

EXAM SHORTCUT

Bayesian: posterior probability statement about θ ; frequentist: coverage in repeated sampling; C is the frequentist interpretation; A is Bayesian

Estimation

CI Wilson Proportion

Hard

Q10.

Wilson CI for proportion p is centered at:

- A. $\hat{p} + z^2 \alpha/2 / (2n)$ (adjusted center)
- B. p (true proportion)
- C. $\hat{p}$
- D. $1/2$

Correct answer: A

EXAM SHORTCUT

Wilson CI: add $z^2/(2n)$ pseudo-observations; center shifts toward $1/2$; better coverage than Wald for small n or extreme p

Estimation

CI Prediction Interval

Medium

Q11.

Prediction interval for future observation $X_{n+1} \sim N(\mu, \sigma^2)$ with σ^2 known is:

- A. $\bar{X} \pm z \alpha/2 \sigma$
- B. $\bar{X} \pm t_{\alpha/2, n-1} S \sqrt{1 + 1/n}$
- C. $\bar{X} \pm z \alpha/2 \sigma / \sqrt{n}$
- D. $\bar{X} \pm z \alpha/2 \sigma \sqrt{1 + 1/n}$

Correct answer: D

EXAM SHORTCUT

Prediction interval wider than CI: $X_{n+1} - \bar{X} \sim N(0, \sigma^2(1 + 1/n))$; B is correct when σ unknown; D when σ known

Estimation

CI Tolerance Interval

Hard

Q12.

A tolerance interval $[L, U]$ with content p and confidence $1 - \alpha$ means:

- A. $P(L \leq X \leq U) = p$
- B. $P(P(L \leq X \leq U) \geq p) \geq 1 - \alpha$
- C. $P(L \leq X_{n+1} \leq U) = p$
- D. $P(L \leq \mu \leq U) = 1 - \alpha$

Correct answer: B

EXAM SHORTCUT

Tolerance interval: probabilistic statement about coverage of future population; A is exact (not achievable for continuous distributions); D is CI for mean; C is prediction interval

Estimation

CI Bootstrap Percentile

Medium

Q13.

Bootstrap percentile CI uses:

- A. $\hat{\theta}^{*}/((\alpha/2), \hat{\theta}^{*}) (1 - \alpha/2)$ — quantiles of bootstrap distribution
- B. $[\hat{\theta} - \hat{\theta}^{*}/((1 - \alpha/2), \hat{\theta} - \hat{\theta}^{*}) (\alpha/2)]$ — reflected
- C. $\hat{\theta} \pm t_{\alpha/2, n-1} SE_{boot}$
- D. $\hat{\theta} \pm z_{\alpha/2} SE_{boot}$

Correct answer: A

EXAM SHORTCUT

Percentile method: A; Basic/Reverse percentile: B; Normal bootstrap: D; BCa (bias-corrected accelerated) is most accurate

Estimation

CI Bootstrap BCa

Hard

Q14.

BCa bootstrap CI corrects for:

- A. Only bias
- B. Bias and skewness of bootstrap distribution
- C. Only skewness
- D. Non-normality of original data

Correct answer: B

EXAM SHORTCUT

BCa = Bias-Corrected and accelerated; bias correction $\hat{z}_0$ and acceleration $\hat{a}$ both applied; most accurate bootstrap CI method

Estimation

CI Jackknife SE

Medium

Q15.

Jackknife standard error estimate is:

- A. $\sqrt{(n-1)/(n) \sum (\theta(i) - \hat{\theta}(\cdot))^2}$
- B. $\sqrt{(1)/(n) \sum (\theta(i) - \hat{\theta}(\cdot))^2}$
- C. $\sqrt{(n-1)/(n^2) \sum (\theta(i) - \hat{\theta}(\cdot))^2}$
- D. $\sqrt{(n-1)/(n) \sum (\theta(i) - \hat{\theta})^2}$

Correct answer: A

EXAM SHORTCUT

Jackknife SE = $\sqrt{(n-1)/(n) \sum_{i=1}^n (\theta(i) - \hat{\theta}(\cdot))^2}$; the $(n-1)/n$ factor (not $1/n$) is crucial

Estimation

CI Optimum Unbiased

Hard

Q16.

Optimum confidence interval (shortest unbiased) for μ in $N(\mu, \sigma^2)$ with σ^2 unknown is:

- A. z-interval using S
- B. Shortest t -interval (not equal-tails)
- C. One-sided t -interval
- D. **Equal-tails t -interval**

Correct answer: D

EXAM SHORTCUT

For t -distribution (symmetric), equal-tails interval IS the shortest unbiased interval; for asymmetric distributions (e.g., χ^2), equal-tails is NOT shortest

Estimation

CI Ratio Two Means

Hard

Q17.

CI for ratio μ_1/μ_2 of two normal means uses:

- A. Fieller's theorem
- B. Log transformation then back-transform
- C. **All of the above are valid approaches**
- D. Delta method approximation

Correct answer: C

EXAM SHORTCUT

Fieller's theorem gives exact CI; delta method gives approximate CI; log transformation is another approximation; all are used in practice

Estimation

CI Nonparametric Median

Hard

Q18.

Distribution-free CI for population median uses:

- A. Sample median $\pm z_{\alpha/2} \cdot \pi/(2n)$ (asymptotic)
- B. Bootstrap only
- C. $\bar{X} \pm z_{\alpha/2} S/\sqrt{n}$
- D. ✓ Order statistics $X(r)$ and $X(s)$ with coverage from binomial distribution

Correct answer: D

EXAM SHORTCUT

Exact nonparametric CI for median: $[X(r), X(s)]$ where r, s chosen so $P(r \leq \text{Bin}(n, 1/2) \leq s-1) \geq 1 - \alpha$; A is asymptotic approximation

Estimation

CI Simultaneous Bonferroni

Medium

Q19.

Bonferroni simultaneous CIs for k parameters at family-wise level $1 - \alpha$ use individual level:

- A. $1 - k\alpha$
- B. $(1 - \alpha)^{1/k}$
- C. $1 - \alpha/2$
- D. ✓ $1 - \alpha/k$

Correct answer: D

EXAM SHORTCUT

Bonferroni: use α/k per interval; B is Šidák correction (exact for independent tests); both are conservative

Estimation

CI Bootstrap Double

Hard

Q20.

Double bootstrap (bootstrap-t) CI:

- A. Applies bootstrap twice to reduce bias
- B. Is equivalent to BCa
- C. ✓ Uses bootstrap to estimate the distribution of $(\theta - \hat{\theta})/SE$
- D. Uses t -distribution with bootstrap SE

Correct answer: C

EXAM SHORTCUT

Bootstrap-t: estimate distribution of studentized statistic via nested bootstrap; computationally expensive but second-order accurate; different from BCa

Estimation

CI Profile Likelihood

Hard

Q21.

Profile likelihood CI for ψ is:

- A. $\hat{\psi} \pm z_{\alpha/2} / \sqrt{nI(\psi)}$
- B. $\{\psi: \ell p(\psi) \geq \ell p(\hat{\psi}) - z^2_{\alpha/2}/2\}$
- C. **Both A and C (equivalent since $\chi^2_{\alpha,1} = z^2_{\alpha/2}$)**
- D. $\{\psi: -2[\ell p(\psi) - \ell p(\hat{\psi})] \leq \chi^2_{\alpha,1}\}$

Correct answer: C

EXAM SHORTCUT

 $\chi^2_{0.05,1} = 3.841 = 1.96^2$; D and B are equivalent; profile likelihood CI is second order accurate unlike Wald CI

Estimation

CI Exponential One Sided

Medium

Q22.

One-sided 95% upper CI for λ in $\text{Exp}(\lambda)$ based on n observations:

- A. $\lambda \leq z_{0.05} / (\bar{X}/n)$
- B. $\lambda \leq \chi^2_{0.05, 2n} / \sum X_i$
- C. $\lambda \leq \chi^2_{0.95, 2n} / (2 \sum X_i)$
- D. **$\lambda \leq \chi^2_{0.05, 2n} / (2 \sum X_i)$**

Correct answer: D

EXAM SHORTCUT

 $P(2\lambda \sum X_i \leq \chi^2_{0.05, 2n}) = 0.05$; upper CI: $\lambda \leq \chi^2_{0.05, 2n} / (2 \sum X_i)$; note $\chi^2_{0.05}$ is LOWER tail quantile giving UPPER bound for λ

Estimation

CI Poisson Exact

Hard

Q23.

Exact CI for Poisson rate λ based on observed count x is:

- A. $[\bar{X} \pm z_{\alpha/2} \sqrt{\bar{X}/n}]$
- B. **$[\chi^2_{\alpha/2, 2x} / (2n), \chi^2_{1-\alpha/2, 2(x+1)} / (2n)]$**
- C. $[\chi^2_{1-\alpha/2, 2x} / (2n), \chi^2_{\alpha/2, 2(x+1)} / (2n)]$
- D. $[x/n \pm z_{\alpha/2} \sqrt{x/n}]$

Correct answer: B

EXAM SHORTCUT

 Exact Poisson CI: lower = $\chi^2_{\alpha/2, 2x} / 2$ (for rate per unit), upper = $\chi^2_{1-\alpha/2, 2(x+1)} / 2$; divide by n for rate; careful with which tail goes where

Estimation

CI Correlation

Hard

Q24.

Fisher's z-transformation CI for correlation ρ :

- A. $\hat{\rho} \pm z_{\alpha/2}/\sqrt{n}$
- B. $\tanh(\tanh^{-1}(\hat{\rho}) \pm z_{\alpha/2}/\sqrt{n-3})$
- C. $\hat{\rho} \pm z_{\alpha/2}\sqrt{(1-\hat{\rho}^2)^2/(n-1)}$
- D. $\hat{\rho} \pm t_{\alpha/2, n-2}\sqrt{(1-\hat{\rho}^2)/(n-2)}$

Correct answer: B

EXAM SHORTCUT

Fisher z: $z = \tanh^{-1}(\hat{\rho}) = (1/2) \log \frac{1+\hat{\rho}}{1-\hat{\rho}}$; $z \sim N(\tanh^{-1}(\rho), 1/(n-3))$; back-transform via $\tanh$

Estimation

CI Difference Proportions

Medium

Q25.

CI for $p_1 - p_2$ (two independent proportions):

- A. **A for CI; B for hypothesis test**
- B. $(\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2}\sqrt{\hat{p}(1-\hat{p})(1/n_1 + 1/n_2)}$ where $\hat{p}$ is pooled
- C. Both A and B are valid CIs
- D. $(\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2}\sqrt{\hat{p}_1(1-\hat{p}_1)/n_1 + \hat{p}_2(1-\hat{p}_2)/n_2}$

Correct answer: A

EXAM SHORTCUT

For CI: use unpooled SE (A); for hypothesis test $H_0: p_1 = p_2$: use pooled SE (B); common exam trap to mix them

Estimation

CI Regression Slope

Medium

Q26.

CI for slope β in simple linear regression:

- A. $\hat{\beta} \pm t_{\alpha/2, n-2} \cdot S/\sqrt{S_{xx}}$ where $S_{xx} = \sum (x_i - \bar{x})^2$
- B. **$\hat{\beta} \pm t_{\alpha/2, n-2} \cdot S_{\hat{\beta}}$ where $S^2(\hat{\beta}) = S^2/S_{xx}$**
- C. $\hat{\beta} \pm z_{\alpha/2} \cdot \sigma/\sqrt{S_{xx}}$
- D. $\hat{\beta} \pm t_{\alpha/2, n-1} \cdot S/\sqrt{S_{xx}}$

Correct answer: B

EXAM SHORTCUT

A and B are equivalent; $n-2$ df (not $n-1$); $S^2 = \text{RSS}/(n-2)$; C uses known σ

Estimation

CI Bootstrap Parametric

Hard

Q27.

Parametric bootstrap CI:

- A. Is equivalent to nonparametric bootstrap
- B. Uses analytical formula with bootstrap SE
- C. Resamples from empirical distribution
- D. Resamples from fitted parametric model $f(x; \hat{\theta})$**

Correct answer: D

EXAM SHORTCUT

Parametric bootstrap: simulate from $f(x; \hat{\theta})$; nonparametric: resample from data; they differ when model is misspecified

Estimation

CI Jackknife Bias Corrected

Hard

Q28.

Jackknife bias-corrected estimator is:

- A. All of the above (equivalent)**
- B. $\hat{\theta} + \text{biasjack}$ where $\text{bias} = (n-1)(\hat{\theta}(\cdot) - \hat{\theta})$
- C. $\hat{\theta} - (n-1)\hat{\theta}(\cdot)$
- D. $\hat{\theta} - (n-1)(\hat{\theta}(\cdot) - \hat{\theta})$

Correct answer: A

EXAM SHORTCUT

Jackknife bias estimate = $(n-1)(\hat{\theta}(\cdot) - \hat{\theta})$; bias-corrected = $\hat{\theta} - \hat{\text{bias}} = \hat{\theta} - (n-1)(\hat{\theta}(\cdot) - \hat{\theta}) = \hat{\theta} - (n-1)\hat{\theta}(\cdot)$; all equivalent

Estimation

CI Subsampling

Hard

Q29.

Subsampling CI differs from bootstrap in that:

- A. It is valid under weaker conditions than bootstrap
- B. It requires correct specification of convergence rate
- C. It samples WITHOUT replacement (subsets of size $b < n$)
- D. All of the above**

Correct answer: D

EXAM SHORTCUT

Subsampling: without replacement, size $b \rightarrow \infty$ with $b/n \rightarrow 0$; valid even when bootstrap fails (e.g., for maximum of i.i.d. uniform); requires knowing rate

Estimation

CI Simultaneous Scheffe

Hard

Q30.

Scheffé simultaneous CI for all linear contrasts $cT\beta$ in regression:

- A. $\hat{cT\beta} \pm t_{\alpha/2, n-p} \cdot SE$
- B. $\hat{cT\beta} \pm \sqrt{F_{\alpha, 1, n-p}} \cdot SE$
- C. $\hat{cT\beta} \pm z_{\alpha/2} \cdot SE$
- D. $\hat{cT\beta} \pm \sqrt{p \cdot F_{\alpha, p, n-p}} \cdot SE$

Correct answer: D

EXAM SHORTCUT

Scheffé: $\sqrt{p \cdot F}$ critical value covers ALL contrasts simultaneously; Bonferroni covers only pre-specified contrasts; p = number of parameters

Estimation

CI Nonparametric Bootstrap Validity

Hard

Q31.

Bootstrap CI is NOT valid when:

- A. ~~Statistic is the sample maximum from $U(0, \theta)$~~
- B. Distribution is skewed
- C. Sample size is small
- D. Distribution has heavy tails

Correct answer: A

EXAM SHORTCUT

Bootstrap fails for non-regular statistics like sample maximum (convergence rate n^{-1} not $n^{-1/2}$); subsampling or m -out-of- n bootstrap needed

Estimation

CI Likelihood Ratio

Hard

Q32.

Likelihood ratio CI has the property of being:

- A. Exact for all distributions
- B. Always equal to Wald CI
- C. ~~Transformation invariant (equivariant under reparametrization)~~
- D. Always shorter than Wald CI

Correct answer: C

EXAM SHORTCUT

LR CI is transformation invariant (key advantage over Wald); Wald CI is NOT invariant; LR CI is generally more accurate than Wald in finite samples

Estimation

CI Bootstrap Iterations

Medium

Q33.

For bootstrap CI, recommended number of bootstrap replications B :

- A. $B = 100$ is always sufficient
- B. $B = n$ (same as sample size)
- C. $B = \infty$ (theoretical limit)
- D. $B \geq 1000$ for percentile CI; $B \geq 2000$ for BCa**

Correct answer: D

EXAM SHORTCUT

Practical guidelines: $B = 1000$ for SE and percentile; $B = 2000-5000$ for BCa; more replications reduce Monte Carlo error in CI endpoints

Estimation

CI Exact Binomial

Hard

Q34.

Clopper-Pearson exact CI for p based on x successes in n trials:

- A. $[p \pm z_{\alpha/2} \sqrt{p(1-p)/n}]$
- B. Both A and C (equivalent via Beta-F relationship)**
- C. $[F_{\alpha/2, 2x, 2(n-x+1)} \cdot x / (n-x+1 + F \cdot x), \dots]$
- D. $[B_{\alpha/2; x, n-x+1}, B_{1-\alpha/2; x+1, n-x}]$ where B is Beta quantile

Correct answer: B

EXAM SHORTCUT

Clopper-Pearson uses Beta distribution; equivalent F-distribution form also exists; both are exact (conservative); A is Wald (approximate)

Estimation

CI Regression Prediction

Medium

Q35.

Prediction interval for new response Y_0 at x_0 in regression:

- A. $\hat{Y}_0 \pm t_{\alpha/2, n-2} S \sqrt{1 + 1/n + (x_0 - \bar{x})^2 / S_{xx}}$**
- B. $\hat{Y}_0 \pm t_{\alpha/2, n-2} S / \sqrt{n}$
- C. $\hat{Y}_0 \pm z_{\alpha/2} \sigma \sqrt{1 + 1/n + (x_0 - \bar{x})^2 / S_{xx}}$
- D. $\hat{Y}_0 \pm t_{\alpha/2, n-2} S \sqrt{1/n + (x_0 - \bar{x})^2 / S_{xx}}$

Correct answer: A

EXAM SHORTCUT

Prediction interval: extra 1 under square root (for new observation variance); D is CI for mean response; A is wider than D

Estimation

CI Variance Ratio

Medium

Q36.

CI for σ^2_1/σ^2_2 (two independent normal samples):

- A. $[S^2_1/S^2_2 \pm z_{\alpha/2}\sqrt{2/(n_1-1) + 2/(n_2-1)}]$
- B. $[S^2_1/(S^2_2 \cdot F_{\alpha/2, n_1-1, n_2-1}), S^2_1 \cdot F_{\alpha/2, n_2-1, n_1-1}/S^2_2]$
- C. **Both A and C (equivalent)**
- D. $[(S^2_1/S^2_2)/F_{1-\alpha/2, n_1-1, n_2-1}, (S^2_1/S^2_2) \cdot F_{1-\alpha/2, n_2-1, n_1-1}]$

Correct answer: C

EXAM SHORTCUT

$F_{1-\alpha/2, m, n} = 1/F_{\alpha/2, n, m}$; both forms are equivalent; note df order reversal in the reciprocal

Estimation

CI Bootstrap Studentized

Hard

Q37.

Studentized bootstrap (bootstrap-t) CI is:

- A. Exact
- B. Less accurate than percentile bootstrap
- C. **Second-order accurate (error $O(n^{-1})$)**
- D. First-order accurate (error $O(n^{-1/2})$)

Correct answer: C

EXAM SHORTCUT

Bootstrap-t is second-order accurate; percentile and basic bootstrap are first-order; BCa is also second-order; studentized bootstrap requires nested bootstrap for SE

Estimation

CI Jackknife Failure

Hard

Q38.

Jackknife fails (is inconsistent) for:

- A. Sample variance
- B. **Sample median (non-smooth functional)**
- C. Sample mean
- D. Regression coefficients

Correct answer: B

EXAM SHORTCUT

Jackknife fails for non-smooth statistics like sample median, sample quantiles; bootstrap is preferred; jackknife works for smooth functionals

Estimation

CI Simultaneous Tukey

Hard

Q39.

Tukey's HSD simultaneous CI for pairwise differences $\mu_i - \mu_j$ in one-way ANOVA uses:

- A. Normal distribution
- B. Studentized range distribution $q_{\alpha, k, N-k}$
- C. t -distribution with Bonferroni correction
- D. F -distribution

Correct answer: B

EXAM SHORTCUT

Tukey uses studentized range (maximum of pairwise t -statistics); exact for balanced designs; Bonferroni is conservative; Tukey is less conservative

Estimation

CI Bootstrap Bias Correction

Hard

Q40.

Bootstrap bias estimate for $\hat{\theta}$ is:

- A. $E[\hat{\theta}] - \theta$
- B. $\hat{\theta} - \theta^*$
- C. $\bar{\theta}^* - \hat{\theta}$ where $\bar{\theta}^*$ is mean of bootstrap estimates
- D. $\hat{\theta}^* - \theta$

Correct answer: C

EXAM SHORTCUT

Bootstrap bias = $E[\bar{\theta}^*] - \hat{\theta}$; bias-corrected estimate = $2\bar{\theta}^* - \hat{\theta}$; B has wrong sign

Estimation

CI Fiducial Location

Hard

Q41.

For location family $f(x - \theta)$, fiducial distribution of θ given $X = x$ is:

- A. Normal distribution always
- B. Posterior with flat prior
- C. $f(x - \theta) \cdot \sqrt{n}$ (scaled)
- D. Both A and B (coincide for location families)

Correct answer: D

EXAM SHORTCUT

For location families, fiducial = posterior with flat (improper) prior = confidence distribution; all three coincide — key result

Estimation

CI Tolerance Normal

Hard

Q42.

For $N(\mu, \sigma^2)$ with both unknown, tolerance interval covering at least proportion p with confidence $1 - \alpha$:

- A. $\bar{X} \pm zp\sigma$
- B. $\bar{X} \pm zp/2S\sqrt{1 + 1/n}$
- C. $\bar{X} \pm zpS$
- D. $\bar{X} \pm k(n, p, \alpha)S$ where k is tolerance factor from tables

Correct answer: D

EXAM SHORTCUT

Tolerance factor k accounts for estimation of both μ and σ ; $k > zp$ always; $k \rightarrow zp$ as $n \rightarrow \infty$; C undercovers

Estimation

CI Bootstrap Smoothed

Hard

Q43.

Smoothed bootstrap adds noise to resampled values to:

- A. Reduce computation time
- B. Handle missing data
- C. **Improve CI coverage for discrete distributions**
- D. Increase bias

Correct answer: C

EXAM SHORTCUT

Smoothed bootstrap: add small random noise (bandwidth h) to discrete resamples; treats discrete data as continuous; improves coverage for discrete statistics

Estimation

CI Confidence Distribution

Hard

Q44.

A confidence distribution $C(\theta)$ satisfies:

- A. **All of the above**
- B. $C(\theta_0) \sim U(0, 1)$ under $H_0: \theta = \theta_0$
- C. $P(\theta \in [C^{-1}(\alpha/2), C^{-1}(1 - \alpha/2)]) = 1 - \alpha$
- D. It is a proper probability distribution over θ

Correct answer: A

EXAM SHORTCUT

Confidence distribution unifies CI, p -values, and point estimates; $C(\theta_0)$ is the p -value for $H_0: \theta \leq \theta_0$; modern framework

Estimation

CI Bootstrap Accelerated

Hard

Q45.

The acceleration constant $\hat{a}$ in BCa bootstrap measures:

- A. Bias of $\hat{\theta}$
- B. Sample size effect
- C. Skewness of bootstrap distribution
- D. Rate of change of SE with respect to θ**

Correct answer: D

EXAM SHORTCUT

$\hat{a} = \frac{\sum(\theta(\cdot) - \theta(i))^3}{6[\sum(\theta(\cdot) - \theta(i))^2]^3/2}$ (jackknife-based); measures how SE changes with θ (influence function skewness)

Estimation

CI Regression Simultaneous

Hard

Q46.

Working-Hotelling simultaneous CI band for regression line $E[Y|x] = \alpha + \beta x$:

- A. $\hat{Y}(x) \pm z_{\alpha/2} \cdot S\sqrt{1/n + (x - \bar{x})^2/S_{xx}}$
- B. $\hat{Y}(x) \pm \sqrt{2F_{\alpha, 2, n-2}} \cdot S\sqrt{1/n + (x - \bar{x})^2/S_{xx}}$**
- C. $\hat{Y}(x) \pm t_{\alpha/2, n-2} \cdot S\sqrt{1/n + (x - \bar{x})^2/S_{xx}}$
- D. $\hat{Y}(x) \pm F_{\alpha, 2, n-2} \cdot S\sqrt{1/n + (x - \bar{x})^2/S_{xx}}$

Correct answer: B

EXAM SHORTCUT

Working-Hotelling: $\sqrt{2F}$ critical value for ALL x simultaneously; C is pointwise CI; $\sqrt{2F_{0.05, 2, n-2}} > t_{0.025, n-2}$ so band is wider

Estimation

CI Bootstrap Monte Carlo Error

Hard

Q47.

Monte Carlo error in bootstrap CI endpoint $\theta^*(\alpha)$ is approximately:

- A. $\sqrt{\alpha(1-\alpha)}/B \cdot \sigma^*$ where σ^* is bootstrap SD**
- B. $\alpha/\sqrt{B}$
- C. $\hat{\sigma}^*/\sqrt{B}$
- D. $1/\sqrt{B}$

Correct answer: A

EXAM SHORTCUT

SE of sample quantile: $\sqrt{p(1-p)/f(\xi_p)}$; for bootstrap distribution, $f(\xi_p) \approx 1/\sigma^*$; so $SE \approx \sigma^* \sqrt{\alpha(1-\alpha)}/B$

Estimation

CI Empirical Likelihood

Hard

Q48.

Empirical likelihood CI (Owen) is:

- A. Requires parametric model
- B. Equivalent to bootstrap percentile CI
- C. **Nonparametric, transformation invariant, Bartlett correctable**
- D. Always wider than bootstrap CI

Correct answer: C

EXAM SHORTCUT

Empirical likelihood: nonparametric analog of likelihood ratio CI; shares properties of parametric LR CI (transformation invariance, Bartlett correction); Wilks theorem holds

Estimation

CI Shortest Chi Square

Hard

Q49.

Shortest CI for σ^2 (not equal-tails) requires finding χ^2_L, χ^2_U satisfying:

- A. χ^2_U / χ^2_L minimized
- B. Equal tail probabilities
- C. **$1/\chi^2_L - 1/\chi^2_U$ minimized subject to $P(\chi^2_L \leq \chi^2_{n-1} \leq \chi^2_U) = 1 - \alpha$**
- D. $\chi^2_{1-\alpha/2, n-1} - \chi^2_{\alpha/2, n-1}$ minimized subject to coverage $1 - \alpha$

Correct answer: C

EXAM SHORTCUT

CI for σ^2 : length $\propto 1/\chi^2_L - 1/\chi^2_U$; minimize this subject to coverage; NOT equal tails; equal-tails is standard but not shortest

Estimation

CI Bootstrap Double Coverage

Hard

Q50.

Double bootstrap corrects coverage error of bootstrap CI from:

- A. $O(n^{-1})$ to $O(n^{-3/2})$
- B. $O(1)$ to $O(n^{-1/2})$
- C. Exact to approximate
- D. **$O(n^{-1/2})$ to $O(n^{-1})$**

Correct answer: D

EXAM SHORTCUT

Single bootstrap CI: coverage error $O(n^{-1/2})$; double bootstrap: $O(n^{-1})$; studentized bootstrap also achieves $O(n^{-1})$; computationally expensive

Hypothesis Testing

15 forecast questions in this topic

Hypothesis Testing

Simple and Composite Hypotheses, Two Kinds of Error, Size and Power

Easy

Q1.

The power of a test at an alternative $\theta \in H_1$ is defined as

A. $\beta(\theta) = P_\theta(\text{accept } H_0)$

B. $\alpha = P(\text{reject } H_0 / H_0 \text{ true})$

C. $1 - \beta(\theta) = P_\theta(\text{reject } H_0)$

D. $1 - \alpha$

Correct answer: C

EXAM SHORTCUT

Power is the probability of correctly rejecting H_0 when the alternative is true - i.e. 1 minus the Type II error probability at that alternative.

Hypothesis Testing

Neyman-Pearson Lemma and Best Critical Region

Medium

Q2.

X is a single observation from $\text{Exponential}(\theta)$ with pdf $f(x, \theta) = \theta e^{-\theta x}$, $x > 0$. To test $H_0: \theta = 1$ against $H_1: \theta = 2$, the test rejects H_0 if $X < 0.5$. The size and power of this test are respectively

A. 0.50, 0.50

B. 0.63, 0.39

C. 0.39, 0.39

D. 0.39, 0.63

Correct answer: D

EXAM SHORTCUT

Size $= P(X < 0.5 / \theta = 1) = 1 - e^{-0.5} \approx 0.39$; power $= P(X < 0.5 / \theta = 2) = 1 - e^{-1} \approx 0.63$.

Q3.

For $X \sim \text{Poisson}(\theta)$, testing $H_0: \theta = \theta_0$ against $H_1: \theta = \theta_1$ with $\theta_1 > \theta_0$, the Neyman-Pearson best critical region is of the form

- A. $X \leq c$
- B. $X = c$ exactly
- C. $|X - \theta_0| \geq c$
- D. $X \geq c$

Correct answer: D

EXAM SHORTCUT

Likelihood ratio $f(x; \theta_0)/f(x; \theta_1) = e^{\theta_1 - \theta_0} (\theta_0/\theta_1)^x$ is decreasing in x since $\theta_0/\theta_1 < 1$; small ratio (reject H_0) means large x .

Q4.

A family with Monotone Likelihood Ratio (MLR) in a statistic T guarantees the existence of a UMP test for

- A. two-sided hypotheses, $H_0: \theta = \theta_0$ vs $H_1: \theta \neq \theta_0$
- B. only point-null versus point-alternative tests
- C. one-sided hypotheses, e.g. $H_0: \theta \leq \theta_0$ vs $H_1: \theta > \theta_0$
- D. any hypothesis, one- or two-sided, without exception

Correct answer: C

EXAM SHORTCUT

MLR guarantees a UMP test exists for ONE-SIDED composite alternatives; for two-sided alternatives no UMP test generally exists, and one instead seeks a UMP-unbiased (UMPU) test.

Q5.

$X \sim \text{Binomial}(10, p)$. To test $H_0: p=0.5$ vs $H_1: p>0.5$ at nominal level $\alpha=0.05$ using the NON-randomized test 'reject H_0 if $X \geq 9$ ', the EXACT size of this test is

- A. 0.0547
- B. 0.1074
- C. **0.0107**
- D. 0.05 exactly

Correct answer: C**EXAM SHORTCUT**

$$P(X \geq 9 / p=0.5) = \frac{\binom{10}{9} + \binom{10}{10}}{2^{10}} = \frac{11}{1024} \approx 0.0107$$
 - discreteness prevents hitting $\alpha=0.05$ exactly without randomizing.

Q6.

$X \sim \text{Poisson}(\lambda)$. To test $H_0: \lambda=2$ vs $H_1: \lambda=3$, the randomized test $\varphi(x)=1$ if $x \geq 4$; $\varphi(x)=0.4$ if $x=3$; $\varphi(x)=0$ if $x \leq 2$ has size and power respectively approximately

- A. 0.442, 0.215
- B. **0.215, 0.442**
- C. 0.323, 0.577
- D. 0.180, 0.224

Correct answer: B**EXAM SHORTCUT**

Size $= P(X \geq 4 / \lambda=2) + 0.4P(X=3 / \lambda=2)$; power is the same formula evaluated at $\lambda=3$.

Q7.

Randomized tests are needed in hypothesis testing primarily because

- A. continuous distributions require randomization to define a critical region
- B. **for discrete distributions, no non-randomized critical region can generally achieve an arbitrary prescribed size α exactly**
- C. they always have strictly higher power than non-randomized tests
- D. randomization removes the need to specify H_1

Correct answer: B

EXAM SHORTCUT

Discreteness of the sample space (Binomial, Poisson, etc.) means the achievable exact sizes form a finite set of values; randomizing at one boundary point lets us hit any α in between exactly.

Q8.

The likelihood ratio test statistic for testing $H_0: \theta \in \Theta_0$ against the general alternative $\theta \in \Theta$ is defined as

- A. **$\lambda = \frac{\sup_{\theta \in \Theta_0} L(\theta)}{\sup_{\theta \in \Theta} L(\theta)}$, and H_0 is rejected for small λ**
- B. λ is always ≥ 1
- C. $\lambda = \frac{\sup_{\theta \in \Theta} L(\theta)}{\sup_{\theta \in \Theta_0} L(\theta)}$, and H_0 is rejected for small λ
- D. $\lambda = L(\theta_0)$ alone, ignoring the alternative

Correct answer: A

EXAM SHORTCUT

LRT statistic $\lambda \in (0, 1]$ compares the best-fit likelihood restricted to H_0 against the best-fit likelihood over the whole parameter space; small λ signals the data fit H_0 poorly relative to the unrestricted fit.

Q9.

Under H_0 , as $n \rightarrow \infty$, $-2 \log \lambda$ (where λ is the LRT statistic) asymptotically follows

- A. t_{n-1}
- B. $N(0,1)$
- C. an F distribution
- D. χ^2_r , where r is the number of restrictions imposed by H_0

Correct answer: D

EXAM SHORTCUT

Wilks' theorem: $-2 \log \lambda \rightarrow d\chi^2_r$ under H_0 , with $r = \dim(\Theta) - \dim(\Theta_0)$, the number of independent restrictions H_0 imposes.

Q10.

A critical region C is called 'similar' with respect to a nuisance parameter if

- A. it has size exactly 0.5
- B. Its size is the same constant for every value of the nuisance parameter, throughout H_0
- C. it is necessarily randomized
- D. it contains only sample points that resemble each other

Correct answer: B

EXAM SHORTCUT

A similar region has rejection probability that does NOT vary with the nuisance parameter anywhere in H_0 - essential when H_0 is composite due to an unknown nuisance parameter.

Q11.

A test φ is said to be UNBIASED for testing H_0 against H_1 if

- A. it has power 1 at every alternative
- B. its power at every alternative in H_1 is at least as large as its size**
- C. its power equals exactly α everywhere
- D. it never commits a Type II error

Correct answer: B

EXAM SHORTCUT

Unbiasedness of a test requires the power function to be $\geq \alpha$ throughout H_1 and $\leq \alpha$ throughout H_0 - the test should never be 'more likely' to reject under a false alternative than under some point of H_0 itself.

Q12.

The power function of a test φ , $\beta(\theta) = E_\theta[\varphi(X)]$, is defined

- A. as the constant $1-\alpha$ everywhere
- B. only at a single point, $\theta = \theta_0$
- C. only for $\theta \in H_1$
- D. for every θ in the full parameter space, both under H_0 and H_1**

Correct answer: D

EXAM SHORTCUT

The power function $\beta(\theta) = P_\theta(\text{reject } H_0)$ is defined over the ENTIRE parameter space: under H_0 it equals the (nominal) size, and under H_1 it equals the power.

Q13.

$X \sim N(\mu, 25)$. A sample of size $n=25$ is used to test $H_0: \mu=50$ against $H_1: \mu=53$ at $\alpha=0.05$ (one-sided, using $z_{0.05}=1.645$). The power of the test is approximately

- A. 0.99
- B. 0.05
- C. 0.50
- D. 0.91

Correct answer: D

EXAM SHORTCUT

Critical value $\bar{x}=50+1.645(5/5)=51.645$; power $=P(\bar{X}>51.645/\mu=53)=P(Z>-1.355)\approx 0.91$.

Q14.

For $X \sim N(\mu, \sigma^2)$ with σ^2 known, testing $H_0: \mu \leq \mu_0$ against $H_1: \mu > \mu_0$, the UMP test rejects H_0 when

- A. $|\bar{X} - \mu_0| > z_{\alpha/2} \sigma / \sqrt{n}$
- B. $\bar{X} < \mu_0 - z_{\alpha} \sigma / \sqrt{n}$
- C. $\bar{X} = \mu_0$
- D. $\bar{X} > \mu_0 + z_{\alpha} \sigma / \sqrt{n}$

Correct answer: D

EXAM SHORTCUT

The standard one-sided UMP test for a normal mean (variance known) rejects for large $\bar{X}$, with the threshold set by the UPPER-tail critical value z_{α} .

Q15.

For testing equality of two normal population variances, $H_0: \sigma_1^2 = \sigma_2^2$, using independent samples of sizes n_1, n_2 , the statistic S_1^2/S_2^2 under H_0 follows

A. F_{n_1-1, n_2-1}

B. $t_{n_1+n_2-2}$

C. $\chi^2_{n_1+n_2-2}$

D. $N(0,1)$

Correct answer: A

EXAM SHORTCUT

The ratio of two independent sample variances (each scaled chi-square) from normal populations with equal true variance follows the F distribution with numerator and denominator df equal to each sample's error df.

Simple and Composite Hypotheses, Two Kinds of Error, Size and Power

1 forecast questions in this topic

Hypothesis Testing

Simple and Composite Hypotheses, Two Kinds of Error, Size and Power

Medium

Q1.

For a composite null hypothesis H_0 , the size of a test with critical region C is defined as

- A.** $\sup_{\theta \in H_0} P_{\theta}(C)$
- B.** the average of $P_{\theta}(C)$ over H_0
- C.** the Type II error probability
- D.** $\inf_{\theta \in H_0} P_{\theta}(C)$

Correct answer: A

EXAM SHORTCUT

For a composite null, size is the SUPREMUM (worst case) of the rejection probability over all $\theta \in H_0$ - guaranteeing the Type I error never exceeds this bound anywhere in H_0 .

UMP, MLR, Unbiased and Similar Tests

1 forecast questions in this topic

Hypothesis Testing

UMP, MLR, Unbiased and Similar Tests

Medium

Q1.

For testing $H_0:\theta=\theta_0$ vs $H_1:\theta\neq\theta_0$ in a one-parameter exponential family, since no UMP test exists in general, one instead seeks a

- A. test with power exactly 0.5 everywhere
- B. ~~UMP~~ Unbiased (UMPU) test**
- C. test most powerful against a single chosen alternative only
- D. randomized test only

Correct answer: B

EXAM SHORTCUT

The standard resolution to 'no UMP test exists for two-sided alternatives' is to restrict attention to UNBIASED tests and find the best one within that class - the UMPU test.

Sequential and Decision Theory

8 forecast questions in this topic

Hypothesis Testing

Wald's SPRT, OC and ASN Functions

Medium

Q1.

In Wald's SPRT for testing H_0 vs H_1 with Type I error α and Type II error β , the approximate stopping bounds A (lower) and B (upper) for the likelihood ratio satisfy

A. $A=\alpha, B=\beta$

B. ✓ $A \approx \frac{\beta}{1-\alpha}, B \approx \frac{1-\beta}{\alpha}$

C. $A \approx \frac{\alpha}{1-\beta}, B \approx \frac{1-\alpha}{\beta}$

D. $A \approx \frac{1-\beta}{\alpha}, B \approx \frac{\beta}{1-\alpha}$

Correct answer: B

EXAM SHORTCUT

Wald's bounds: continue sampling while $A < LR < B$, with $A \approx \beta/(1-\alpha)$ (lower) and $B \approx (1-\beta)/\alpha$ (upper) - getting these two the wrong way round is the classic slip.

Hypothesis Testing

Wald's SPRT, OC and ASN Functions

Medium

Q2.

In SPRT, the Operating Characteristic function $L(\theta)$ gives the probability of accepting H_0 . At $\theta=\theta_0$ (i.e. under H_0), $L(\theta_0)$ approximately equals

A. ~~$1-\alpha$~~

B. $1-\beta$

C. α

D. β

Correct answer: A

EXAM SHORTCUT

$L(\theta_0)$ is the probability of (correctly) accepting H_0 when H_0 is true, which is 1 minus the Type I error rate, i.e. $1-\alpha$.

Q3.

The Average Sample Number (ASN) of Wald's SPRT, compared to a fixed-sample-size Neyman-Pearson test achieving the SAME α, β , is generally

- A. always larger than the fixed-sample test
- B. **smaller - SPRT is more efficient on average, often needing roughly half the expected sample size**
- C. exactly equal to the fixed sample size
- D. infinite

Correct answer: B

EXAM SHORTCUT

SPRT's key advantage (Wald-Wolfowitz optimality) is achieving the same (α, β) error rates with a SMALLER expected sample size than any fixed-sample test.

Q4.

In decision-theoretic estimation, the risk function $R(\theta, \delta)$ of a decision rule δ is defined as

- A. the maximum possible loss over all θ
- B. **$\mathbb{E}_{\theta}[L(\theta, \delta(X))]$**
- C. $\text{Var}_{\theta}[\delta(X)]$
- D. $L(\theta, \delta(X))$ for a single realization of X

Correct answer: B

EXAM SHORTCUT

Risk is the EXPECTED loss, averaged over the sampling distribution of X at each fixed θ - not the loss for one particular realization.

Q5.

A decision rule that minimizes the MAXIMUM risk over all θ is called

- A. admissible
- B. UMVUE
- C. Bayes
- D. **Minimax**

Correct answer: D

EXAM SHORTCUT

'Minimax' literally means minimizing the maximum (worst-case) risk across the parameter space - a conservative, distribution-free criterion.

Q6.

A decision rule δ is INADMISSIBLE if

- A. it has the single highest risk at exactly one value of θ
- B. it is unbiased
- C. **There exists δ' with $R(\theta, \delta') \leq R(\theta, \delta)$ for every θ , and strictly less for at least one θ**
- D. it is a randomized rule

Correct answer: C

EXAM SHORTCUT

Inadmissibility means another rule DOMINATES it: never worse anywhere, and strictly better somewhere - so there is no reason to ever use the dominated rule.

Q7.

Under squared error loss, the Bayes estimator of θ given data x is

- A. the maximum likelihood estimate
- B. the prior mean, ignoring the data
- C. the posterior mode
- D. **the posterior mean $E[\theta | x]$**

Correct answer: D

EXAM SHORTCUT

For squared error loss specifically, the Bayes estimator (minimizer of posterior expected loss) is exactly the posterior mean - a standard, frequently-tested result.

Q8.

Under Wald's SPRT, if θ is the true parameter value (with $\theta_0 \neq \theta_1$ and finite, nonzero boundaries A, B), the probability that the procedure eventually terminates with a decision is

- A. α
- B. **1 - the SPRT terminates with probability 1**
- C. β
- D. 0

Correct answer: B**EXAM SHORTCUT**

A fundamental SPRT result: as long as $\theta_0 \neq \theta_1$, the procedure terminates (reaches a decision) with probability 1, regardless of the true θ .

Neyman–Pearson Lemma, UMP Tests & Power Functions

50 forecast questions in this topic

Hypothesis Testing

NP Lemma Statement

Medium

Q1.

Neyman-Pearson lemma: Most powerful test of size α for $H_0: \theta = \theta_0$ vs $H_1: \theta = \theta_1$ rejects when:

- A. $X > k$ always
- B. $\Lambda = f(x; \theta_0)/f(x; \theta_1) > k$
- C. $\Lambda = f(x; \theta_1)/f(x; \theta_0) > k$ where k chosen so $P_{H_0}(\Lambda > k) = \alpha$
- D. $p\text{-value} < \alpha$

Correct answer: C

EXAM SHORTCUT

Ratio is f_1/f_0 (alternative over null); reject for LARGE ratio; B has ratio inverted; k determined by size constraint

Hypothesis Testing

UMP Exponential Family

Hard

Q2.

For one-parameter exponential family with monotone likelihood ratio in $T(x)$, UMP test of $H_0: \theta \leq \theta_0$ vs $H_1: \theta > \theta_0$:

- A. Rejects for large $T(x)$
- B. Rejects for small $T(x)$
- C. Rejects for $|T(x)| > k$
- D. Depends on direction of monotonicity

Correct answer: D

EXAM SHORTCUT

If LR is increasing in T : reject for large T ; if decreasing: reject for small T ; must check direction

Hypothesis Testing

UMP Existence Two Sided

Hard

Q3.

UMP test for two-sided $H_1: \theta \neq \theta_0$ against one-sided $H_0: \theta = \theta_0$:

A. ~~Generally does NOT exist~~

B. Exists only for normal distribution

C. Exists when distribution is symmetric

D. Always exists for exponential families

Correct answer: A

EXAM SHORTCUT

UMP for two-sided alternatives generally fails; the test that is MP against $\theta_1 > \theta_0$ differs from MP against $\theta_1 < \theta_0$; UMPU tests exist instead

Hypothesis Testing

NP Lemma Normal Mean

Medium

Q4.

For $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ with σ^2 known, MP test of $H_0: \mu = \mu_0$ vs $H_1: \mu = \mu_1 > \mu_0$:

A. Reject if $\bar{X} > \mu_1$

B. ~~Reject if $\bar{X} > c$ where $c = \mu_0 + z\alpha\sigma/\sqrt{n}$~~

C. Reject if $\bar{X} < c$

D. Reject if $|\bar{X} - \mu_0| > c$

Correct answer: B

EXAM SHORTCUT

LR = $\exp[n(\mu_1 - \mu_0)\bar{X}/\sigma^2 - \text{const}]$; increasing in $\bar{X}$ (since $\mu_1 > \mu_0$); reject for large $\bar{X}$; c from size constraint

Hypothesis Testing

Power Function Definition

Medium

Q5.

Power function $\beta(\theta)$ of a test is:

A. $P_\theta(\text{accept } H_0)$ for $\theta \in H_1$

B. $P(\text{Type I error})$ under H_0

C. ~~$P_\theta(\text{reject } H_0)$ for all θ~~

D. $1 - P(\text{Type II error})$ only under H_1

Correct answer: C

EXAM SHORTCUT

Power function defined for ALL θ ; under H_0 : $\beta(\theta) = \alpha$ (size); under H_1 : $\beta(\theta)$ = power; D is incomplete

Q6.

For testing $H_0: \mu = \mu_0$ vs $H_1: \mu \neq \mu_0$ (normal, σ known), critical region is:

- A. $\{X < \mu_0 - z\alpha\sigma/\sqrt{n}\}$
- B. $\{|X - \mu_0| > z\alpha/2\sigma/\sqrt{n}\}$
- C. $\{X > \mu_0 + z\alpha\sigma/\sqrt{n}\}$
- D. $\{(X - \mu_0)^2 > z^2\alpha\sigma^2/n\}$

Correct answer: B

EXAM SHORTCUT

Two-sided test: B and D are equivalent; C is one-sided right; A is one-sided left; for two-sided, use $z\alpha/2$ not $z\alpha$

Q7.

A critical region C is similar of size α if:

- A. $P_\theta(X \in C) = \alpha$ for all $\theta \in H_0$
- B. $P_\theta(X \in C) = \alpha$ for all θ
- C. $P_\theta(X \in C) \geq \alpha$ for all $\theta \in H_1$
- D. $P_\theta(X \in C) \leq \alpha$ for all $\theta \in H_0$

Correct answer: A

EXAM SHORTCUT

Similar region: EXACTLY α for all $\theta \in H_0$ (not just $\leq \alpha$); level α test has $\leq \alpha$; similar tests are useful for composite H_0

Q8.

A test is unbiased if:

- A. Type I error equals Type II error
- B. $\beta(\theta) = \alpha$ for all θ
- C. Power equals size
- D. $\beta(\theta) \geq \alpha$ for all $\theta \in H_1$ and $\beta(\theta) \leq \alpha$ for all $\theta \in H_0$

Correct answer: D

EXAM SHORTCUT

Unbiased: power under H_1 is at least as large as size; ensures test is not “worse than random”; B is similar test; A is unrelated

Hypothesis Testing

Randomized Test

Hard

Q9.

A randomized test with critical function $\phi(x)$ means:

- A. Randomly choose between two tests
- B. Randomly permute data
- C. **Reject with probability $\phi(x) \in [0, 1]$**
- D. Use random sample size

Correct answer: C

EXAM SHORTCUT

Randomized test: $\phi(x) = P(\text{reject} | X = x)$; needed to achieve exact size α for discrete distributions; $\phi(x) \in \{0, 1\}$ for non-randomized

Hypothesis Testing

NP Lemma Poisson

Medium

Q10.

For $X \sim \text{Poisson}(\lambda)$, MP test of $H_0: \lambda = \lambda_0$ vs $H_1: \lambda = \lambda_1 > \lambda_0$:

- A. Reject for $X = 0$
- B. **Reject for large X (i.e., $X \geq c$)**
- C. Reject for $|X - \lambda_0| > c$
- D. Reject for small X

Correct answer: B

EXAM SHORTCUT

LR = $(\lambda_1/\lambda_0)^x e^{-(\lambda_1 - \lambda_0)}$; increasing in x (since $\lambda_1 > \lambda_0$); reject for large X ; randomization at boundary for exact size

Hypothesis Testing

UMP Normal Variance

Hard

Q11.

For $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ with μ known, UMP test of $H_0: \sigma^2 \leq \sigma_0^2$ vs $H_1: \sigma^2 > \sigma_0^2$:

- A. Reject for small $\sum (X_i - \mu)^2$
- B. **Reject for large $\sum (X_i - \mu)^2$**
- C. Reject for large $|X - \mu|$
- D. UMP does not exist

Correct answer: B

EXAM SHORTCUT

$\sum (X_i - \mu)^2 / \sigma^2 \sim \chi^2_n$; MLR in $\sum (X_i - \mu)^2$; reject for large values; μ known is crucial

Q12.

For $H_0: \mu = 0$ vs $H_1: \mu = \delta > 0$ with σ known, power of z-test at level α is:

A. ~~Both A and C (equivalent)~~

B. $\Phi(-z_\alpha + \delta\sqrt{n}/\sigma)$

C. $1 - \Phi(z_\alpha - \delta\sqrt{n}/\sigma)$

D. $\Phi(z_\alpha - \delta\sqrt{n}/\sigma)$

Correct answer: A

EXAM SHORTCUT

Power = $P(X > z_\alpha \sigma / \sqrt{n} | \mu = \delta) = P(Z > z_\alpha - \delta\sqrt{n}/\sigma) = 1 - \Phi(z_\alpha - \delta\sqrt{n}/\sigma) = \Phi(\delta\sqrt{n}/\sigma - z_\alpha)$; B and C equivalent

Q13.

For $X_1, \dots, X_n \sim \text{Exp}(\theta)$, MP test of $H_0: \theta = \theta_0$ vs $H_1: \theta = \theta_1 < \theta_0$ (smaller rate = larger mean):

A. ~~Both A and C (equivalent)~~

B. Reject for large $\bar{X}$

C. Reject for small $\sum X_i$

D. Reject for large $\sum X_i$

Correct answer: A

EXAM SHORTCUT

LR = $(\theta_1/\theta_0)^n \exp[-(\theta_1 - \theta_0) \sum x_i]$; since $\theta_1 < \theta_0$: $-(\theta_1 - \theta_0) > 0$; LR increasing in $\sum x_i$; reject for large $\sum X_i$

Q14.

For $H_0: \theta \leq \theta_0$ vs $H_1: \theta > \theta_0$ in exponential family with MLR in $T(x)$:

A. UMP does not exist for composite H_0

B. UMP test rejects for $T(x) < c$

C. UMP rejects for $|T(x)| > c$

D. ~~UMP test rejects for $T(x) > c$ where c satisfies $\sup_{\theta \leq \theta_0} P_\theta(T > c) = \alpha$~~

Correct answer: D

EXAM SHORTCUT

MLR property: test that is MP for each simple $\theta_1 > \theta_0$ is the same test; this common test is UMP; size achieved at $\theta = \theta_0$ (boundary)

Q15.

When $PH_0(\Lambda > k) = \alpha_1 < \alpha < \alpha_2 = PH_0(\Lambda \geq k)$, the NP test randomizes at $\Lambda = k$ with probability:

- A. $\gamma = \alpha/\alpha_2$
- B. $\gamma = \alpha$
- C. $\gamma = (\alpha_2 - \alpha)/(\alpha_2 - \alpha_1)$
- D. $\gamma = (\alpha - \alpha_1)/(\alpha_2 - \alpha_1)$

Correct answer: D

EXAM SHORTCUT

Randomization probability γ chosen so total size $= \alpha_1 + \gamma(\alpha_2 - \alpha_1) = \alpha$; solving gives D; needed for discrete distributions

Q16.

For UMP test in exponential family with MLR, power function $\beta(\theta)$ is:

- A. Constant
- B. **Monotone increasing in θ (for one-sided $H_1: \theta > \theta_0$)**
- C. Monotone decreasing
- D. Non-monotone (U-shaped)

Correct answer: B

EXAM SHORTCUT

MLR implies monotone power function; this is the key property that makes the test UMP; power increases as θ moves further into H_1

Q17.

A test is most powerful of level α (not size α) if:

- A. It minimizes Type II error
- B. $\beta(\theta_0) = \alpha$ exactly
- C. **$\sup_{\theta \in H_0} \beta(\theta) \leq \alpha$ and power is maximized**
- D. Power equals 1 $-\beta$ for all $\theta \in H_1$

Correct answer: C

EXAM SHORTCUT

Level α : size $\leq \alpha$ (inequality); size α : size $= \alpha$ (equality); MP of level α may have size $< \alpha$ (conservative)

Hypothesis Testing

NP Lemma Uniform

Hard

Q18.

For $X \sim U(0, \theta)$, testing $H_0: \theta = 1$ vs $H_1: \theta = 2$, MP test:

- A. Rejects if $X < c$
- B. Always rejects (since H_1 support contains H_0 support)
- C. Rejects if $X > 1$ (impossible under H_0)
- D. **Rejects if $X > c$ for some $c \in (0, 1)$**

Correct answer: D

EXAM SHORTCUT

$f(x; 2)/f(x; 1) = 1/2$ for $x \in (0, 1)$ and ∞ for $x \in (1, 2)$; always reject if $X > 1$; for $X \leq 1$, LR = $1/2$ (constant); randomize to achieve exact size α

Hypothesis Testing

UMP Bernoulli

Medium

Q19.

For $X_1, \dots, X_n \sim \text{Bernoulli}(p)$, UMP test of $H_0: p \leq p_0$ vs $H_1: p > p_0$:

- A. UMP does not exist
- B. **Reject for large $\sum X_i$**
- C. Reject for $|\sum X_i - np_0| > c$
- D. Reject for small $\sum X_i$

Correct answer: B

EXAM SHORTCUT

Bernoulli is exponential family with MLR in $\sum X_i$; UMP rejects for large $\sum X_i$; critical value from $\text{Bin}(n, p_0)$ distribution

Hypothesis Testing

Power Envelope

Hard

Q20.

The power envelope is:

- A. Power of the UMP test
- B. Maximum achievable power at each $\theta \in H_1$ over all level- α tests
- C. **Both A and B when UMP exists**
- D. Power of the likelihood ratio test

Correct answer: C

EXAM SHORTCUT

Power envelope = upper bound on power; when UMP exists, it achieves the envelope; when UMP doesn't exist, no single test achieves the envelope everywhere

Hypothesis Testing

NP Lemma Composite H_1

Hard

Q21.

NP lemma directly applies to:

- A. Simple H_0 vs composite H_1
- B. Composite H_0 vs composite H_1
- C. **Simple H_0 vs simple H_1 only**
- D. Any hypothesis testing problem

Correct answer: C

EXAM SHORTCUT

NP lemma: both hypotheses must be simple (point hypotheses); extension to composite uses UMP theory (MLR, exponential family); LRT handles general composite

Hypothesis Testing

Similar Test Construction

Hard

Q22.

For composite $H_0: \theta \in \Theta_0$, similar tests of size α are constructed using:

- A. Rao-Blackwell theorem
- B. Bootstrap
- C. **Neyman structure: $E_{\theta}[\phi(X)|T(X)] = \alpha$ for sufficient statistic T under H_0**
- D. Likelihood ratio

Correct answer: C

EXAM SHORTCUT

Neyman structure: conditional on sufficient statistic for nuisance parameter, test has constant conditional size α ; this gives similar tests

Hypothesis Testing

UMP Unbiased Two Sided

Hard

Q23.

UMPU test for $H_0: \theta = \theta_0$ vs $H_1: \theta \neq \theta_0$ in exponential family:

- A. **Rejects for $T < c_1$ or $T > c_2$ with c_1, c_2 chosen for size and unbiasedness**
- B. Does not exist
- C. Is the same as the two-sided z-test always
- D. Rejects for $|T - \theta_0| > c$

Correct answer: A

EXAM SHORTCUT

UMPU: two critical values $c_1 < c_2$; conditions: $E_{\theta_0}[\phi] = \alpha$ (size) and $E_{\theta_0}[T\phi] = \alpha E_{\theta_0}[T]$ (unbiasedness); for normal, reduces to equal-tails z-test

Hypothesis Testing

Power Sample Size

Medium

Q24.

To achieve power $1 - \beta$ at $\mu = \mu_1$ for one-sided z-test at level α , required n is:

A. $n = \sigma^2(z_\alpha + z_\beta)^2 / (\mu_1 - \mu_0)^2$

B. $n = \sigma^2(z_\alpha - z_\beta)^2 / (\mu_1 - \mu_0)^2$

C. $n = \sigma^2 z_\alpha^2 / (\mu_1 - \mu_0)^2$

D. $n = \sigma^2(z_{\alpha/2} + z_\beta)^2 / (\mu_1 - \mu_0)^2$

Correct answer: A

EXAM SHORTCUT

Power = $\Phi(\delta\sqrt{n}/\sigma - z_\alpha) = 1 - \beta$; so $\delta\sqrt{n}/\sigma = z_\alpha + z_\beta$; squaring; A; D uses $z_{\alpha/2}$ (two-sided)

Hypothesis Testing

NP Lemma Sufficient Stat

Hard

Q25.

When sufficient statistic T exists, NP lemma critical region can be expressed in terms of T because:

A. T is complete

B. T has minimum variance

C. **Likelihood ratio is a function of T only**

D. T is ancillary

Correct answer: C

EXAM SHORTCUT

By factorization theorem, $f(x; \theta) = g(T; \theta)h(x)$; LR = $g(T; \theta_1)/g(T; \theta_0)$; depends on x only through T

Hypothesis Testing

Power Function Beta Distribution

Hard

Q26.

For $X \sim \text{Beta}(\theta, 1)$, testing $H_0: \theta = 1$ vs $H_1: \theta = 2$, power of MP test at level α :

A. α^2

B. $2\alpha - \alpha^2$

C. $1 - \alpha^2$

D. **$1 - (1 - \alpha)^2$**

Correct answer: D

EXAM SHORTCUT

$f(x; \theta) = \theta x^{\theta-1}$; LR = $2x/1 = 2x$; reject for large x ; $P(X > c | \theta = 1) = 1 - c = \alpha$; $c = 1 - \alpha$; power = $P(X > 1 - \alpha | \theta = 2) = 1 - (1 - \alpha)^2$

Q27.

For scale family $(1/(\theta f))(x/\theta)$ with f log-concave, UMP test of $H_0: \theta \leq \theta_0$ vs $H_1: \theta > \theta_0$:

- A. Rejects for large $X(n)$ (if f is uniform)
- B. ~~Depends on specific f~~
- C. Rejects for large $\sum X_i$ (if f is exponential)
- D. Does not exist in general

Correct answer: B

EXAM SHORTCUT

Sufficient statistic and rejection region depend on the specific scale family; exponential: $\sum X_i$; uniform: $X(n)$; no universal answer

Q28.

For composite $H_0: \mu \geq \mu_0$ vs $H_1: \mu < \mu_0$ (normal, σ known), level- α test:

- A. Rejects for $\bar{X} < \mu_0 - z_{\alpha/2}\sigma/\sqrt{n}$
- B. Rejects for $\bar{X} > \mu_0 + z_{\alpha}\sigma/\sqrt{n}$
- C. ~~Rejects for $\bar{X} < \mu_0 - z_{\alpha}\sigma/\sqrt{n}$~~
- D. Rejects for $|\bar{X} - \mu_0| > z_{\alpha/2}\sigma/\sqrt{n}$

Correct answer: C

EXAM SHORTCUT

One-sided left test; size achieved at $\mu = \mu_0$ (boundary of H_0); use z_{α} not $z_{\alpha/2}$; B is right-sided; D is two-sided

Q29.

Power function $\beta(\theta)$ of a test based on continuous data is:

- A. Continuous only for exponential families
- B. May be discontinuous for discrete distributions
- C. Always differentiable
- D. ~~Always continuous in θ~~

Correct answer: D

EXAM SHORTCUT

For continuous data, power function is continuous (and often differentiable) in θ ; for discrete data, power function may have jumps; D is correct for continuous case

Hypothesis Testing

NP Lemma Multiparameter

Hard

Q30.

For multiparameter family, NP lemma:

A. Applies directly to simple vs simple hypotheses

B. Gives UMP test always

C. Does not apply

D. Requires sufficient statistic

Correct answer: A

EXAM SHORTCUT

NP lemma applies to any simple vs simple hypothesis regardless of dimension; the LR test is MP; extension to composite is harder in multiparameter case

Hypothesis Testing

UMP Exponential Rate

Medium

Q31.

For $X_1, \dots, X_n \sim \text{Exp}(\theta)$ (rate), UMP test of $H_0: \theta = \theta_0$ vs $H_1: \theta > \theta_0$:

A. UMP does not exist

B. Reject for large $\sum X_i$

C. Reject for small $\sum X_i$ (large rate = small mean)

D. Reject for large $\bar{X}$

Correct answer: C

EXAM SHORTCUT

Large rate θ means small mean $1/\theta$; LR increasing in $-\sum X_i$ (or decreasing in $\sum X_i$); reject for SMALL $\sum X_i$; common sign error

Hypothesis Testing

Power Function Normal Two Sided

Hard

Q32.

Power function of two-sided z-test for $H_0: \mu = 0$ vs $H_1: \mu \neq 0$:

A. $\beta(\mu) = \Phi(-z\alpha/2 - \mu\sqrt{n}/\sigma) + 1 - \Phi(z\alpha/2 - \mu\sqrt{n}/\sigma)$

B. $\beta(\mu) = 1 - \Phi(z\alpha/2 - \mu\sqrt{n}/\sigma)$

C. $\beta(\mu) = \Phi(\mu\sqrt{n}/\sigma - z\alpha/2)$

D. $\beta(\mu) = 2\Phi(\mu\sqrt{n}/\sigma - z\alpha/2) - 1$

Correct answer: A

EXAM SHORTCUT

Two-sided: power = P(reject left) + P(reject right); for $\mu > 0$: left tail negligible; A is exact; C and B are one-sided approximations

Q33.

In exponential family $f(x; \theta, \lambda) = h(x) \exp[\eta_1(\theta)T_1 + \eta_2(\lambda)T_2 - B]$, similar test for $H_0: \theta = \theta_0$ (nuisance λ) conditions on:

- A. (T_1, T_2) jointly
- B. Neither
- C. T_1 (sufficient for θ)
- D. T_2 (sufficient for λ)

Correct answer: D

EXAM SHORTCUT

Condition on sufficient statistic for nuisance parameter λ ; conditional distribution of $T_1|T_2$ is free of λ ; gives similar test

Q34.

For $X_1, \dots, X_n \sim \text{Laplace}(\theta, 1)$, UMP test of $H_0: \theta = 0$ vs $H_1: \theta > 0$:

- A. Rejects for large sample median
- B. Rejects for large $\sum X_i$
- C. UMP does not exist
- D. Rejects for large $\sum |X_i|$

Correct answer: C

EXAM SHORTCUT

Laplace is NOT exponential family (absolute value in exponent); no MLR in simple statistic; UMP does not exist; sign test or Wilcoxon are used

Q35.

For $X \sim \text{Bin}(10, p)$, test $H_0: p = 0.5$ vs $H_1: p > 0.5$, reject if $X \geq 8$. Power at $p = 0.7$:

- A. $1 - P(X \leq 7 | p = 0.7)$
- B. Both A and C (equivalent)
- C. $P(X \geq 8 | p = 0.5)$
- D. $P(X \geq 8 | p = 0.7) = C(10, 8)(0.7)^8(0.3)^2 + C(10, 9)(0.7)^9(0.3) + C(10, 10)(0.7)^{10}$

Correct answer: B

EXAM SHORTCUT

Power at $p = 0.7$: compute under H_1 ; D and A are equivalent expressions; C is the size (under H_0)

Q36.

For $X_1, \dots, X_n \sim \Gamma(\alpha, \theta)$ with α known, MP test of $H_0: \theta = \theta_0$ vs $H_1: \theta = \theta_1 < \theta_0$:

- A. Both A and C (equivalent)
- B. Reject for large $\sum X_i$
- C. Reject for large $\bar{X}$
- D. ~~Reject~~ for small $\sum X_i$**

Correct answer: D

EXAM SHORTCUT

$\Gamma(\alpha, \theta)$ with scale θ : $f \propto \theta^{-\alpha} \exp(-\sum x_i / \theta)$; LR = $(\theta_0 / \theta_1)^\alpha \exp[-\sum x_i (1/\theta_1 - 1/\theta_0)]$; since $\theta_1 < \theta_0$: $1/\theta_1 > 1/\theta_0$; LR decreasing in $\sum x_i$; reject for SMALL $\sum X_i$

Q37.

For $N(\mu, \sigma^2)$ with σ^2 unknown, UMP test of $H_0: \mu \leq \mu_0$ vs $H_1: \mu > \mu_0$:

- A. UMP does not exist (nuisance σ^2)
- B. ~~Rejects~~ for large $t = (X - \mu_0) / (S / \sqrt{n})$**
- C. Rejects for large $\bar{X}$
- D. Rejects for large $\bar{X} / S$

Correct answer: B

EXAM SHORTCUT

With nuisance σ^2 , UMP doesn't exist in general; BUT the t -test is UMPU (uniformly most powerful unbiased); B is UMPU, not UMP; A is technically correct for UMP

Q38.

For one-sided test in exponential family, power function is strictly increasing because:

- A. ~~MLR~~ property implies power is monotone in θ**
- B. Sufficient statistic is complete
- C. MLE is consistent
- D. CRLB is attained

Correct answer: A

EXAM SHORTCUT

MLR $\Rightarrow$ monotone power function; this is the fundamental connection between MLR and UMP tests; completeness is separate

Hypothesis Testing

NP Lemma Composite Reduction

Hard

Q39.

For composite $H_1: \theta > \theta_0$ in exponential family, UMP test is found by:

- A. Rao-Blackwell theorem
- B. Maximizing power at $\theta_1 = \theta_0 + 1$
- C. Applying NP lemma to each simple $H_1: \theta = \theta_1$ and noting the MP test is the same for all $\theta_1 > \theta_0$
- D. Using LRT

Correct answer: C

EXAM SHORTCUT

Key insight: MLR ensures the NP critical region (reject for large T) is the same for ALL $\theta_1 > \theta_0$; this common test is UMP

Hypothesis Testing

Critical Region Randomized Discrete

Hard

Q40.

For $X \sim \text{Bin}(5, p)$, testing $H_0: p = 0.5$ at level $\alpha = 0.05$, exact size requires:

- A. Reject if $X \geq 4$ (size = $6/32 > 0.05$, liberal)
- B. Reject if $X \geq 5$ (size = $1/32 < 0.05$, conservative)
- C. Randomization at boundary since $P(X \geq 5 | p = 0.5) = 1/32 < 0.05 < P(X \geq 4 | p = 0.5) = 6/32$
- D. Use normal approximation

Correct answer: C

EXAM SHORTCUT

Exact size 0.05 requires randomization: reject if $X = 5$; reject with prob γ if $X = 4$; $\gamma = (0.05 - 1/32)/(6/32 - 1/32) = (0.05 - 0.03125)/0.15625$

Hypothesis Testing

UMP Negative Binomial

Hard

Q41.

For $X_1, \dots, X_n \sim \text{NegBin}(r, p)$ with r known, UMP test of $H_0: p \geq p_0$ vs $H_1: p < p_0$:

- A. Reject for small $\sum X_i$
- B. Reject for large $\bar{X}$
- C. Reject for large $\sum X_i$
- D. Both A and C (equivalent)

Correct answer: D

EXAM SHORTCUT

NegBin: $E[X] = r(1-p)/p$; smaller p means larger mean; LR in $\sum X_i$; for $p_1 < p_0$: reject for large $\sum X_i$; C and B equivalent

Q42.

Power envelope at θ_1 for testing $H_0: \theta = \theta_0$ at level α is:

- A. ~~All of the above (equivalent)~~
- B. Power of NP MP test at θ_1
- C. $1 - \beta$ where β is Type II error of best test
- D. $\sup_{\theta \in \Theta} \text{level-}\alpha \text{ tests } \beta(\theta_1)$

Correct answer: A

EXAM SHORTCUT

All three definitions are equivalent; NP lemma gives the test achieving the envelope; power envelope is the theoretical maximum

Q43.

For $N(\mu, \sigma^2)$ with both unknown, similar test of $H_0: \mu = \mu_0$ conditions on:

- A. $X_{(1)}$
- B. $\sum X_i^2$
- C. $\bar{X}$
- D. ~~S^2 (sufficient for σ^2 under H_0)~~

Correct answer: D

EXAM SHORTCUT

Under $H_0: \mu = \mu_0$, S^2 is sufficient for σ^2 ; conditioning on S^2 gives t -statistic with $n-1$ distribution free of σ^2 ; this gives the t -test as UMPU

Q44.

For $X_1, \dots, X_n \sim \text{Pareto}(\theta)$, $f(x; \theta) = \theta/x^{\theta+1}$, $x > 1$, UMP test of $H_0: \theta \leq \theta_0$ vs $H_1: \theta > \theta_0$:

- A. Reject for large $\prod X_i$
- B. Both B and C (equivalent since $\log \prod = \sum \log$)
- C. Reject for large $\sum \log X_i$
- D. ~~Reject for small $\sum \log X_i$~~

Correct answer: D

EXAM SHORTCUT

$\sum \log X_i \sim \Gamma(n, 1/\theta)$; larger θ means smaller $\sum \log X_i$; MLR in $-\sum \log X_i$; reject for small $\sum \log X_i$ (= large $-\sum \log X_i$)

Q45.

For $X \sim \text{Beta}(\theta, 1)$, UMP test of $H_0: \theta \leq 1$ vs $H_1: \theta > 1$ rejects for:

- A. Small $\sum \log X_i$ (more negative)
- B. Large $\sum X_i$
- C. **Large $\sum \log X_i$ (less negative, closer to 0)**
- D. Large $X(n)$

Correct answer: C

EXAM SHORTCUT

$\log X_i < 0$ for $X_i \in (0, 1)$; $-\log X_i \sim \text{Exp}(\theta)$; large θ means small $E[-\log X_i] = 1/\theta$; so large θ means $\sum \log X_i$ closer to 0 (less negative); reject for large (less negative) $\sum \log X_i$

Q46.

For location family $f(x-\theta)$ with f symmetric and log-concave, MP test of $H_0: \theta = 0$ vs $H_1: \theta = \delta > 0$:

- A. **Depends on specific f**
- B. Rejects for large $\sum X_i$ (if f is normal)
- C. Always rejects for large $\bar{X}$
- D. Rejects for large $\sum \text{sgn}(X_i)$ (if f is Laplace)

Correct answer: A

EXAM SHORTCUT

MP test depends on f ; normal: reject for large $\bar{X}$; Laplace: reject for large $\sum \text{sgn}(X_i)$ (sign test); Cauchy: complex; C is wrong in general

Q47.

For Hypergeometric(N, M, n) with N, n known, UMP test of $H_0: M \leq M_0$ vs $H_1: M > M_0$:

- A. Reject for small X
- B. **Reject for large X (number of successes in sample)**
- C. Reject for $X > M_0 n / N$
- D. UMP does not exist (not exponential family)

Correct answer: B

EXAM SHORTCUT

Hypergeometric has MLR in X ; UMP exists and rejects for large X ; not exponential family but MLR still holds

Q48.

For composite $H_0 : \theta \in [0, 1]$ vs $H_1 : \theta > 1$, size of test is:

- A. $\beta(1)$
- B. $\inf_{\theta \in [0,1]} \beta(\theta)$
- C. $\beta(0)$
- D. $\sup_{\theta \in [0,1]} \beta(\theta)$

Correct answer: D

EXAM SHORTCUT

Size = supremum of power function over H_0 ; for monotone power function, supremum achieved at boundary $\theta = 1$; so $D = \beta(1)$ in this case

Q49.

If T is sufficient and NP critical region is $\{x: \Lambda(x) > k\}$, then equivalently:

- A. $\{x: g(T(x)) > k'\}$ for some function g and constant k'
- B. The critical region cannot be simplified
- C. $\{x: T(x) > k'\}$ always
- D. $\{x: T(x) > k'\}$ always

Correct answer: A

EXAM SHORTCUT

LR is a function of T (by factorization); so critical region is a function of T ; D is true only if LR is monotone in T ; A is the general statement

Q50.

UMP test exists for $H_0 : \theta = \theta_0$ vs $H_1 : \theta \neq \theta_0$ (two-sided) when:

- A. Distribution is from exponential family
- B. Never (for two-sided alternatives in general)
- C. Distribution is normal
- D. Distribution has MLR

Correct answer: B

EXAM SHORTCUT

UMP for two-sided alternatives generally does NOT exist; UMPU exists for exponential families; this is a fundamental limitation of UMP theory

Likelihood Ratio Tests, SPRT & Decision Theory

55 forecast questions in this topic

Hypothesis Testing

LRT Definition

Medium

Q1.

Likelihood ratio test statistic Λ is:

A. ~~Both A and C~~ (equivalent)

B. $\Lambda = \sup_{\theta \in \Theta_0} L(\theta) / \sup_{\theta \in \Theta} L(\theta)$

C. $\Lambda = \sup_{\theta \in \Theta} L(\theta) / \sup_{\theta \in \Theta_0} L(\theta)$

D. $\Lambda = L(\hat{\theta}_0) / L(\hat{\theta})$ where $\hat{\theta}_0$ is restricted MLE

Correct answer: A

EXAM SHORTCUT

$\Lambda \in [0, 1]$; reject for SMALL Λ (equivalently large $-2 \log \Lambda$); C is ≥ 1 always; B and D are equivalent

Hypothesis Testing

LRT Wilks Theorem

Medium

Q2.

Under H_0 and regularity conditions, $-2 \log \Lambda \xrightarrow{d} :$

A. ~~χ^2_r~~ where $r = \dim(\Theta) - \dim(\Theta_0)$

B. χ^2_n

C. $F_{r, n-r}$

D. $N(0, 1)$

Correct answer: A

EXAM SHORTCUT

Wilks' theorem: $r =$ number of restrictions (difference in parameter dimensions); for testing k parameters: $r = k$; requires regularity conditions

Q3.

For $N(\mu, \sigma^2)$ with both unknown, LRT of $H_0: \mu = \mu_0$ vs $H_1: \mu \neq \mu_0$:

- A. Reduces to Z-test
- B. Has no closed form
- C. **Reduces to t -test: reject for large $|t| = |\bar{X} - \mu_0|/(S/\sqrt{n})$**
- D. Reduces to χ^2 test

Correct answer: C

EXAM SHORTCUT

LRT statistic: $-2 \log \Lambda = n \log[1 + t^2/(n-1)]$ which is monotone in $|t|$; LRT is equivalent to t -test; $r = 1$ df

Q4.

Wald's SPRT continues sampling while:

- A. $\Lambda_m > A$
- B. $\Lambda_m < B$
- C. **$B < \Lambda_m < A$ where $\Lambda_m = \prod_{i=1}^m f(X_i; \theta_1)/f(X_i; \theta_0)$**
- D. $m < n_0$ (fixed sample size)

Correct answer: C

EXAM SHORTCUT

SPRT: continue if $B < \Lambda_m < A$; stop and reject H_0 if $\Lambda_m \geq A$; stop and accept H_0 if $\Lambda_m \leq B$

Q5.

Wald's approximate SPRT boundaries are:

- A. $A = 1 - \beta, B = \alpha$
- B. $A = 1/\alpha, B = \beta$
- C. **$A \approx (1 - \beta)/\alpha, B \approx \beta/(1 - \alpha)$**
- D. $A \approx (1 - \alpha)/\beta, B \approx \alpha/(1 - \beta)$

Correct answer: C

EXAM SHORTCUT

$A = (1 - \beta)/\alpha$ (upper boundary for rejecting H_0); $B = \beta/(1 - \alpha)$ (lower boundary for accepting H_0); these are approximate (ignore overshoot)

Q6.

For $X_i \sim N(\theta, 1)$, SPRT of $H_0: \theta = 0$ vs $H_1: \theta = \delta$, log-LR after m observations:

- A. $\delta \sum_{i=1}^m X_i - \delta^2/2$
- B. $\delta \sum_{i=1}^m X_i - m\delta^2/2$
- C. $\delta \sum_{i=1}^m X_i^2$
- D. $\sum_{i=1}^m X_i^2$

Correct answer: B

EXAM SHORTCUT

$\log \Lambda_m = \sum [\log f(X_i; \delta) - \log f(X_i; 0)] = \delta \sum X_i - m\delta^2/2$; linear in $\sum X_i$; SPRT reduces to boundary on $\sum X_i$

Q7.

Operating Characteristic (OC) function $L(\theta)$ of SPRT is:

- A. $P_{\theta}(\text{reject } H_0)$
- B. $P_{\theta}(\text{accept } H_0)$
- C. $E_{\theta}[N]$ (expected sample size)
- D. Power function

Correct answer: B

EXAM SHORTCUT

OC function = probability of accepting H_0 ; $L(\theta_0) = 1 - \alpha$; $L(\theta_1) = \beta$; power function = $1 - L(\theta)$; C is ASN function

Q8.

ASN (Average Sample Number) function $E_{\theta}[N]$ for SPRT satisfies Wald's identity:

- A. $E_{\theta}[N] = n(\text{fixed})$
- B. $E_{\theta}[N] \approx L(\theta) \log(B/A) + (1 - L(\theta)) \log(A/B) / E_{\theta}[\log f(X; \theta_1)/f(X; \theta_0)]$ (Wald's approximation)
- C. $E_{\theta}[N] = \log(A/B) / E_{\theta}[\log \Lambda_1]$
- D. $E_{\theta}[N] = 1/\alpha$

Correct answer: B

EXAM SHORTCUT

Wald's approximation for ASN uses OC function $L(\theta)$ and expected log-LR per observation; C is a special case; exact formula involves overshoot correction

Hypothesis Testing

SPRT Optimality

Hard

Q9.

Wald-Wolfowitz theorem states SPRT is optimal in that:

- A. SPRT minimizes maximum expected sample size
- B. **Among all tests with error probabilities $\leq \alpha$ and $\leq \beta$, SPRT minimizes $E\theta_0[N]$ and $E\theta_1[N]$ simultaneously**
- C. SPRT is UMP among sequential tests
- D. SPRT minimizes total error probability

Correct answer: B

EXAM SHORTCUT

Wald-Wolfowitz: SPRT minimizes ASN at BOTH θ_0 and θ_1 simultaneously; no other test with same error bounds can have smaller ASN at either point

Hypothesis Testing

LRT Composite Both

Hard

Q10.

LRT for $H_0: \theta \in \Theta_0$ vs $H_1: \theta \in \Theta_1 = \Theta \setminus \Theta_0$:

- A. $\Lambda = L(\theta_0)/L(\theta)$; reject for small Λ
- B. $\Lambda = \sup_{\Theta_1} L / \sup_{\Theta_0} L$; reject for large Λ
- C. **Both A and B (equivalent)**
- D. $\Lambda = \sup_{\Theta_0} L / \sup_{\Theta} L$; reject for small Λ

Correct answer: C

EXAM SHORTCUT

D and A are equivalent; $\sup_{\Theta} L = \max(\sup_{\Theta_0} L, \sup_{\Theta_1} L)$; when $\theta \in \Theta_0$, $\sup_{\Theta} L = L(\theta)$; reject for small Λ (large $-2 \log \Lambda$)

Hypothesis Testing

Decision Theory Loss

Medium

Q11.

In decision theory, risk function $R(\theta, d)$ is:

- A. $\sup_{\theta} L(\theta, d(X))$
- B. $L(\theta, d(X))$ — loss function
- C. **$E_{\theta}[L(\theta, d(X))]$ — expected loss**
- D. $P_{\theta}(\text{wrong decision})$

Correct answer: C

EXAM SHORTCUT

Risk = expected loss; loss function is random (depends on X); risk is its expectation; D is risk for 0-1 loss only

Q12.

Estimator d_1 dominates d_2 if:

- A. $R(\theta, d_1) \leq R(\theta, d_2)$ for all θ with strict inequality for some θ
- B. $\sup_{\theta} R(\theta, d_1) < \sup_{\theta} R(\theta, d_2)$
- C. $R(\theta, d_1) \leq R(\theta, d_2)$ for all θ
- D. $R(\theta, d_1) < R(\theta, d_2)$ for all θ

Correct answer: A

EXAM SHORTCUT

Dominance: $\leq$ everywhere AND $<$ somewhere; D is too strong (strict everywhere); C allows equality everywhere (no dominance); B is minimax comparison

Q13.

Minimax estimator minimizes:

- A. $\int R(\theta, d) \pi(\theta) d\theta$ — Bayes risk
- B. $R(\theta_0, d)$ at a specific θ_0
- C. $E_{\theta}[d(X)]$ — expected estimate
- D. $\sup_{\theta} R(\theta, d)$ — worst-case risk

Correct answer: D

EXAM SHORTCUT

Minimax: minimize maximum risk; Bayes: minimize average risk (B); minimax estimator is often Bayes w.r.t. least favorable prior

Q14.

Bayes estimator under squared error loss is:

- A. Posterior median
- B. Posterior mode (MAP)
- C. Prior mean
- D. Posterior mean $E[\theta|X]$

Correct answer: D

EXAM SHORTCUT

Squared error: posterior mean; absolute error: posterior median; 0-1 loss: posterior mode; these are fundamental results

Hypothesis Testing

SPRT Poisson

Hard

Q15.

For $X_i \sim \text{Poisson}(\lambda)$, SPRT of $H_0: \lambda = \lambda_0$ vs $H_1: \lambda = \lambda_1 > \lambda_0$, log-LR:

- A. $-X \log(\lambda_1/\lambda_0)$
- B. $m(\lambda_1 - \lambda_0)$
- C. $\sum X_i \log(\lambda_1/\lambda_0)$
- D. $-\sum X_i \log(\lambda_1/\lambda_0) - m(\lambda_1 - \lambda_0)$

Correct answer: D

EXAM SHORTCUT

$\log \Lambda = \sum [x_i \log(\lambda_1/\lambda_0) - (\lambda_1 - \lambda_0)] = \sum X_i \log(\lambda_1/\lambda_0) - m(\lambda_1 - \lambda_0)$; linear in $\sum X_i$; SPRT boundary on $\sum X_i$

Hypothesis Testing

LRT Independence

Hard

Q16.

LRT for independence in $r \times c$ contingency table: $-2 \log \Lambda \xrightarrow{d} \chi^2_{(r-1)(c-1)}$ because:

- A. Degrees of freedom = rc
- B. Wilks theorem gives χ^2_{r-1} always
- C. $\dim(\Theta) = rc - 1$, $\dim(\Theta_0) = (r-1) + (c-1)$; $r = (rc-1) - (r+c-2) = (r-1)(c-1)$
- D. Degrees of freedom = $r + c - 2$

Correct answer: C

EXAM SHORTCUT

Saturated model: $rc - 1$ free parameters; independence model: $(r-1) + (c-1)$ free parameters; difference = $(r-1)(c-1)$; this is the df for χ^2

Hypothesis Testing

Decision Theory Bayes Risk

Hard

Q17.

Bayes risk of estimator d w.r.t. prior π is:

- A. $r(\pi, d) = E[L(\theta, d(X))]$ over joint distribution
- B. $r(\pi, d) = \int R(\theta, d) \pi(\theta) d\theta = E\pi[R(\theta, d)]$
- C. $r(\pi, d) = R(\theta_0, d)$ at prior mean
- D. $r(\pi, d) = \sup_{\theta} R(\theta, d)$

Correct answer: A

EXAM SHORTCUT

B and A are equivalent (both are expectation of loss over joint distribution of (θ, X)); D is minimax risk; Bayes estimator minimizes B=A

Q18.

For $X_i \sim \text{Exp}(\theta)$ (rate), SPRT of $H_0: \theta = \theta_0$ vs $H_1: \theta = \theta_1 > \theta_0$, SPRT boundary on $\sum X_i$:

A. ~~Accept H_0 if $\sum X_i > b$, reject if $\sum X_i < a$ (for some $a < b$)~~

B. Reject for small $\sum X_i$

C. Reject for large $\sum X_i$

D. Accept H_0 if $\sum X_i < a$, reject if $\sum X_i > b$

Correct answer: A

EXAM SHORTCUT

Large θ (rate) means small mean; $\log\text{-LR} = n \log(\theta_1/\theta_0) - (\theta_1 - \theta_0) \sum X_i$; decreasing in $\sum X_i$; large $\sum X_i$ favors H_0 (small rate); accept H_0 for large $\sum X_i$

Q19.

LRT for $H_0: \sigma^2 = \sigma_0^2$ vs $H_1: \sigma^2 \neq \sigma_0^2$ (normal, μ unknown):

A. Has no closed form

B. Reduces to Z-test

C. Reduces to F-test

D. ~~Reduces to χ^2 test: reject for large or small $(n-1)S^2/\sigma_0^2$~~

Correct answer: D

EXAM SHORTCUT

LRT statistic: $-2 \log \Lambda = n[\log(\sigma^2/\sigma_0^2) + \sigma_0^2/(2\sigma^2) - 1]$ which is monotone function of $(n-1)S^2/\sigma_0^2$; equivalent to χ^2 test

Q20.

For SPRT of $H_0: \theta = 0$ vs $H_1: \theta = \delta$ with $X_i \sim N(\theta, 1)$, OC function $L(\theta)$ satisfies:

A. $L(\theta) = \Phi(-\theta/\sqrt{n})$

B. ~~$L(\theta) = Ah(\theta) - 1$ where $h(\theta)$ solves $E\theta[\exp(h \log \Lambda)] = 1$~~

C. $L(\theta) = \beta$ for all $\theta \in H_1$

D. $L(\theta) = 1 - \Phi(\theta/\sqrt{n})$

Correct answer: B

EXAM SHORTCUT

Wald's OC approximation uses moment generating function of $\log\text{-LR}$; $h(\theta)$ is the non-trivial root of $E\theta[\Lambda(h)/(1)] = 1$; complex but standard formula

Q21.

Complete class theorem states:

- A. Every admissible estimator is Bayes (or limit of Bayes)
- B. Admissible estimators form a complete class
- C. Every Bayes estimator is admissible
- D. Both A and B**

Correct answer: D

EXAM SHORTCUT

C: Bayes estimators with proper priors are admissible; A: admissible estimators are Bayes (or generalized Bayes); together they characterize admissibility via Bayes

Q22.

LRT for $H_0: \beta_1 = \dots = \beta_k = 0$ in multiple regression (normal errors):

- A. Has no closed form
- B. Reduces to χ^2 -test
- C. Reduces to t -test
- D. Reduces to F -test: $F = (RSS_0 - RSS_1) / (k\sigma^2)$**

Correct answer: D

EXAM SHORTCUT

$LRT = -2 \log \Lambda = n \log(RSS_0/RSS_1)$; for normal errors, this is monotone in F -statistic; LRT and F -test are equivalent

Q23.

For $X_i \sim \text{Bernoulli}(p)$, SPRT of $H_0: p = p_0$ vs $H_1: p = p_1 > p_0$, log-LR:

- A. $m \cdot \log[(1-p_1)/(1-p_0)]$
- B. $\sum X_i \log[p_1(1-p_0)/(p_0(1-p_1))] + m \cdot \log[(1-p_1)/(1-p_0)]$**
- C. $\bar{X} \cdot \log(p_1/p_0) + \log[(1-p_1)/(1-p_0)]$
- D. $\sum X_i \log(p_1/p_0)$

Correct answer: B

EXAM SHORTCUT

$\log \Lambda_m = \sum X_i \log(p_1/p_0) + \sum (1-X_i) \log[(1-p_1)/(1-p_0)] = \sum X_i \log[p_1(1-p_0)/(p_0(1-p_1))] + m \cdot \log[(1-p_1)/(1-p_0)]$; linear in $\sum X_i$.

Q24.

For $X \sim N(\theta, 1)$, minimax estimator of θ under squared error loss is:

- A. 0 (constant)
- B. $\bar{X}$ (since it is admissible and has constant risk)
- C. James-Stein estimator
- D. $\bar{X}$ (sample mean)

Correct answer: B

EXAM SHORTCUT

For $p=1$: $\bar{X}$ is minimax (constant risk = $1/n$, admissible); for $p \geq 3$: James-Stein dominates $\bar{X}$ in MSE but $\bar{X}$ is still minimax; C is wrong for $p=1$

Q25.

LRT for goodness of fit (multinomial): $G^2 = -2 \log \Lambda = 2 \sum O_i \log(O_i/E_i) \xrightarrow{d} \chi^2_{k-p}$:

- A. χ^2_k
- B. $F_{k-1, n-k}$
- C. χ^2_{k-1-p} where k = cells, p = estimated parameters
- D. χ^2_{k-1}

Correct answer: C

EXAM SHORTCUT

G^2 (log-likelihood ratio) and Pearson χ^2 are asymptotically equivalent; $df = k-1-p$ (subtract 1 for total constraint and p for estimated parameters)

Q26.

Minimax estimator equals Bayes estimator w.r.t. least favorable prior π^* when:

- A. Prior is uniform
- B. Bayes risk equals minimax risk: $r(\pi^*, d^*) = \sup_{\theta} R(\theta, d^*)$
- C. Loss function is squared error
- D. Sample size is large

Correct answer: B

EXAM SHORTCUT

Least favorable prior: prior that maximizes Bayes risk; if Bayes estimator has constant risk, it is minimax; key connection between Bayes and minimax

Q27.

SPRT has smaller ASN than fixed sample test because:

- A. It uses sufficient statistics
- B. It has smaller error probabilities
- C. **It stops early when evidence is strong (in either direction)**
- D. It uses more information per observation

Correct answer: C

EXAM SHORTCUT

SPRT's efficiency comes from early stopping; when data strongly favors one hypothesis, no need to continue; fixed sample test always uses n observations

Q28.

For exponential family, LRT of $H_0: \eta = \eta_0$ vs $H_1: \eta \neq \eta_0$:

- A. $-2 \log \Lambda = 2 \sum \log f(X_i; \hat{\eta}) / f(X_i; \eta_0)$
- B. **All equivalent asymptotically**
- C. $-2 \log \Lambda = n(\hat{\eta} - \eta_0)' I(\eta_0) (\hat{\eta} - \eta_0)$
- D. $-2 \log \Lambda = 2n[B(\hat{\eta}) - B(\eta_0) - (\hat{\eta} - \eta_0)' \nabla B(\eta_0)]$ where B is log-partition

Correct answer: B

EXAM SHORTCUT

All three are asymptotically equivalent (Wald, LRT, Score); D is exact LRT; C is Wald; they differ in finite samples

Q29.

Regret of decision d at θ is:

- A. $R(\theta, d)$ — total risk
- B. $E[L(\theta, d(X))]$
- C. **$R(\theta, d) - \inf_{d'} R(\theta, d')$ — excess risk over best possible**
- D. $\sup_{\theta} R(\theta, d) - R(\theta, d)$

Correct answer: C

EXAM SHORTCUT

Regret = how much worse than oracle (who knows θ); minimax regret minimizes worst-case regret; different from minimax risk

Q30.

LRT for $H_0: \mu_1 = \mu_2$ (two independent normal samples, equal σ^2 unknown):

A. Reduces to two-sample t -test

B. Has no closed form

C. Reduces to Z-test

D. Reduces to F -test

Correct answer: A

EXAM SHORTCUT

LRT statistic is monotone in t^2 ; equivalent to two-sample t -test with pooled variance; $r = 1$ df for Wilks

Q31.

For $X_i \sim \text{Geometric}(p)$, SPRT of $H_0: p = p_0$ vs $H_1: p = p_1 < p_0$, log-LR:

A. $-\sum X_i \log[(1-p_1)/(1-p_0)] + m \cdot \log(p_1/p_0)$

B. $\sum X_i \log(p_1/p_0)$

C. $m \cdot \log(p_1/p_0)$

D. $\bar{X} \cdot \log[(1-p_1)/(1-p_0)]$

Correct answer: A

EXAM SHORTCUT

$\log f(x;p) = \log p + (x-1)\log(1-p)$; $\log \Lambda_m = m \cdot \log(p_1/p_0) + \sum (X_i - 1) \log[(1-p_1)/(1-p_0)] = \sum X_i \log[(1-p_1)/(1-p_0)] + m \cdot \log(p_1/p_0)$.

Q32.

James-Stein estimator $\hat{\theta}_{JS} = (1 - \frac{p-2}{n\|\bar{X}\|^2})\bar{X}$ for $p \geq 3$:

A. Is minimax

B. Dominates $\bar{X}$ and is admissible

C. Is admissible

D. Dominates $\bar{X}$ in total MSE but is inadmissible itself

Correct answer: D

EXAM SHORTCUT

James-Stein dominates $\bar{X}$ for $p \geq 3$ (Stein's paradox); but JS itself is inadmissible — the positive-part JS estimator $\hat{\theta}_{JS+} = \max(0, 1 - \frac{p-2}{n\|\bar{X}\|^2})\bar{X}$ dominates JS.

Hypothesis Testing

LRT Homogeneity

Hard

Q33.

LRT for homogeneity of k binomial proportions $H_0: p_1 = \dots = p_k$:

- A. $G^2 \xrightarrow{d} \chi^2_{2k-2}$
- B. $G^2 \xrightarrow{d} F_{k-1, n-k}$
- C. $G^2 = 2 \sum_{i,j} O_{ij} \log(O_{ij}/E_{ij}) \xrightarrow{d} \chi^2_{k-1}$
- D. $G^2 \xrightarrow{d} \chi^2_k$

Correct answer: C

EXAM SHORTCUT

k proportions, 1 constraint (common p): $r = k-1$ df; same as Pearson χ^2 for $2 \times k$ table

Hypothesis Testing

SPRT Truncated

Hard

Q34.

Truncated SPRT (with maximum sample size $n_{\max}$):

- A. Is always more efficient than SPRT
- B. Is equivalent to fixed sample test
- C. Has same error probabilities as SPRT
- D. Stops at $n_{\max}$ and makes decision based on accumulated evidence

Correct answer: D

EXAM SHORTCUT

Truncated SPRT: if boundaries not crossed by $n_{\max}$, make decision at $n_{\max}$; error probabilities change slightly; practical compromise between SPRT and fixed sample

Hypothesis Testing

Decision Theory Posterior Risk

Hard

Q35.

Posterior expected loss (posterior risk) of decision d given $X = x$ is:

- A. $\rho(\pi, d|x) = \int L(\theta, d) \pi(\theta) d\theta$
- B. $\rho(\pi, d|x) = R(\theta, d)$
- C. $\rho(\pi, d|x) = \int L(\theta, d) \pi(\theta|x) d\theta$
- D. $\rho(\pi, d|x) = E[L(\theta, d)|X = x]$

Correct answer: D

EXAM SHORTCUT

C and D are equivalent (posterior expectation of loss); A uses prior not posterior; Bayes estimator minimizes posterior risk for each x

Q36.

For nested models $M_0 \subset M_1$, LRT statistic $-2 \log \Lambda$:

- A. **All of the above**
- B. Is always non-negative
- C. Equals $2[\ell(\theta_1) - \ell(\theta_0)]$ where ℓ is log-likelihood
- D. Has χ^2 distribution under M_0

Correct answer: A

EXAM SHORTCUT

All three: B (log-likelihood increases with more parameters); C (definition); D (Wilks theorem); all hold under regularity

Q37.

For composite hypotheses $H_0: \theta \leq \theta_0$ vs $H_1: \theta \geq \theta_1$ ($\theta_0 < \theta_1$), SPRT:

- A. Does not exist
- B. **Uses $\Lambda = f(X; \theta_1)/f(X; \theta_0)$ (simple vs simple at boundaries)**
- C. Uses generalized likelihood ratio
- D. Uses MLE at each stage

Correct answer: B

EXAM SHORTCUT

For composite hypotheses with indifference zone (θ_0, θ_1) , SPRT uses simple hypotheses at boundaries; error probabilities guaranteed for all $\theta \leq \theta_0$ and $\theta \geq \theta_1$

Q38.

Invariant decision rules under group G of transformations satisfy:

- A. $d(gx) = g'd(\theta)$ for all $g \in G$ (equivariance)
- B. **Both A and C**
- C. Risk is constant over orbits of G
- D. $d(gx) = d(x)$ for all $g \in G$

Correct answer: B

EXAM SHORTCUT

Equivariant estimators have constant risk over orbits; best equivariant estimator minimizes risk within equivariant class; Pitman estimator is best location equivariant

Q39.

Bartlett correction for LRT improves χ^2 approximation by:

- A. Adding b/n to $-2 \log \Lambda$
- B. Using χ^2_{r+1} instead of χ^2_r
- C. Multiplying by $n/(n-r)$
- D. **Dividing $-2 \log \Lambda$ by $1 + b/n$ where b is Bartlett factor**

Correct answer: D

EXAM SHORTCUT

Bartlett: $(-2 \log \Lambda)/(1 + b/n) \sim \chi^2_r$ to order $O(n^{-2})$; improves from $O(n^{-1})$ error; b computed from cumulants of log-likelihood

Q40.

SPRT efficiency relative to fixed sample test (same error probabilities) is measured by:

- A. α/β
- B. Power ratio
- C. $E\theta[N]/n$
- D. **$n/E\theta[N]$ — ratio of fixed to expected sequential sample size**

Correct answer: D

EXAM SHORTCUT

Efficiency > 1 means SPRT uses fewer observations on average; for normal testing, SPRT efficiency $\approx 50\%$ savings at H_0 and H_1 ; efficiency < 1 would mean SPRT is worse

Q41.

Under 0-1 loss $L(\theta, d) = I(\theta \neq d)$, Bayes estimator is:

- A. **Posterior mode (MAP)**
- B. Prior mode
- C. Posterior median
- D. Posterior mean

Correct answer: A

EXAM SHORTCUT

0-1 loss: minimize $P(\theta \neq d|X) = 1 - P(\theta = d|X)$; maximize $P(\theta = d|X)$; this is the posterior mode; for continuous θ , 0-1 loss requires careful definition

Hypothesis Testing

LRT Multivariate Normal

Hard

Q42.

LRT for $H_0: \mu = \mu_0$ in $Np(\mu, \Sigma)$ with Σ known:

A. $-2 \log \Lambda = n(X - \mu_0)^T S^{-1} (X - \mu_0) \sim \chi^2_p$

B. $-2 \log \Lambda = (X - \mu_0)^T \Sigma^{-1} (X - \mu_0) \sim \chi^2_p$

C. $-2 \log \Lambda = n(X - \mu_0)^T \Sigma^{-1} (X - \mu_0) \sim \chi^2_p$

D. $-2 \log \Lambda \sim F_{p, n-p}$

Correct answer: C

EXAM SHORTCUT

$n(X - \mu_0)^T \Sigma^{-1} (X - \mu_0) = T^2$ (Hotelling's T^2 when Σ known); $\sim \chi^2_p$; A uses S (unknown Σ case) giving $T^2 \sim (p(n-1))/(n-p) F_{p, n-p}$

Hypothesis Testing

SPRT Normal Variance

Hard

Q43.

For $X_i \sim N(0, \sigma^2)$, SPRT of $H_0: \sigma^2 = \sigma^2_0$ vs $H_1: \sigma^2 = \sigma^2_1 > \sigma^2_0$, log-LR:

A. $m \cdot \log(\sigma_1/\sigma_0)$

B. $\sum X_i^2 / \sigma^2$

C. $\sum X_i^2 (1/\sigma^2_0 - 1/\sigma^2_1)$

D. $(m/2) \log(\sigma^2_0/\sigma^2_1) + (\sum X_i^2)(1/\sigma^2_0 - 1/\sigma^2_1)$

Correct answer: D

EXAM SHORTCUT

$\log \Lambda_m = \Sigma [\log f(X_i; \sigma^2_1) - \log f(X_i; \sigma^2_0)] = (m/2) \log(\sigma^2_0/\sigma^2_1) + (\sum X_i^2)(\sigma^{-2}_0 - \sigma^{-2}_1)$; linear in $\sum X_i^2$.

Hypothesis Testing

Decision Theory Shrinkage

Hard

Q44.

Shrinkage estimators are preferred in high dimensions because:

A. They reduce total MSE by trading bias for variance reduction

B. They attain CRLB

C. They are minimax

D. They are always unbiased

Correct answer: A

EXAM SHORTCUT

Shrinkage: introduce bias but reduce variance; total MSE decreases; Stein paradox shows this is beneficial for $p \geq 3$; C: James-Stein is minimax but not all shrinkage estimators are

Hypothesis Testing

LRT Poisson Regression

Hard

Q45.

LRT for $H_0: \beta = 0$ in Poisson regression $\log \mu_i = \alpha + \beta x_i$:

- A. $-2 \log \Lambda \xrightarrow{d} N(0, 1)$
- B. $-2 \log \Lambda \xrightarrow{d} \chi^2_2$
- C. $-2 \log \Lambda = 2[\ell(\hat{\alpha}, \hat{\beta}) - \ell(\alpha_0, 0)] \xrightarrow{d} \chi^2_1$
- D. $-2 \log \Lambda \xrightarrow{d} F_{1, n-2}$

Correct answer: C

EXAM SHORTCUT

One restriction ($\beta = 0$): $r = 1$ df; Wilks theorem applies; χ^2_1 ; equivalent to Wald test $(\beta/\text{SE})^2$ asymptotically

Hypothesis Testing

SPRT Stopping Time

Hard

Q46.

SPRT stopping time N satisfies:

- A. $N \rightarrow \infty$ as $\alpha, \beta \rightarrow 0$
- B. $E\theta[N] < \infty$ for all θ under regularity conditions
- C. Both A and B
- D. $N = n_0$ always (fixed)

Correct answer: C

EXAM SHORTCUT

B: SPRT terminates with probability 1 (Wald); A: as error bounds tighten, more observations needed; both are true

Hypothesis Testing

Decision Theory Empirical Bayes

Hard

Q47.

Empirical Bayes estimates hyperparameters of prior by:

- A. Bootstrap
- B. Maximizing marginal likelihood $m(x|\lambda) = \int f(x|\theta)\pi(\theta|\lambda)d\theta$
- C. Cross-validation
- D. Using flat prior

Correct answer: B

EXAM SHORTCUT

Empirical Bayes: estimate λ from data (marginal likelihood or method of moments); then use posterior with estimated prior; James-Stein is an empirical Bayes estimator

Hypothesis Testing

LRT Survival

Hard

Q48.

Log-rank test for comparing two survival curves is:

A. Asymptotically equivalent to LRT under proportional hazards

B. Exact LRT for exponential survival

C. Equivalent to t -test on survival times

D. Equivalent to Wilcoxon test

Correct answer: A

EXAM SHORTCUT

Log-rank test is score test (not LRT) under Cox model; asymptotically equivalent to LRT; exact LRT for exponential gives different statistic; D: Wilcoxon is different weighting

Hypothesis Testing

SPRT Curtailed

Hard

Q49.

Curtailed sampling (stop when outcome is determined) differs from SPRT in that:

A. Curtailment uses likelihood ratio boundaries

B. Curtailment and SPRT are equivalent

C. Curtailment is optimal

D. Curtailment stops when final decision is certain regardless of remaining observations

Correct answer: D

EXAM SHORTCUT

Curtailment: stop when remaining observations cannot change decision; used in clinical trials; SPRT uses probabilistic boundaries; different stopping rules

Hypothesis Testing

Decision Theory Posterior Odds

Hard

Q50.

Posterior odds ratio for H_1 vs H_0 is:

A. $P(X|H_1)/P(X|H_0)$ (Bayes factor only)

B. $P(H_1|X)$ alone

C. $P(H_1|X)/P(H_0|X) = (P(X|H_1)/P(X|H_0)) \cdot P(H_1)/P(H_0)$ (Bayes factor $\times$ prior odds)

D. $1 - p$ -value

Correct answer: C

EXAM SHORTCUT

Posterior odds = Bayes factor $\times$ prior odds; Bayes factor alone (B) ignores prior; p -value is frequentist (D); C is the complete Bayesian update

Q51.

For curved exponential family (e.g., $N(\theta, \theta^2)$), Wilks theorem:

- A. Requires larger sample size
- B. **Still holds: $-2 \log \Lambda \xrightarrow{d} \chi^2_r$**
- C. Fails completely
- D. Holds with modified df

Correct answer: B

EXAM SHORTCUT

Wilks theorem holds for curved exponential families under regularity; r = number of restrictions; the curvature affects finite-sample performance but not asymptotic distribution

Q52.

For testing among $k > 2$ hypotheses sequentially, SPRT generalizes to:

- A. Sequential probability ratio test with $k(k-1)/2$ pairwise boundaries
- B. **Simultaneous SPRT (Armitage)**
- C. Group sequential design
- D. Repeated significance testing

Correct answer: B

EXAM SHORTCUT

Armitage's simultaneous SPRT: continue until one hypothesis is accepted; boundaries involve all k hypotheses; more complex than pairwise

Q53.

For normal location problem with squared error loss, complete class consists of:

- A. All linear estimators
- B. All unbiased estimators
- C. **All Bayes estimators w.r.t. normal priors and their limits**
- D. Only the sample mean

Correct answer: C

EXAM SHORTCUT

For normal location, complete class = Bayes estimators w.r.t. normal priors (conjugate family) and limits; includes James-Stein type estimators; D is too restrictive

Q54.

When true parameter is on boundary of H_0 (e.g., $H_0: \theta \geq 0$, true $\theta = 0$), $-2 \log \Lambda \xrightarrow{d} :$

- A. χ^2_0 (degenerate at 0)
- B. $N(0, 1)$
- C. χ^2_1
- D. $(1)/(2\chi^2_0) + (1)/(2\chi^2_1)$ (mixture)

Correct answer: D

EXAM SHORTCUT

Boundary case: $-2 \log \Lambda$ is 0 with prob 1/2 (when $\hat{\theta} < 0$, constrained MLE = 0) and χ^2_1 with prob 1/2; mixture distribution

Q55.

SPRT has power one property: for any $\theta \neq \theta_0$:

- A. $P_\theta(\text{SPRT eventually terminates}) = 1$
- B. $P_\theta(\text{reject } H_0) \rightarrow 1 \text{ as } n \rightarrow \infty$
- C. Both A and B
- D. Neither (SPRT may not terminate)

Correct answer: C

EXAM SHORTCUT

SPRT terminates with probability 1 (A); and is consistent (B); both follow from Wald's fundamental identity and the strong law of large numbers applied to the log-likelihood ratio (incomplete in source — trap explanation cuts off here)